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APUL

NEET FLIX

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Exercise – I
Breathing and Exchange of Gases

1. Read the following statements and select the incorrect ?
(1) Respiratory rhythm centre present in medulla region.
(2) A chemosensitive area is highly sensitive to CO_2 & H^+.
(3) Pneumotaxic centre can reduce the duration of inspiration and there by decrease the respiratory rate.
(4) The role oxygen in the regulation of respiratory rhythm is quite insignificant.

BBG319

2. Read the following statements:-
(A) It is a chronic disorder
(B) Major cause of it is cigarette smoking.
(C) Alveolar walls are damaged
(D) Respiratory surface is decreased.
(E) It is due to inflammation of bronchi & bronchioles.
How many of the above statements are correct about "emphysema" disorder ?
(1) Two (2) Three
(3) Four (4) All

BBG320

3. What will happen when pneumotaxic centre provides weak signal?
(1) Respiratory rate will be increased
(2) Respiratory rate will be decreased
(3) Respiratory rate will not change
(4) Respiratory rate will be more increased

BBG321

4. Match the following column-I and column-II and select the correct option?

Column-I		Column-II	
(A)	Eupnoea	(i)	O_2 conc. Low at tissue level
(B)	Hypoxia	(ii)	Decreased respiratory rate
(C)	Anoxia	(iii)	Normal respiratory rate
(D)	Bradypnoea	(iv)	O_2 conc. Absent at tissue level

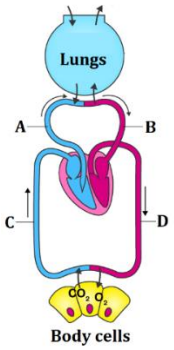
(1) A-iv, B-i, C-iii, D-ii
(2) A-i, B-iii, C-ii, D-iv
(3) A-iii, B-i, C-iv, D-ii
(4) A-ii, B-iii, C-iv, D-i

BBG322

5. At the alveolar site where different CO_2 compounds like carbamino compound, bicarbonates etc. are dissociated and release the CO_2. Find out the suitable factor which is responsible for dissociation for CO_2-compounds?
(1) PCO_2 is high
(2) PCO_2 is low
(3) Bohr's effect
(4) High H^+ concentration

BBG323

6. Given below the diagram represents the exchange of gases at the alveolar level and tissue level in which several parts are labelled by alphabets A, B, C & D. Select the correct option regarding them.



(1) A is the pulmonary vein in which PO_2 & PCO_2 are respectively 95 & 45 mmHg.
(2) B is the systemic artery in which PO_2 & PCO_2 are quite similar with PO_2 & PCO_2 pulmonary artery respectively.
(3) C is the systemic vein in which PCO_2 & PO_2 are 45 mmHg and 40 mmHg respectively
(4) D is the pulmonary artery in which PO_2 & PCO_2 are 97 mmHg and 40 mmHg. respectively.

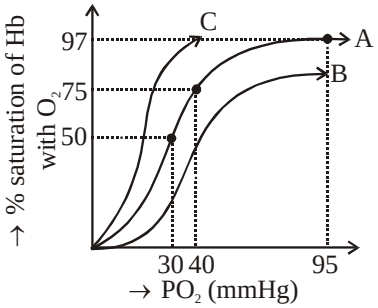
BBG324

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7. During the Normal conditions 100 ml oxygenated blood transport the .....(A).... ml of O_2 while .....(B)..... ml O_2 delivers to the body tissues.
Fill in the blanks and select the correct option?
(1) A = 20 ml and B = 5 ml
(2) A = 20 ml and B = 15 ml.
(3) A = 100 ml and B = 25 ml.
(4) A = 25 ml and B = 4 ml.

BBG325

8. Given below the three oxyhaemoglobin dissociation curves A, B and C are plotted at different conditions of body temperature.



On the basis of above 'ODC' which is the curve plotted at the condition the body-temperature below to the 37° C ?
(1) Curve - A (2) Curve - B
(3) Curve - C (4) Both curve A & B

BBG326

9. Which pulmonary capacity is correctly matched with its formula and value ?

	Respiratory capacity	Formula	Value
(1)	Functional residual capacity	ERV + TV	2300 ml.
(2)	Expiratory capacity	TV + ERV	1200 ml.
(3)	Vital capacity	ERV + IRV + TV	4000 ml.
(4)	Total lung capacity	RV + IRV + ERV	5800 ml.

BBG327

10. Respiratory rate increases during :-
(a) The time of fever.
(b) The time of fear
(c) High CO_2 concentration in venous blood.
(d) Low pH of arterial blood.
Out of these, which are correct situation?
(1) Only a & b (2) Only c & d
(3) a, b & d (4) a, b & c

BBG328

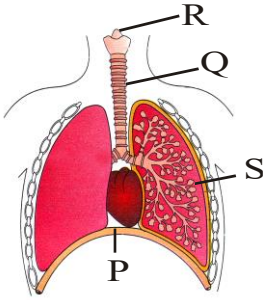
11. The contraction of ..... muscles lifts up the ribs and outward movement of sternum causing an increase in the volume of thoracis chamber at the dorso-ventral axis.
Which muscles are responsible for above statement ?
(1) External intercostal muscles.
(2) internal intercostal muscles.
(3) Phrenic muscles.
(4) Abdominal muscles.

BBG329

12. Which of the following has least diameter?
(1) Respiratory bronchioles
(2) Terminal bronchioles
(3) Alveolar duct
(4) Secondary bronchi

BBG330

13. Given figure shows respiratory system of human in which some parts are labelled through P, Q, R and S. Select the option in which labelled part is not correctly matched with their function or property ?



(1) R = Epiglottis : It is a cartilagenous flap like structure and covers the glottis during deglutination.
(2) Q = Tracheal Rings: They prevents the collapsing of trachea.
(3) P= Diaphragm :- Muscular septum and main muscles of breathing mechanism.
(4) S = Alveoli :- Forms the conduction zone and functional unit of lungs.

BBG331

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14. What will be the in the atmospheric air compared to those in the alveolar air?
(1) lesser, higher (2) higher, lesser
(3) higher, higher (4) lesser, lesser

BBG332

15. Which of the following statement is correct?
(1) Flatworms exchange Oxygen with carbondioxide by simple diffusion over their entire body surface.
(2) Larynx is a bony box which help in sound production.
(3) Epiglottis is made up of collagen fibrous connective tissue.
(4) In birds larynx is a sound producing organ.

BBG333

16. What is the function of pleural fluid?
(1) reduces friction on the lung surface.
(2) protection from external shocks.
(3) provide moisture.
(4) all of the above.

BBG334

17. Which of the following part of respiratory system form exchange part?
(1) external nostrils to terminal bronchioles.
(2) alveoli and their ducts.
(3) larynx to primary bronchi.
(4) primary bronchi to terminal bronchioles.

BBG335

18. Select the correct option for conducting system of respiratory system?
(1) It is non-collapsable part.
(2) It is the site of actual diffusion of oxygen and carbon dioxide between blood and atmospheric air.
(3) it clears the air from foreign particles and humidifies.
(4) Both 1 and 3.

BBG336

19. Thoracic chamber is formed by :-

	Dorsally	Ventrally	Laterally
(1)	Sternum	Ribs	Ribs
(2)	Vertebral column	Sternum	Diaphragm
(3)	Vertebral column	Sternum	Ribs
(4)	Sternum	Vertebral column	Ribs

BBG337

20. Inspiration can occur if :-
(1) intrapulmonary pressure is higher than the atmospheric pressure.
(2) intrapulmonary pressure and atmospheric pressure are equal.
(3) intrapulmonary pressure is less than atmospheric pressure.
(4) muscles of diaphragm relaxes.

BBG338

21. Select the correct option for expiration?
(1) expiration is a passive process.
(2) during this process diaphragm and sternum returns to their normal position.
(3) intrapulmonary pressure is slightly higher to the atmospheric pressure.
(4) all of the above.

BBG339

22. In which of the following respiratory volume is correctly match with their value?

	Volume	Value (in ml)
(1)	IRV	250 – 300
(2)	ERV	1000 – 1100
(3)	TV	1100 – 1200
(4)	RV	500

BBG340

23. (A) inspiratory capacity.
(B) expiratory capacity.
(C) functional residual capacity.
Choose the correct option for A, B and C ?
(1) A–TV+ERV, B–TV+IRV, C–ERV+RV
(2) A–TV+ERV, B–TV+ERV, C–TV+IRV
(3) A–TV+IRV, B–TV+ERV, C–ERV+RV
(4) A–TV+ERV, B–TV+IRV, C–ERV+RV

BBG341

24. Which of the following statement is correct about alveoli?
(1) alveoli are primary sites for gaseous exchange.
(2) its walls are one celled thick.
(3) partial pressure of oxygen in alveoli is 104 mm Hg.
(4) all of the above.

BBG342

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25. How much amount of oxygen deliver to tissue under normal physiological condition by every 100 ml of oxygenated blood ?
(1) 5 ml (2) 25 ml
(3) less than 1 ml (4) more than 50 ml
BBG343

26. Consider the given statements and select correct statement for transport of oxygen?
(a) seventy percent of oxygen is carried as bicarbonates.
(b) about ninty seven percent of oxygen is transported by RBCs in blood.
(c) it is carried by haemoglobin as carb-amino haemoglobin.
(1) a, b (2) b, c
(3) only b (4) only c
BBG344

27. Respiration is the physiological process in which :-
(1) breathing occur.
(2) breathing and external respiration occurs.
(3) only inspiration occurs.
(4) breathing, external respiration and cellular respiration occurs.
BBG345

28. Neither the trachea nor the bronchi contain :-
(1) hyaline cartilage.
(2) goblet cells.
(3) ciliated columnar epithelium.
(4) simple squamous epithelium.
BBG346

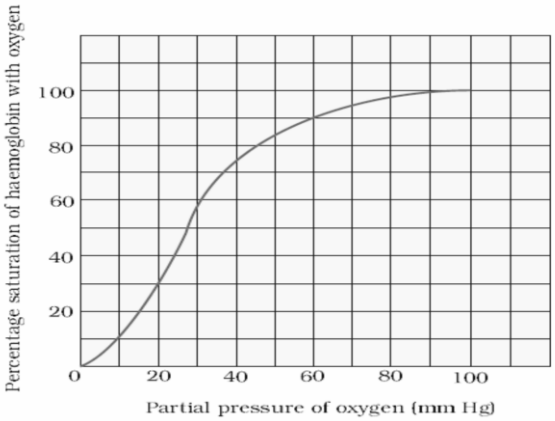
29. Which of the following structure is present inside the larynx of the respiratory system?
(1) Glottis (2) Epiglottis
(3) Vocal cords (4) Gullet
BBG347

30. If a large number of people are enclosed in a room then :-
(1) oxygen decreases and CO_2 increases.
(2) oxygen increases and CO_2 decreases.
(3) oxygen and CO_2 decreases.
(4) oxygen and CO_2 increases.
BBG348

31. Identify the type of pulmonary volume / capacity on the basis of quantity of air present in the lungs given below.
(I) 1100 ml to 1200 ml.
(II) 500 ml.
(III) 5000 ml to 6000 ml.
Choose the correct option ?
(1) I – Vital capacity , II – Functional residual capacity , III – Residual volume
(2) I – Residual volume , II – Tidal volume , III – Total lung capacity
(3) I – Expiratory capacity , II – Inspiratory capacity , III – Residual volume
(4) I – Tidal volume , II – Inspiratory reserve volume , III – Expiratory reserve volume
BBG349

32. How many layers are present in the diffusion membrane of alveolar capillary ?
(1) 5 (2) 3
(3) 2 (4) 4
BBG350

33. Shifting of the curve to right takes place in the case of :



(1) raise in pCO_2
(2) fall in pH.
(3) raise in temperature.
(4) All of the above.
BBG351

34. The mucus produced by the goblet cells
(a) moistens the incoming air
(b) trap the fine dust particles
(c) filter out large dust particles
(d) warms the incoming air

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(1) a , b , c are correct.
(2) only a is correct.
(3) b and d are correct.
(4) a and c are incorrect.

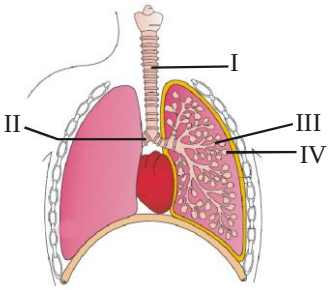
BBG352

35. How much amount of carbondioxide is transported by plasma?

(1) 75 % to 78% (2) 5% to 7%
(3) 20% to 25% (4) 93%

BBG353

36. The given diagram represents the human respiratory system with few structures labelled as I, II, III and IV.

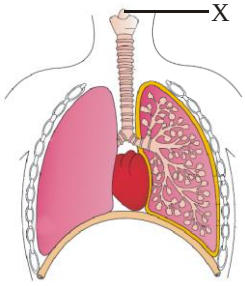


The exchange of gases takes place in which labelled structure?

(1) I (2) II (3) III (4) IV

BBG354

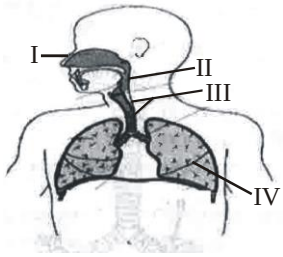
37. In the given diagram of human respiratory system what is the function of structure marked as X ?



(1) To prevent the entry of food in to the larynx.
(2) To filter and warm the air.
(3) To help in exchange of gases.
(4) To catch dust and bacteria.

BBG355

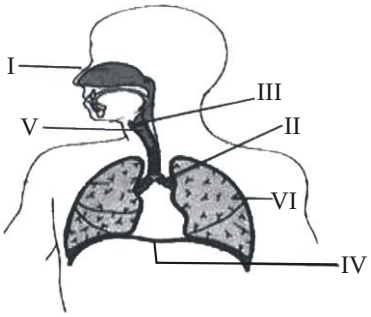
38. In the given diagram of human respiratory system which marked label (I, II, III & IV) is the common passageway where the nasal and oral cavities meet?



(1) I (2) II (3) III (4) IV

BBG356

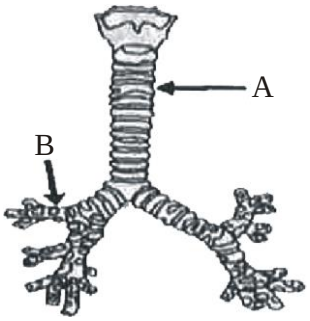
39. In the given diagram of human respiratory system, few parts are marked as I, II, III, IV, V & VI. Choose the correct combination of labelling from the given options?



(1) I-Nose, II-Bronchus, III-Larynx, IV-Diaphragm, V-Trachea, VI-Lung
(2) I-Nose, II-Larynx, III-Bronchus, IV-Lung, V-Diaphragm, VI-Trachea
(3) I-Mouth, II-Trachea, III-Larynx, IV-Lung, V-Diaphragm, VI-Bronchus
(4) I-Mouth, II-Diaphragm, III-Trachea, IV-Bronchi, V-Larynx, VI-Lung

BBG357

40. In the given figure, label A represents _____ while label B represents _____.



(1) A-Trachea, B-Bronchus
(2) A-Alveolus, B-Bronchiole
(3) A-Bronchiole, B-Trachea
(4) A-Trachea, B-Bronchiole

BBG358

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41. **Assertion:** Fishes use tracheal tubes for respiration
Reason: Fishes are amphibian
(1) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(2) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both assertion and reason are false

BBG359

42. **Assertion:** During inspiration, pressure of air falls in the thorax.
Reason: There is a rise in volume of thorax during inspiration.
(1) Assertion is true but the reason is false.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(4) Both assertion and reason are false

BBG360

43. **Assertion:** 70 per cent of CO_2 is carried as bicarbonate ion in plasma.
Reason: If it is transported directly then pH of blood becomes acidic.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both assertion and reason are false
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

BBG361

44. **Assertion:** Emphysema is a chronic disorder.
Reason: One of the major cause of emphysema is cigarette smoking.
(1) Assertion is true but the reason is false.
(2) Both assertion and reason are false
(3) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(4) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

BBG362

45. **Assertion:** Pneumotaxic centre can alter the respiratory rate.
Reason: Neural signal from pneumotaxic centre can reduce the duration of inspiration.
(1) Assertion is true but the reason is false.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(4) Both assertion and reason are false.

BBG363

Exercise – I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	2	3	2	3	1	3	3	3	1	3	4	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	4	3	3	4	2	3	4	1	3	4	4	3	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	2	4	1	1	4	1	2	1	1	4	3	1	4	3

Exercise – II
Breathing and Exchange of Gases

1. Match the following correctly.

	Animals		Respiratory organs
A.	Earthworms	I	Lungs
B.	Most aquatic arthropods	II	Trachea
C.	Fishes	III	Gills
D.	Birds/Reptiles	IV	Moist cuticle
E.	Insects		

(1) A-IV, B and C-III, D-I, E-II
(2) A-IV, B-III C and D-I, E-III
(3) A-II, B and C-III, D-I, E-IV
(4) A-III, B and C-I, D-II, E-IV

BBG364

2. Which is the correct sequence of air passage in man?
(1) External Nostril → Nasal passage → Internal nostril → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli
(2) Nose → Larynx → Pharynx → Bronchioles → Bronchi → Alveoli
(3) Nose → Pharynx → Trachea → Larynx → Bronchi → Bronchioles → Alveoli
(4) Nose → Larynx → Bronchi → Pharynx → Trachea → Bronchioles → Alveoli

BBG365

3. I. It is double layered and covers the lungs.
II. Fluid between the layers reduces friction on lung-surface
III. Outer layer is in contact with thoracic wall
IV. Inner layer is in contact with lungs
The above features refer to-
(1) Pericardium (2) Peritoneum
(3) Pleura (4) None

BBG366

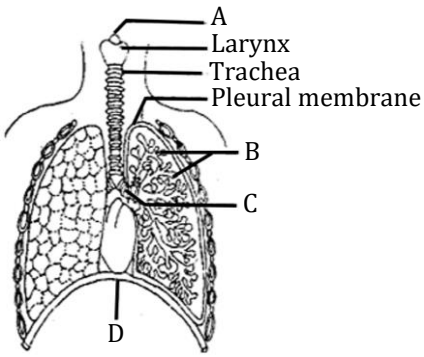
4. The chamber formed dorsally by the vertebral column, ventrally by sternum laterally by ribs and on the lower side by dome-shaped diaphragm is-
(1) Abdominal cavity
(2) Thoracic cavity
(3) Pelvic cavity
(4) Cranial cavity

BBG367

5. Respiration involves following steps-
I. Diffusion of gases O_2 and CO_2 across alveolar membrane
II. Transport of gases by blood
III. Utilization of O_2 by cell for catabolic reactions and resultant release of CO_2
IV. Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out
V. Diffusion of O_2 and CO_2 between blood and tissues.
The correct sequence of steps is-
(1) I→ II→ III→ IV → V
(2) V→ IV→ III→ II→ I
(3) IV→ I→ II→ V→ III
(4) III→ II→ V→ I→ IV

BBG368

6. Study the given diagram and identify A, B C and D.



(1) A-Epiglottis, B-Alveoli, C-Bronchus, D-Diaphragm
(2) A-Epiglottis, B-Alveoli, C-Bronchioles, D-Diaphragm
(3) A-Sound box, B-Alveoli, C-Bronchus, D-Diaphragm
(4) A-Sound box, B-Alveoli, C-Bronchioles, D-Diaphragm

BBG369

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7. Which of the following statements about the mechanism of ventilation/breathing is false?
(1) As the diaphragm relaxes, air is expelled from the respiratory system
(2) During inspiration the lungs act as suction pump
(3) Inspiration is a passive and expiration is an active process
(4) For quiet breathing external intercostal muscles and diaphragm play an important role

BBG370

8. Inspiration occurs when there is a negative pressure in the lungs with respect to atmospheric pressure. This negative pressure is achieved when-
(1) Intrapulmonary pressure is less than the atmospheric pressure
(2) Intra pulmonary pressure is greater than the atmospheric pressure
(3) Intrapulmonary pressure is equal to the atmospheric pressure
(4) Intrapleural pressure becomes more than the intra-alveolar pressure

BBG371

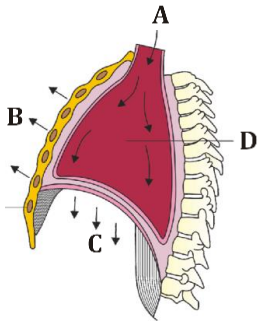
9. Which of the following sequences is correct to initiate inspiration?
I. The contraction of external intercostal muscles raises the ribs and sternum
II. Volume of thorax increases in the dorso-ventral axis
III. Intrapulmonary pressure decreases
IV. Diaphragm contraction
V. Air rushes into lungs
VI. Volume of thorax increases in the anterio-posterior axis
(1) I, II, IV, V III, VI (2) I, II, III, IV, V
(3) I, II, IV, VI, III, V (4) VI, I, II, III, V

BBG372

10. Which of the following sequences is correct to initiate expiration?
I. Relaxation of external intercostal muscles and return of diaphragm and sternum to their normal position
II. Air expelled from lungs
III. Volume of thorax decreases
IV. Intrapulmonary pressure increases
(1) I, III, IV, II
(2) II, IV, III, I
(3) IV, III, II, I
(4) I, II, III, IV

BBG373

11. Following illustration depicts the mechanism of breathing. In which of the following option all the parts A, B, C and D are correctly labelled?



(1) A-Air entering into lungs; B-Ribs and sternum raised; C-Diaphragm contracted; D-Volume of thorax raised
(2) A-Air expelled from lungs; B-Ribs and sternum return to original position; C-Diaphragm relaxed; D-Volume of thorax decreased
(3) A-air expelled from lungs; B-Ribs and sternum raised; C-Diaphragm relaxed; D-Volume of thorax decreased
(4) A-air expelled from lungs; B-Ribs and sternum raised; C-Diaphragm contracted; D-Volume of thorax decreased

BBG374

12. Match the following-

	Column A		**Column B**
1.	Tidal volume	A.	Tidal volume and inspiratory reserve volume and expiratory reserve volume

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2.	Residual volume	B.	Additional amount of air inhaled beyond tidal volume when taking a very deep breath
3.	Expiratory reserve volume	C.	Amount of air remaining in lungs after expiratory reserve volume is expelled
4.	Inspiratory reserve volume	D.	Tidal volume and inspiratory reserve volume
5.	Inspiratory capacity	E.	Volume of air in one breath
6.	Vital capacity	F.	Amount of air exhaled in forced exhalation

(1) 1-C, 2-E, 3-B, 4-F, 5-D, 6-A
(2) 1-E, 2-F, 3-C, 4-B, 5-A, 6-D
(3) 1-E, 2-C, 3-F, 4-B, 5-D, 6-A
(4) 1-E, 2-C, 3-B, 4-F, 5-A, 6-D

BBG375

13. Match the following-

	Column A		**Column B**
1.	Tidal volume	A.	2500-3000 ml of air
2.	Inspiratory reserve volume	B.	1000 ml of air
3.	Expiratory reserve volume	C.	500 ml of air
4.	Residual volume	D.	3400-4800 ml of air
5.	Vital capacity	E.	1200 ml of air

(1) 1-C, 2-D, 3-B, 4-A, 5-E
(2) 1-C, 2-A, 3-B, 4-E, 5-D
(3) 1-C, 2-A, 3-D, 4-E, 5-B
(4) 1-E, 2-A, 3-B, 4-E, 5-D

BBG376

14. Arrange the following in order of increasing volume-
1. Tidal volume
2. Residual volume
3. Expiratory reserve volume
4. Vital capacity

(1) 1 < 2 < 3 < 4 (2) 1 < 4 < 3 < 2
(3) 1 < 3 < 2 < 4 (4) 1 < 4 < 2 < 3

BBG377

15. The maximum volume of air you can forcefully exhale after taking the deepest possible breath is called
(1) Tidal volume
(2) Total respiratory volume
(3) Residual volume
(4) Vital capacity

BBG378

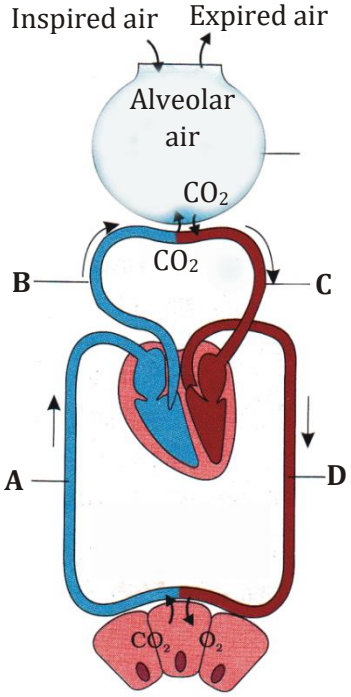
16. Which of the following statements about the partial pressure of CO_2 is true?
(1) It is higher in alveoli than in pulmonary artery
(2) It is higher in the systemic arteries than in tissues
(3) It is higher in systemic veins than in systemic arteries
(4) It is higher in the pulmonary veins than in pulmonary arteries

BBG379

17. The partial pressure of CO_2 in the venous blood is-
(1) Greater than in the tissue spaces
(2) Lesser than in the tissue spaces
(3) Equal to tissue space
(4) Less than in alveoli

BBG380

18. Name the blood vessels A to D-

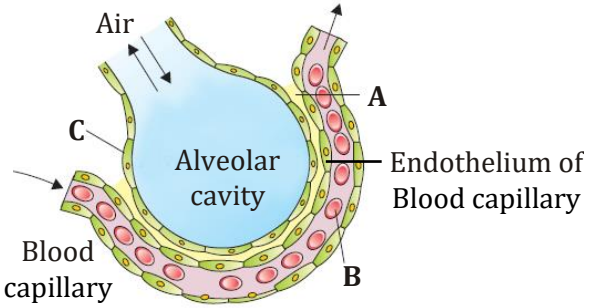


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	A	B	C	D
(1)	Systemic vein	Pulmonary artery	Pulmonary vein	Systemic artery
(2)	Systemic artery	Systemic vein	Pulmonary artery	Systemic vein
(3)	Pulmonary artery	Systemic vein	Pulmonary vein	Systemic artery
(4)	Systemic vein	Pulmonary vein	Pulmonary artery	Systemic artery

BBG381

19. Study the given figure and identify A to C.



	A	B	C
(1)	Basement membrane	RBC	Alveolar wall
(2)	O_2	CO_2	Alveolar O_2
(3)	Pleura	RBC	Pericardium
(4)	Pleura	WBC	Pulmonary vein

BBG382

20. Besides RBC blood plasma also carries O_2 in solution. The percentage is-

(1) 3% (2) 97%

(3) 49% (4) 100%

BBG383

21. Blood carries the CO_2 in 3 forms. The correct percentages of CO_2 in these forms are-

	As carbaminohaemoglobin in RBC	As bicarbonates	Dissolved from in plasma
(1)	20-25%	70%	7%
(2)	70%	20-25%	7%
(3)	20-25%	7%	70%
(4)	7%	20-25%	70%

BBG384

22. Dissociation curve is associated with-

(1) Carbonic anhydrase

(2) CO

(3) $CHCl_3$

(4) Oxyhaemoglobin

BBG385

23. Binding of CO_2 with haemoglobin is primary related to-

(1) pO_2 (2) H + conc

(3) Temperature (4) All

BBG386

24. Which of the following statements wrong?

(1) O_2 binds with haemoglobin in a reversible manner to form oxyhaemoglobin.

(2) Every 100 ml of oxygenated blood can deliver around 5ml of O_2 to the tissue

(3) Occupational respiratory disorder are characterised by fibrosis (proliferation of fibrous tissues)

(4) None

BBG387

25. What will be the pO_2 and pCO_2 in the atmospheric air compared to those in the alveolar?

(1) pO_2 lesser, pCO_2 higher

(2) pO_2 higher, pCO_2 lesser

(3) pO_2 higher, pCO_2 higher

(4) pO_2 lesser, pCO_2 lesser

BBG388

26. Which of the following would have the same O_2 content?

(1) Blood entering the lungs-blood leaving the lungs

(2) Blood entering the right side of the heart-blood leaving the right side of the heart

(3) Blood entering the right side of the heart-blood leaving the left side of the heart

(4) Blood entering the tissue capillaries-blood leaving the tissue capillaries

BBG389

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27. CO_2 dissociates from carbaminohaemoglobin when-

(1) $pO_2\downarrow$, $pCO_2\downarrow$ (2) $pO_2\uparrow$, $pCO_2\uparrow$

(3) $pO_2\downarrow$, $pCO_2\uparrow$ (4) $pO_2\uparrow$, $pCO_2\downarrow$

BBG390

28. Which of the following situations would result in the greatest degree of O_2 saturation for haemoglobin, assuming pO_2 remains constant-

(1) Increased CO_2 levels, decreased temperature

(2) Increased CO_2 levels, increased temperature

(3) Decreased CO_2 levels, decreased temperature

(4) Decreased CO_2 levels, increased temperature

BBG391

29. Which of the following factors favour the formation of oxyhaemoglobin in lungs?

(1) $pO_2\downarrow$, $pCO_2\uparrow$, $H+\downarrow$, Temperature$\uparrow$

(2) $pO_2\uparrow$, $pCO_2\uparrow$, $H+\downarrow$, Temperature$\uparrow$

(3) $pO_2\uparrow$, $pCO_2\downarrow$, $H+\downarrow$, Temperature$\downarrow$

(4) $pO_2\downarrow$, $pCO_2\uparrow$, $H+\uparrow$, Temperature$\downarrow$

BBG392

30. All of the following favour the dissociation of oxyhaemoglobin to deliver O_2 to tissue except-

(1) $pO_2\uparrow$ (2) pCO_2 OR $H+\uparrow$

(3) Temperature $\uparrow$ (4) $PO_2\downarrow$

BBG393

31. Which of the following statements is false?

(1) pO_2 is the major factor which affects the binding of CO_2 with haemoglobin.

(2) pCO_2 is low and pO_2 is high as in the tissues, more binding of CO_2 with Hb occurs.

(3) RBC contains a very high conc. of carbonic anhydrase and minute quantities of the same in the plasma.

(4) Every 100 ml of deoxygenated blood delivers approximately 4 ml of CO_2 to the alveoli.

BBG394

32. Which of the following equation is correct?

(1) $CO_2 \rightarrow H_2CO_3 \rightarrow HCO_3^- + H^+$

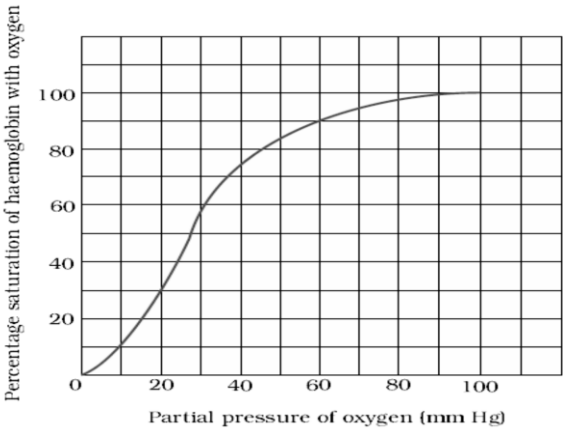
(2) $CO_2 + H_2O \xrightleftharpoons{\text{Carbonic anhydrase}} H_2CO_3 \xrightleftharpoons{\text{Carbonic anhydrase}} H^+ + HCO_3^-$

(3) $CO_2 \rightarrow H_2O \rightarrow CH_4 + 2O_2$

(4) $CO_2 \rightarrow H_2O \rightleftharpoons CO + H_2O_2$

BBG395

33. Which of the following is incorrect about the given graph.



(1) Increase in partial pressure of CO_2 shift the curve to right

(2) At low temperature the curve shifts to left

(3) At high pH the curve shifts to right

(4) Decrease in partial pressure of oxygen shifts the curve to right

BBG396

34. **Assertion :-** The maximum pO_2 in alveoli is considerably less than in the atmospheric.

Reason :- Lungs in mammals do not completely empty with each breath and inhalation occurs through the same airways as exhalation, so each inhalation mixes fresh air with oxygen depleted residual air.

(1) Both assertion and reason are true and reason is correct explanation of assertion

(2) Both assertion and reason are true and reason is not correct explanation of assertion

(3) Assertion is true but reason is false

(4) Both assertion and reason are false

BBG397

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35. How does an increase in the CO_2 concentration in the blood affect the pH of CSF?
(1) pH↑
(2) pH↓
(3) pH remains same
(4) pH may increase or decrease

BBG398

36. **Assertion:-** A drop in the blood pH causes an increase in heart rate.
Reason:- Increased Heart rate increases the rate at which CO_2 is delivered to the lungs, where CO_2 is removed.
(1) Both assertion and reason are true and reason is correct explanation of assertion
(2) Both assertion and reason are true and reason is not correct explanation of assertion
(3) Assertion is true but reason is false
(4) Both assertion and reason are false

BBG399

37. Respiratory process is regulated by certain specialized centres in the brain. One of the following listed centres can reduce the inspiratory duration upon stimulation-
(1) Medullary inspiratory centre
(2) Pneumotaxic centre
(3) Chemosynthetic centre
(4) Apneustic centre

BBG400

38. Receptors associated with aortic arch and carotid artery can recognise changes in __and__conc. and send necessary signal to__ for remedial action.
(1) O_2, CO_2, Pneumothorax
(2) CO_2, H+, rhythm centre
(3) CO_2, H+, apneustic centre
(4) O_2, H+, Pneumothorax

BBG401

39. Respiratory centre of brain is stimulated by-
(1) CO_2 content in various blood
(2) CO_2 content in arterial blood
(3) O_2 content in arterial blood
(4) O_2 content in venous blood

BBG402

40. Why do human beings have difficulty breathing at high elevations?
(1) O_2 makes up lower percentage of air there
(2) The temperature is lower there
(3) The barometric pressure is higher there
(4) pO_2 is lower there

BBG403

41. **Assertion:** Larynx is called sound box.
Reason: Larynx helps in production of sound.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both assertion and reason are false
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

BBG404

42. **Assertion:** About 97% of O_2 is transported by RBCs.
Reason: RBCs has haemoglobin.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Both assertion and reason are false
(4) Assertion is true but the reason is false.

BBG405

43. **Assertion:** The maximum amount of CO_2 is converted into bicarbonate ion in RBCs.
Reason: RBCs contain a very high concentration of the enzyme, carbonic anhydrase.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both assertion and reason are false
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

BBG406

44. **Assertion:** We can increase strength of inspiration and expiration.
Reason: We can use additional muscle of thoracic and abdominal region
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both assertion and reason are false
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

BBG407

45. **Assertion:** Alveoli and their duct form respiratory part of respiratory tract.
Reason: This part is actual site of exchange of O_2 and CO_2 between blood and atmospheric air.
(1) Assertion is true but the reason is false.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Both assertion and reason are false
(4) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.

BBG408

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Exercise – II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	3	2	3	1	3	1	3	1	1	3	2	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	1	1	1	1	4	1	4	2	2	4	3	3	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	2	3	2	2	1	2	2	2	4	1	1	1	1	4

Exercise – I
Body Fluids and Circulation

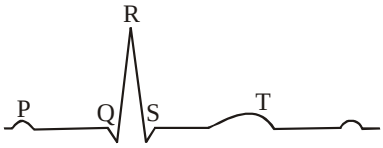
1. Which is required for the conversion of inactive prothrombin into active thrombin during coagulation of blood ?
(1) Fibrinogen
(2) Thromboplastin
(3) Thrombokinase
(4) Antihaemophilic globulin

BFC520

2. "Erythroblastosis foetalis" is a disorder in which:
(1) Foetus undergoes blood cancer
(2) Foetus would be die due to excessive weight gain
(3) Foetus would be suffered by severe anaemia and jaundice
(4) Adult would be suffered by severe anaemia and jaundice

BFC521

3. The given figure represents diagrammatic presentation of an ECG. Each peak in the ECG is identified with a letter from P to T that corresponds to a specific electrical activity of the heart.



Which of the following letter shows the incorrect activity of heart ?
(1) QRS - Depolarization of the ventricles
(2) R wave - Marks the beginning of the systole
(3) P - Electrical excitation (or depolarization) of the atria
(4) T wave - Return of the ventricles from excited to normal state (repolarization)

BFC522

4. (a) It is colourless fluid containing specialized lymphocytes.
(b) It is important carrier for nutrients, hormones etc.
(c) It has clotting capacity.
(d) It does not possess RBC, WBC and platelets.
(e) Exchange of nutrients, gases etc. between the blood and the cells always occurs through this fluid.
(f) It has more No. of WBC than blood.
How many statements are correct for lymph?
(1) Two (2) Three
(3) Four (4) Five

BFC523

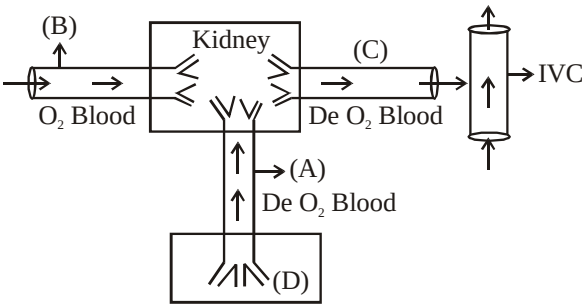
5. Lymphatic system of human and frog are respectively :-
(1) Open type and close type
(2) Closed type and open type
(3) Open type and open type
(4) Close type and close type

BFC524

6. "Lacteals" are the lymph capillaries which are found in :-
(1) Large intestine (2) Small intestine
(3) Neck region (4) Right leg

BFC525

7. In given below figure, identify A, B, C and D:-



	A	B	C	D
(1)	Renal portal vein	Hepatic artery	Renal vein	Abdominal parts & hind limbs
(2)	Renal portal vein	Renal artery	Renal vein	Alimentary canal
(3)	Renal portal vein	Renal artery	Renal vein	Abdominal parts & hind limbs
(4)	Renal artery	Renal vein	Renal portal vein	Gut wall

BFC526

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8. Select the correct pathway of pulmonary circulation ?

(1) Right ventricle $\xrightarrow[CO_2\ Blood]{Pulmonary\ vein}$ Lungs $\xrightarrow[O_2\ Blood]{Pulmonary\ artery}$ Left atrium.

(2) Right ventricle $\xrightarrow[CO_2\ Blood]{Pulmonary\ artery}$ Lungs $\xrightarrow[O_2\ Blood]{Pulmonary\ vein}$ Left atrium.

(3) Left ventricle $\xrightarrow[O_2\ Blood]{Aorta}$ Body tissues $\xrightarrow[CO_2\ Blood]{Vena\ cava}$ Right ventricle.

(4) Left atrium $\rightarrow$ Left Ventricle $\xrightarrow{O_2\ Blood}$ Body tissues $\xrightarrow{CO_2\ Blood}$ Right atrium $\rightarrow$ Right ventricle.

BFC527

9. A special system of blood vessels is present in our body exclusively for the circulation of blood to and from the cardiac musculature. Such type of blood circulation is known as :-

(1) Pulmonary circulation
(2) Systemic circulation
(3) Coronary circulation
(4) Double circulation

BFC528

10. **Assertion:** Individuals with 'O' blood group are considered to be universal donor.
Reason: Blood of 'O' blood group individual do not contain antigens on their RBCs.

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BFC529

11. Match the column-I and column-II and choose the correct option ?

	Column-I		Column-II
(A)	Cardiac arrest	(i)	Heart is not pumping blood effectively
(B)	Heart failure	(ii)	Heart muscles are suddenly damaged
(C)	Heart attack	(iii)	Acute chest pain
(D)	Angina pectoris	(iv)	Heart stop beating

Option :-

(1) A-(i), B-(ii), C-(iii), D-(iv)
(2) A-(iv), B-(i), C-(ii), D-(iii)
(3) A-(iv), B-(ii), C-(i), D-(iii)
(4) A-(iii), B-(iv), C-(i), D-(ii)

BFC530

12. High blood pressure affects our vital organs like:-

(a) Heart (b) Kidney
(c) Spleen (d) Brain
(e) Liver

Which are mainly affected ?

(1) a, b & d (2) a, b & c
(3) c, d & e (4) b, c & d

BFC531

13. An artificial pacemaker is implanted under subcutaneous tissue and connected to the heart in patients of :-

(1) Having 90% blockage in the two main coronary arteries
(2) Having a very high blood pressure
(3) With irregularity in the heart rhythm
(4) Suffering from arteriosclerosis

BFC532

14. Neural signals through parasympathetic nervous system can :-

(a) Increase the heart beat rate.
(b) Decrease the heart beat rate.
(c) Increase the strength of ventricular contraction.
(d) Decrease the speed of conduction of action potential.
(e) Increase cardiac output.
(f) Decrease cardiac output.

Option :-

(1) b, c & e (2) a, d & f
(3) b, d & f (4) a, c & e

BFC533

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15. Which hormone decreases heart beat rate thereby decreases the cardiac output ?
(1) Thyroxin (2) Sympathin
(3) Acetylcholine (4) Adrenaline
BFC534

16. The component of cardiac tissue having the highest propagation velocity is :-
(1) Purkinje fibres (2) AV node
(3) Bundle of His (4) AV bundle
BFC535

17. ECG depicts the depolarization and repolarization processes during the cardiac cycle. In the ECG of a normal healthy individual one of following waves is not represented.
(1) Depolarisation of atria
(2) Repolarisation of atria
(3) Depolarisation of ventricles
(4) Repolarisation of ventricles
BFC536

18. Match column I with column II:

	Column-I		Column-II
(a)	Eosinophils	(i)	Coagulation
(b)	RBC	(ii)	Universal Recipient
(c)	AB Group	(iii)	Resist Infections
(d)	Platelets	(iv)	Contraction of Heart
(e)	Systole	(v)	Gas transport

	a	b	c	d	e
(1)	iii	v	ii	i	iv
(2)	iii	v	i	ii	iv
(3)	iii	v	ii	iv	i
(4)	iii	ii	iv	v	i

BFC537

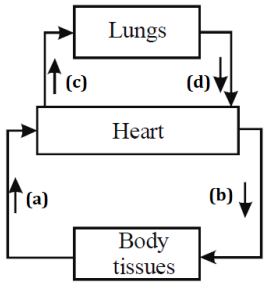
19. Open type of circulatory system is found in
(1) Toad (2) Prawn
(3) Lizard (4) Pigeon
BFC538

20. Blood will lose maximum Oxygen when pass through
(1) alveolar capillaries
(2) left atrium
(3) arteries
(4) tissue capillaries
BFC539

21. As compared to artery, a vein has
(1) thinner walls (2) larger lumen
(3) less musculature (4) all of the above
BFC540

22. Normally, the greater portion of circulating blood is present in the
(1) Heart (2) Large arteries
(3) Capillaries (4) Large veins
BFC541

23. The given figure represents the pathway of blood throughout the body-



Identify the correct of marked a , b , c and d-
(1) a – artery
(2) b – pulmonary vein
(3) c – pulmonary artery
(4) d – systemic vein
BFC542

24. If due to some injury chordae tendenae of the bicuspid valve of the human heart is partially non-functional, what will be the immediate effect?
(a) the flow of blood into the pulmonary artery will be reduced
(b) the flow of blood into the aorta will be slowed down
(c) the pacemaker will stop working
(d) the blood will tend to flow back into the left atrium
(1) a and b (2) b and c
(3) b and d (4) c and d
BFC543

25. Which of the following events take place in cardiac cycle after the end of first heart sound and before the onset of second heart sound?
(1) atrial systole and ventricular systole
(2) atrial systole and ventricular diastole
(3) atrial diastole and ventricular systole
(4) joint diastole
BFC544

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26. In order for the blood to flow from right ventricle to left ventricle in mammalian heart, it must flow through :-
(I) RIGHT VENTRICLE
(II) PULMONARY VEINS
(III) LEFT ATRIUM
(IV) LUNGS
(V) PULMONARY ARTERIES
(1) I–V–IV–II–III
(2) I–II–III–IV–V
(3) III–V–I–II–IV
(4) III–II–I–IV–V

BFC545

27. Which of the following statements are incorrect?
(1) among all the four chambers of heart left ventricle has the thickest walls
(2) contraction of the ventricle in the heart begins by the command from SA node
(3) QRS complex in ECG is related with ventricular contraction and atrial relaxation
(4) human heart is myogenic

BFC546

28. Match the following columns and choose the correct answer from the options given below

	Column-I		Column-II
(A)	Erythrocytes	(i)	Most abundant white blood cells and the phagocytic in nature
(B)	Eosinophils	(ii)	Least abundant white blood cells
(C)	Monocyte	(iii)	Resist infections and are associated with allergic reactions
(D)	Neutrophils	(iv)	Blood cells that contain haemoglobin and transport oxygen
(E)	Basophils	(v)	Macro policeman

	A	B	C	D	E
(1)	iv	iii	v	i	ii
(2)	i	ii	iii	iv	v
(3)	ii	iii	i	v	iv
(4)	iv	i	ii	iii	v

BFC547

29. Given below are four statements regarding human blood vascular system :-
(a) there is no role of Calcium ions in blood clotting
(b) cardiac output is the amount of blood pumped by each ventricle in one second
(c) arteries are thin walled and have narrow lumen as compared to veins
(d) the stage of cardiac cycle where all the four chambers are in relaxed stage is called joint Diastole
How many of the above statements are correct?
(1) one (2) two
(3) three (4) four

BFC548

30. Blood clotting is initiated by
(1) Ca^{2+} ions and thromboplastin
(2) Prothrombin and thromboplastin
(3) Ca ions and fibrinogen
(4) fibrinogen and thromboplastin

BFC549

31. Total number of valves in human heart are
(1) four (2) six
(3) seven (4) eleven

BFC550

32. Significance of heart valves is
(1) to allow the flow of blood in one direction, i.e. from ventricles to atria and from the ventricles to the pulmonary artery or aorta
(2) to allow the flow of blood in one direction, i.e. from atria to ventricles and from the ventricles to the pulmonary artery or aorta
(3) to allow the flow of blood in one direction, i.e. from atria to ventricles and from the pulmonary artery or aorta to the ventricles
(4) to allow the flow of blood in one direction, i.e. from ventricles to atria and from the pulmonary artery or aorta to the ventricles

BFC551

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33. Systemic heart comprises
(1) Left atrium and left ventricle
(2) Left atrium and right atrium
(3) Right atrium and left ventricle
(4) Left atrium and right ventricle

BFC552

34. For safe blood transfusions :-
(1) Donors RBC should not contain antibodies against recipient serum
(2) Recepient serum should not contain antigens against donors antibodies
(3) Recepient serum should not contain antibodies against donors RBCs
(4) Recepient RBCs should not contain antibodies against donors antigens

BFC553

35. Ruptured RBCs cannot be trapped at one of the following sites :
(1) Blood (2) Liver
(3) Bone marrow (4) Spleen

BFC554

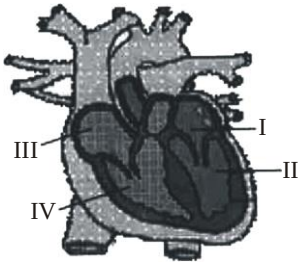
36. When the heart muscle is suddenly damaged by an inadequate blood supply. This condition is called :-
(1) Angina pectoris (2) Heart failure
(3) Heart attack (4) Heart arrest

BFC555

37. First heart sound is associated with the closure of ____valves whereas second heart sound is associated with the closure of _____valves.
(1) semilunar, atrioventricular
(2) atrioventricular, semilunar
(3) bicuspid, tricuspid
(4) tricuspid, bicuspid

BFC556

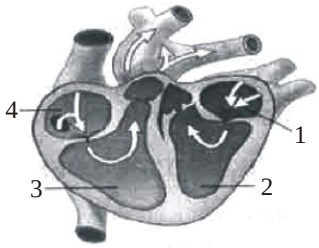
38. The given diagram represents human heart with four chambers labelled as I, II, III & IV ?



Which labelled structure receives carbon dioxide rich blood from the body ?
(1) I (2) II (3) III (4) IV

BFC557

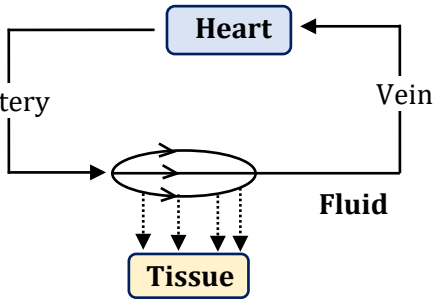
39. Refer the given figure of human heart and identify the chamber (marked as 1, 2, 3 & 4) which receives most of the blood returning from the brain :-



(1) 1 (2) 2 (3) 3 (4) 4

BFC558

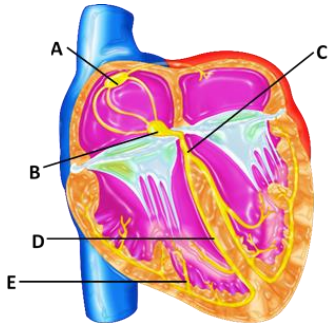
40.



Observe the given diagram and choose the incorrect statement about it.
(1) The fluid released out has same mineral distribution as in plasma.
(2) Exchange of nutrient, gases occurs through this fluid.
(3) The fluid here is blood.
(4) This fluid is drained back into major veins.

BFC559

41. Observe the given diagram and choose the correct pair of label and mentioned information about it :-

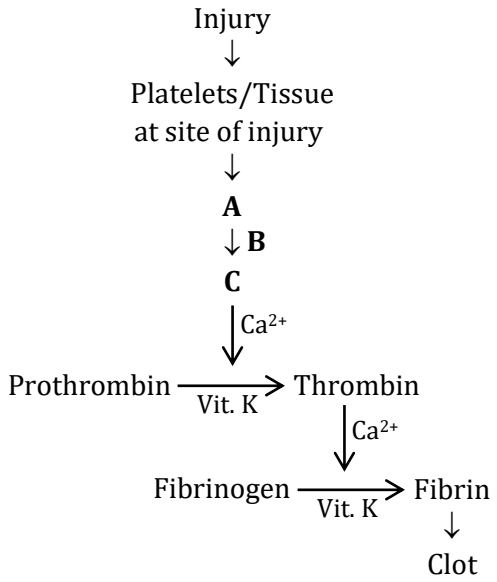


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(1) B- It is a component of bundle of HIS
(2) C- Continues from the AV node and divide into the AV bundle
(3) E-They spread throughout the ventricular musculature
(4) A- Reserve pacemaker of the heart

BFC560

42. Below mentioned is a flowchart representing blood coagulation process choose the option that recognize the labels correctly.



(1) A—Thrombokinase, C—Prothrombinase
(2) A—Thromboplastin, B—Cascade process
(3) A—Thromboplastin, B—Thrombokinase
(4) A—Thrombokinase, B—Cascade process

BFC561

43. **Assertion :** AB blood group is called as universal recipient.
Reason : AB blood group person lacks both A & B antibody in plasma.
(1) Both (A) & (R) are correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) are incorrect.
(3) (A) is correct but (R) is incorrect.
(4) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

BFC562

44. **Assertion :** Tricuspid and Bicuspid valve are pushed open during ventricular diastole.
Reason : Pressure was exerted by the blood in the atria which was being emptied into them by the veins.
(1) Both (A) & (R) are true and (R) is the not correct explanation of (A).
(2) (A) is true but (R) is false.
(3) (A) is not false but (R) is true.
(4) Both (A) & (R) are true and (R) is the correct explanation of (A).

BFC563

45. **Assertion:** Closed circulatory system is more effective than open type.
Reason: The closed circulatory system considerably enhances the speed, precision and efficiency of circulation.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BFC564

46. **Assertion:** An artificial pacemaker can replace the sinoatrial node of heart.
Reason: This is because an artificial pacemaker is capable of stimulating the heart electrically to maintain its beats.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BFC565

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47. **Assertion:** When heart stops beating it is known as cardiac arrest.
Reason: Heart attack occur due to cardiac arrest.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BFC566

48. **Assertion:** A reduction in number of platelets can lead to clotting disorders.
Reason: Platelets release a variety of substances most of which are involved in the clotting of blood.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BFC567

49. **Assertion:** An injury or trauma stimulates the platelets to release certain factors which regulate the mechanism of coagulation.
Reason: Calcium iron is very important for clotting of blood.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BFC568

50. **Assertion:** The number of eosinophils increases in patients suffering from asthma.
Reason: Asthma is an allergic disease.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BFC569

Exercise – I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	2	3	3	2	3	2	3	1	2	1	3	3	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	1	2	4	4	4	3	3	3	1	3	1	1	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	2	1	3	1	3	2	3	4	3	3	2	1	4	1
Question	46	47	48	49	50										
Answer	1	3	1	2	1										

Exercise – II
Body Fluids and Circulation

1. Blood, a special type of connective tissue
(1) Consists of a fluid matrix (Plasma)
(2) Has formed elements
(3) Is the most commonly used body fluid by most of the higher organisms
(4) All

BFC570

2. I. Proteins contribute 6-8% of the blood plasma
II. Plasma contains very high amount of minerals
III. Plasma without the clotting factors is called serum
IV. Glucose, amino acids, lipids, etc., are also present in the plasma as they are always in transit in the body
Of the above statements-
(1) All are correct
(2) Only III is false
(3) Only I, III, IV is correct
(4) All are false

BFC571

3. Formed elements of blood include-
(1) RBC, WBC and blood platelets
(2) All solutes present in blood
(3) Proteins present in blood
(4) All minerals (elements)

BFC572

4. What is the amount of haemoglobin present in 100 ml blood of human blood?
(1) 45g (2) 18-20g
(3) 12-16g (4) 10-12g

BFC573

5. Mammalian RBCs are in shape-
(1) Oval
(2) Biconvex
(3) Biconcave
(4) Sickle like

BFC574

6. All of the following are granulocytes except-
(1) Neutrophils
(2) Eosinophils
(3) Basophils only
(4) Lymphocytes and monocytes

BFC575

7. Match list I with list II correctly :

	List I (Types of leucocytes/WBCs)		List II (Their % (of total WBC)
I.	Neutrophils	A.	20-25
II.	Basophils	B.	2 – 3
III.	Monocytes	C.	6 – 8
IV.	Eosinophils	D.	0.5 – 1
V.	Lymphocytes	E.	60 - 65

(1) I-E, II-D, III-C, IV-A, V-B
(2) I-A, II-B, III-C, IV-E, V-D
(3) I-E, II-D, III-C, IV-B, V-A
(4) I-B, II-D, III-A, IV-C, V-A

BFC576

8. Match the following :

	Column I		Column II
I.	Basophils	A	Phagocytes
II.	Neutrophils	B	Secrete histamine, serotonin, heparin and involved in inflammatory response
III.	Monocytes	C	Resist infections and are also involved in allergic reaction
IV.	Eosinophils	D	Immunity
V.	Lymphocytes		

(1) I-B, II and III-A, IV-C, V-D
(2) I-B, II and III-C, IV-A, V-D
(3) I-C, II and III-A, IV-B, V-D
(4) I-D, II and III-C, IV-A, V-B

BFC577

9. During blood clotting, platelets release-
(1) Thrombin
(2) Fibrinogen
(3) Prothrombin
(4) Thromboplastin and other blood clotting factors

BFC578

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10. Find the correct descending order of percentage proportion of leucocytes in human blood.

(1) Neutrophils → Basophils → Lymphocytes → Acidophils (Eosinophils) Monocytes

(2) Neutrophils → Monocytes → Lymphocytes → Acidophils → Basophils

(3) Neutrophils → Lymphocytes → Monocytes → Acidophils → Basophils

(4) Neutrophils→ Acidophils → Basophils → Lymphocytes→ Monocytes

BFC579

11. **Assertion**:- A physician might order a white cell count for a patient with symptoms of an infection.

Reason:-An increase in the number of white blood cells (leukocytes) may indicate that the person is combating an infection.

(1) Both assertion and reason are true and reason is correct explanation of assertion

(2) Both assertion and reason are true and reason is not correct

(3) Assertion is true but reason is false

(4) Both assertion and reason are false

BFC580

12. ABO blood grouping is based on the presence or absence of surface antigens

(1) 2 (2) 3 (3) 6 (4)12

BFC581

13. Which one indicates B.P. or hypertension?

(1) 120/80 mmHg

(2) 80/120 mmHg

(3) 140/90 mmHg or higher

(4) 40/60 mmHg

BFC582

14. Rh factor is responsible for-

(1) Sickle cell anemia

(2) Erythroblastosis foetalis

(3) AIDS

(4) Turner's syndrome

BFC583

15. In erythroblastosis foetalis, which of the following factor passes through placenta into foetus-

(1) Rh antigens (2) Rh antibodies

(3) Agglutinins (4) ABO antibodies

BFC584

16. Which of the following is expected if husband is Rh+ and wife is Rh-?

(1) No problem with 1st pregnancy

(2) Problem would be expected with 2nd and subsequent pregnancies

(3) Both (1) and (2)

(4) No problem could be expected in any pregnancy

BFC585

17. What is the correct order of these events?

1. Conversion of fibrinogen of fibrin
2. Clot retraction and leakage of serum
3. Thromboplastin formation
4. Conversion of prothrombin to thrombin

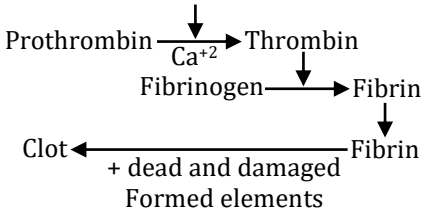
(1) 3,2,1,4 (2) 3,4,1,2

(3) 3,4,2,1 (4) 4,1,3,2

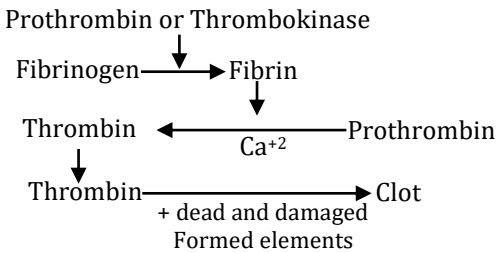
BFC586

18. Which of the following pathways is correct for blood clotting

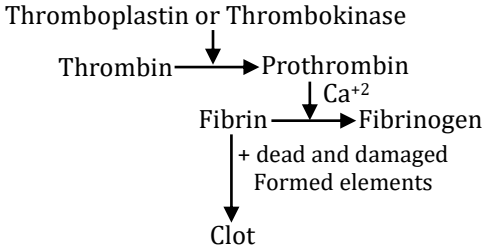
(1)



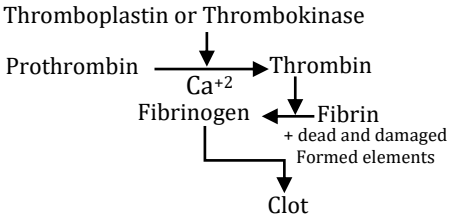
(2)



(3)



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(4)


BFC587

19. White of following statements is wrong about lymph?

I. Lymph is colourful as it has haemoglobin but no RBC

II. The fluid present in lymphatic system is called lymph

III. It contains specialized lymphocytes which are responsible for immunity of the body

IV. Lymph is an important carrier for nutrients and hormones

V. Fats are absorbed through lymph in the lacteals present in the intestinal villi

(1) Only I (2) III and IV

(3) II and III (4) Only IV

BFC588

20. Closed circulatory system is found in-

(1) Arthropod and chordates

(2) Molluscs and chordates

(3) Amphibians and molluscs

(4) Annelids and chordates

BFC589

21. Advantages of closed circulatory system over open circulatory system includes which of the following?

(1) Closed system can direct blood to specific tissues

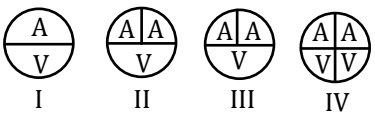
(2) Exchange occurs more rapidly

(3) Close circulatory system can support higher levels of metabolic activity

(4) All of these

BFC590

22. Following are figures of hearts in different animals



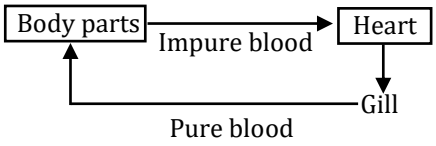
A = Auricle , V = Ventricle

Identify with their characteristic hearts-

	I	II	III	IV
(1)	Fishes	Reptiles	All reptiles	Birds, Mammals
(2)	Fishes	Birds	Reptiles	Mammal
(3)	Fishes	Amphibians	Reptiles	Crocodiles, Birds, Mammals
(4)	Fishes	Crocodiles	Amphibians, Reptiles	Birds, Mammals

BFC591

23. In fishes the blood circulation is represented as-



The above flow of blood indicates it is a:

(1) Double circulation

(2) Single circulation

(3) Incomplete single circulation

(4) Incomplete double circulation

BFC592

24. Bicuspid valve/mitral valve is found between:

(1) Left atrium and left ventricle

(2) Right atrium and right ventricle

(3) Right atrium and right ventricle

(4) Left atrium and right ventricle

BFC593

25. Ventricles are thick-walled as compared to atrium because-

(1) They are to receive blood from atria

(2) They are present on the posterior side

(3) They are to pump blood

(4) None

BFC594

26. Match the following

	Column I		Column II
A.	Superior vena cava	p.	Carries deoxygenated blood to lungs
B.	Inferior vena cava	q.	Carries oxygenated blood from lungs
C.	Pulmonary artery	r.	Brings deoxygenated blood from lower part of body to right atrium

D.	Pulmonary vein	s.	Bring deoxygenated blood from upper part of body to right atrium

(1) A-q, B-s, C-r, D-p
(2) A-s, B-p, C-q, D-r
(3) A-s, B-r, C-p, D-q
(4) A-s, B-p, C-r, D-q

BFC595

27. Origin of heart beat and its conduction is represented by-
(1) SA-node → Purkinje fibres → AV node → Bundle of his
(2) AV node → Bundle of His → SA-node → Purkinje fibres
(3) Purkinje fibres → AV node → SA-node → Bundle of His
(4) SA-node → AV node → Bundle of His → Purkinje fibres

BFC596

28. SA node is called pace maker of the heart. Why?
(1) It can change contractile activity generated by AV node
(2) It delays the transmission of impulse between the atria and ventricles
(3) It gets stimulated when it receives neural signal
(4) It initiates and maintains the rhythmic contractile activity of heart

BFC597

29. Sino-atrial node (SAN) can generate impulses-
(1) 70-75 min^{-1} (2) 50-55 min^{-1}
(3) 35-40 min^{-1} (4) 100-150 min^{-1}

BFC598

30. Bundle of His is a group of-
(1) Ganglia
(2) Nerve fibres
(3) Muscular fibres
(4) Connective tissue

BFC599

31. Atria-ventricular node (AVN) is situated in
(1) Lower left corner of left auricle, close to AV-septum
(2) Lower left corner of right auricle, close to AV septum
(3) Upper left corner of right auricle, close to AV-septum
(4) Upper left corner of left auricle, close to AV septum

BFC600

32. The chordae tendineae-
(1) Close the AV-valves
(2) Prevent the AV-valves flaps from everting
(3) Open semilunar valves
(4) Are present in auricles

BFC601

33. An atrioventricular valve prevents the back flow or leakage of blood from-
(1) The right ventricular into the right atrium
(2) The left atrium into the left ventricle
(3) The aorta into the left ventricle
(4) The pulmonary vein into the right atrium

BFC602

34. **Assertion**-If you trace the path of a molecule of carbon dioxide that starts in an arteriole in the right thumb and leaves the body in exhaled air, the minimum number of capillary beds the molecule encountered is 2.
Reason-The molecule of carbon dioxide would need to enter a capillary bed in the thumb before returning to the right atrium and ventricle, then travel to the lung and enter a capillary from which it would diffuse into an alveolus and be available to be exhaled.
(1) Both assertion and reason are true and reason is correct explanation of assertion.
(2) Both assertion and reason are true and reason is not correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.

BFC603

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35. The duration of cardiac cycle in a normal man is :-
(1) 0.8 seconds (2) 80 seconds
(3) 60 seconds (4) 72 seconds
BFC604

36. During ventricular systole-
(1) Oxygenated blood is pumped into the aorta and deoxygenated blood is pumped into the pulmonary artery
(2) Oxygenated blood is pumped into the pulmonary artery and deoxygenated blood is pumped into the artery
(3) Oxygenated blood is pumped into aorta and deoxygenated blood is pumped into pulmonary vein
(4) Oxygenated blood is pumped into pulmonary vein and deoxygenated blood is pumped into pulmonary artery
BFC605

37. When ventricular systole occurs-
(1) Auricular diastole coincides
(2) Tricuspid and bicuspid valves close
(3) Semilunar valves guarding pulmonary artery and aorta are forced to open
(4) All of these
BFC606

38. Which of the following events do not occur during joint diastole?
I. All 4 chambers of heart are in relaxed state
II. Tricuspid and bicuspid valves open
III. Action potential is conducted from SAN to AVN
IV. Blood from the pulmonary veins and vena cava flows into the left and right ventricles respectively through the left and right atria.
V. The semilunar valves are closed.
(1) Only V (2) Only III
(3) Only IV (4) Only I and II
BFC607

39. Cardiac output is determined by-
(1) Heart rate (2) Stroke volume
(3) Blood flow (4) Both (1) and (2)
BFC608

40. During cardiac cycle each ventricle pumps out about 70 ml of blood which is called-
(1) Stroke volume (2) Cardiac output
(3) Tidal volume (4) Residual volume
BFC609

41. A red blood cell, entering the right side of the heart passes by or through the following structures-
1. Atrioventricular valves
2. Semi-lunar valves 3. Right atrium
4. Right ventricle 5. SAN
(1) 2→3→1→4→5 (2) 3→1→5→2→4
(3) 3→5→1→2→4 (4) 5→3→1→4→2
BFC610

42. Cardiac output is-
(1) Stroke volume (SV) x Heart rate (HR) =5L/min
(2) SV x HR = 500 ml
(3) SV x HR = 72 ml/min
(4) SV x HR = 70 ml/min
BFC611

43. First cardiac sound (lub) is associated with-
(1) Closure of tricuspid and bicuspid valves
(2) Opening of tricuspid and bicuspid valves
(3) Closure of semilunar valves
(4) Opening of semilunar valves
BFC612

44. **Assertion-**After exercising regularly for several months, our resting heart rate decreases, but our cardiac output at rest in unchanged.
Reason-The heart, like any other muscle, becomes stronger through regular exercise. The stronger heart would have a lesser stroke volume, which would allow for the decrease in heart rate.
(1) Both assertion and reason are true and reason is correct explanation of assertion
(2) Both assertion and reason are true and reason is not correct explanation of assertion
(3) Assertion is true but reason is false
(4) Both assertion and reason are false
BFC613

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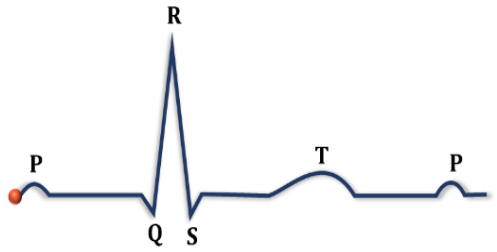
45. Electrocardiogram is a measure of-
(1) Heart rate
(2) Ventricular contraction
(3) Volume of blood pumped
(4) Electrical activity of heart

BFC614

46. QRS complex represents the-
(1) Depolarization of ventricles
(2) Repolarization of ventricles
(3) Repolarization of atria
(4) Depolarization of atria

BFC615

47. The below figure is the diagrammatic representation of standard ECG.



	Column I		Column II
A.	P-wave	I.	Ventricular depolarization followed by ventricular contraction
B.	QRS complex	II.	Atrial depolarization followed by systole of both atria
C.	T-wave	III.	Ventricular repolarization followed by ventricular relaxation

(1) A-I, B-II, C-III (2) A-III, B-II, C-I
(3) A-II, B-I, C-III (4) A-II, B-III, C-I

BFC616

48. Match the column I with column II.

	Column I		Column II
A.	Counting the number of QRS complex in a given time period	I.	A detailed evaluation of the heart function
B.	Potential generate by the recovery of ventricle from the depolarization state	II.	Determination of heart beat
C.	Multiple leads are attached to the chest region	III	T-wave

(1) A-II, B-III, C-I (2) A-III, B-II, C-I
(3) A-I, B-II, C-III (4) A-III, B-I, C-v

BFC617

49. Note the following blood vessels-
A. Arteriole B. Capillary
C. Aorta D. Muscular artery
E. Vein F. Venule
Choose the correct path that lists the blood vessels in order, blood passes through them as it leaves the heart, travels to tissue and returns to heart-
(1) C, D, B, A, E, F (2) C, D, A, B, F, E
(3) D, C, A, B, E, F (4) D, C, B, A, E, F

BFC618

50. The blood circulation which starts and ends into capillaries is-
(1) Portal circulation
(2) Renal circulation
(3) Hepatic circulation
(4) Lymphatic circulation

BFC619

51. Vascular connection between digestive tracts and liver is called-
(1) Hepatic circulation
(2) Hepatic portal system
(3) Both (1) and (2)
(4) Hepatic sinusoid

BFC620

52. A special coronary system of blood vessels present in our body exclusively for the circulation of blood to and from the-
(1) Corneocytes
(2) Cornea
(3) Cori cycle
(4) Heart/Cardiac musculature

BFC621

53. **Assertion-**There is low velocity of blood flow in the capillaries.
Reason-There is large total cross sectional area of the capillaries.
(1) Both assertion and reason are true and reason is correct explanation of assertion
(2) Both assertion and reason are true and reason is not correct explanation of assertion
(3) Assertion is true but reason is false
(4) Both assertion and reason are false

BFC622

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54. Cardiac centre lies in
(1) Medulla oblongata (2) Pons
(3) Cerebrum (4) Epithalamus
BFC623

55. Neural signal through the sympathetic nerve (part of ANS) increases cardiac output because of-
(1) Increasing the rate of heart beat
(2) Increasing the strength of ventricular contraction
(3) Both (1) and (2)
(4) Increasing the stimulation of vagus nerve
BFC624

56. Heart beat increases:
(1) On stimulation of sympathetic nerves
(2) On stimulation of vagus nerve (para sympathetic nerve)
(3) By adrenaline secreted by adrenal medulla
(4) Both (1) and (3)
BFC625

57. In adult, normal blood pressure is-
(1) 80/120 mmHg (2) 100/80 mmHg
(3) 120/80mmHg (4) 100/120mmHg
BFC626

58. Cardiac cycle comprises of systole and diastole in chambers of the heart In which of the below mentioned phases, ventricular filling does not occur
(1) Atrial systole
(2) Atrial diastole
(3) Ventricular systole
(4) Ventricular diastole
BFC627

59. **Statement I** : AV node is present in lower left corner of the right atrium.
Statement II : Only SAN has the ability to initiate and generate impulses among the nodal tissue hence SAN is called as the pacemaker of heart
(1) Both statement I and II are correct
(2) Statement I is correct, II is incorrect
(3) Statement II is correct, I is incorrect
(4) Both statement I and II are incorrect
BFC628

60. Match column I with column II & choose the correct option.

	Column I		Column II
A.	Acute chest pain	i.	Coronary artery disease
B.	Makes lumen of arteries narrower	ii.	Angina
C.	Can lead to brain haemorrhage/ stroke/ CVA	iii.	Myocardial infarction
D.	Heart muscles suddenly damaged due to inadequate blood supply	iv.	Hypertension

(1) A-i, B-ii, C-iii, D-iv
(2) A-iii, B-i, C-iv, D-ii
(3) A-ii, B-i, C-iv, D-iii
(4) A-iii, B-iv, C-ii, D-i
BFC629

61. Match column I with column II and choose the correct option.

	Column I		Column II
A.	AV bundle	i.	Ventricular musculature
B.	Purkinje fibers	ii.	Three bundle of muscle fibres
C.	Internodal pathways	iii.	Delays impulse transmission to ventricles
D.	AV node	iv.	Interventricular septum

(1) A-ii, B-i, C-iii, D-iv
(2) A-iv, B-i, C-ii, D-iii
(3) A-ii, B-iv, C-ii, D-iii
(4) A-iv, B-ii, C-i, D-iii
BFC630

62. **Statement I :** A patch of nodal tissue is present in the right upper corner of the right atrium called the sino atrial node.
Statement II : The right and left bundle along with the purkinje fibres is known as bundle of HIS.
(1) Statement I correct and II are incorrect
(2) Both statement I and II is correct
(3) Statement I is incorrect, II is correct
(4) Both statement I and II are incorrect
BFC631

63. Match column I with column II and choose the correct option.

	Column I		Column II
A.	Mitral valve	i.	Thin, muscular
B.	Tricuspid valve	ii.	Thick, fibrous
C.	AV septum	iii.	Is in contact with deoxygenated blood
D.	Inter atrial septum	iv.	Is located near left AV septum

(1) A-iv, B-iii, C-ii, D-i
(2) A-iv, B-iii, C-i, D-ii
(3) A-iii, B-iv, C-i, D-ii
(4) A-iv, B-iii, C-ii, D-i

BFC632

64. **Assertion:** Erythroblastosis foetalis is usually not observed during first pregnancy although foetus is Rh positive and mother is Rh negative.
Reason: Rh negative blood of mother do not get exposed to Rh antigens of foetus in first pregnancy as the two bloods are well separated by placenta.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BFC633

65. **Assertion:** If we cut nerve supply to human heart, there will be no effect on heart beat.
Reason: Human heart is myogenic in nature.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BFC634

Exercise – II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	1	3	3	4	3	1	4	3	1	1	3	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	2	1	1	4	4	3	2	1	3	3	4	4	1	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	2	1	1	1	1	4	2	4	1	4	1	1	3	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	3	1	2	1	2	4	1	1	3	4	1	3	2	3
Question	61	62	63	64	65										
Answer	2	2	4	1	1										

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Exercise – I
Locomotion and Movement

1. The number of bones in the lower jaw of human is :
(1) 1 (2) 4 (3) 6 (4) 8
BLC301

2. Out of 'X' pairs of ribs in humans only 'Y' pairs are true ribs. Select the option that correctly represents values of X and Y and provides their explanation.

(1)	X=12, Y=7	True ribs are attached dorsally to vertebral column and ventrally to the sternum
(2)	X=12, Y=5	True ribs are attached dorsally to vertebral column and sterbyn on the two ends
(3)	X=24, Y=7	True ribs are dorsally attached to vertebral column but are free on ventral side
(4)	X=24, Y=12	True ribs are dorsally attached to vertebral column but are free on ventral side

BLC302

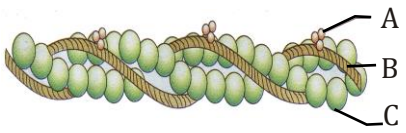
3. Symphysis contains :-
(1) Hyaline cartilage (2) Fibrous cartilage
(3) Calcified cartilage (4) Elastic cartilage
BLC303

4. Choose the correct option regarding a normal human :-
A. The skull is dicondylic
B. Metacarpals are five in numbers
C. Patella is a cup-shaped bone covering the knee dorsally
D. Scapula is a large triangular flat bone, situated on the ventral side of the thorax
E. The pelvic girdle has two coxal bones
(1) A and E are wrong
(2) A and B are wrong
(3) B and E are wrong
(4) C and D are wrong
BLC304

5. The pectoral and pelvic girdles and the bones of limb form :-
(1) Axial skeleton
(2) Appendicular skeleton
(3) Visceral skeleton
(4) Outer skeleton
BLC305

6. When the head of humerus fits into glenoid cavity, joint is :-
(1) Ball and socket joint (2) Hinge joint
(3) Pivot joint (4) Saddle joint
BLC306

7. See the figure of actin (thin) filaments. Identify A, B and C



(1) A – Troponin, B – Tropomyosin, C– F-actin
(2) A–Troponin, B–Tropomyosin, C–F-Myosin
(3) A – Troponin, B – Myosin, C– F-Tropomyosin
(4) A – Tropomyosin, B – Troponin, C– F-actin
BLC307

8. The dark bands (Black bands) of a skeletal muscle are known as :-
(1) Anisotropic or A-band
(2) Isotropic or I-band
(3) Z-band
(4) Both in A and I-band
BLC308

9. Match the following and mark the correct option:-

Column-I	**Column-II**
A. Sternum	i. Synovial fluid
B. Glenoid Cavity	ii. Vertebrae
C. Freely movable joint	iii. Pectoral girdle
D. Cartilagenous joint	iv. Flat bones

Option :-
(1) A-ii, B-i, C-iii, D-iv
(2) A-iv, B-iii, C-i, D-ii
(3) A-ii, B-i, C-iv, D-iii
(4) A-iv, B-i, C-ii, D-iii
BLC309

10. Which one of the following is not a disorder of bone ?
(1) Arthritis (2) Osteoporosis
(3) Rickets (4) Atherosclerosis
BLC310

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11. Which one of the following statement is incorrect?
(1) Heart muscles are striated and involuntary.
(2) The muscles of hands and legs are striated and voluntary.
(3) The muscles located in the inner walls of alimentary canal are striated and involunatry.
(4) Muscles located in the reproductive tracts are unstriated and involuntary.

BLC311

12. Which one of the following statements is true?
(1) Head of humerus bone articulates with acetabulum of pectoral girdle
(2) Head of humerus bone articulates with glenoid cavity of pectoral girdle
(3) Head of humerus bone articulates with a cavity called acetabulum of pelvic girdle
(4) Head of humerus bone articulates with glenoid cavity of pelvic girdle

BLC312

13. A sarcomere is best described as a :-
(1) movable structural unit within a myofibril bounded by H zones.
(2) fixed structural unit within a myofibril bounded By Z lines.
(3) fixed structural unit within a myofibril bounded by A bands.
(4) movable structural unit within a myofibril bounded by Z lines.

BLC313

14. Function of long bones in adult mammals is to provide :-
(1) support only
(2) support and produce RBCs only
(3) support and produce WBCs only
(4) support and produce RBCs and WBCs

BLC314

15. Which of the following vertebrae are fused ?
(1) Cervical (2) Sacrum
(3) Lumbar (4) Thoracic

BLC315

16. Part of the body having a single pair of bones is called :-
(1) pelvic girdle (2) external ear
(3) wrist (4) lower jaw

BLC316

17. Which of the following is a single U-shaped bone, present at the base of the buccal cavity and it is also included in the skull ?
(1) Hyoid (2) Malleus
(3) Sacrum (4) Scapula

BLC317

18. Which of the following statement is incorrect?
(1) All movements lead to locomotion
(2) Ciliary movement help in passage of ova through female reproductive tract
(3) Microfilaments are involved in amoeboid movement
(4) In *Paramecium*, the cilia help in movement of food through cytopharynx and in locomotion as well

BLC318

19. Which of the following is not the feature of red muscle fibres ?
(1) They have plenty of mitochondria
(2) They have high content of myoglobin
(3) They have high amount of sarcoplasmic reticulum
(4) They are called aerobic muscles

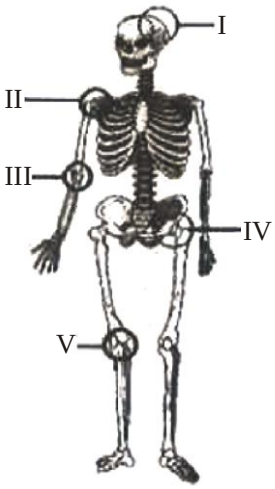
BLC319

20. Read the following statements (A to D) and select the one option that contains both correct statements.
A. Z-line is present in the centre of the light band.
B. Thin filaments are firmly attached to the M-line.
C. The central part of thick filaments, not overlapped by thin filaments is called Z-band.
D. Light band contains only thin filaments.
(1) A and D (2) B and C
(3) A and C (4) B and D

BLC320

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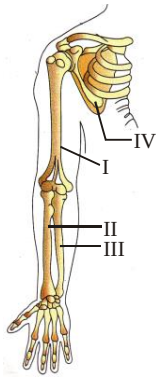
21. The given diagram of human skeleton system shows different type of joints. Identify which of the following from I, II, III, IV and V as marked below are ball and socket joints.



(1) I and II
(2) II and IV
(3) III and IV
(4) IV and V

BLC321

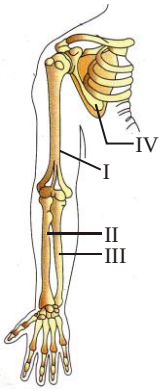
22.



	I	II	II	IV
(1)	Clavicle	Ulna	Radius	Humerus
(2)	Humerus	Radius	Ulna	Scapula
(3)	Scapula	Radius	Ulna	Clavicle
(4)	Humerus	Ulna	Radius	Scapula

BLC322

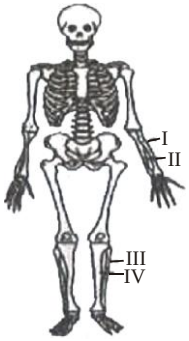
23. Which two bones in the given figure form a hinge joint ?



(1) I and II
(2) I and III
(3) Both (1) and (2)
(4) II and III

BLC323

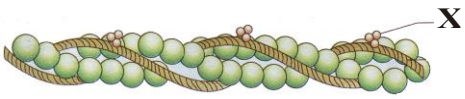
24. The given diagram shows a human skeleton. Which of the following correctly identifies the bones marked as I, II, III & IV ?



	I	II	III	IV
(1)	Tibia	Fibula	Radius	Ulna
(2)	Radius	Ulna	Fibula	Tibia
(3)	Ulna	Radius	Tibia	Fibula
(4)	Fibula	Tibia	Radius	Ulna

BLC324

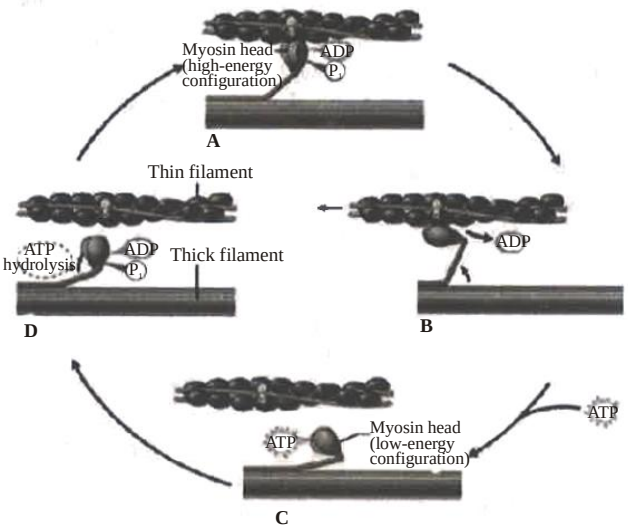
25. The label X in the given figure of an actin filament represents



(1) actin
(2) myosin
(3) tropomyosin
(4) troponin

BLC325

26. The given figure represents the cross bridge cycle in skeletal muscle. What does the step B in the figure represents ?



(1) Attachment of myosin head to actin forming cross bridge.
(2) Release of phosphate, Myosin changes shape to pull actin.

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(3) Attachment of new ATP to myosin head. The cross bridge detaches.
(4) Splitting of ATP into ADP and Pi. Myosin cocks into its high energy conformation.

BLC326

27. **Assertion:** Repeated activation of skeletal muscle can lead to fatigue.
Reason: Accumulation of lactic acid occur due to anaerobic respiration.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BLC327

28. **Assertion:** Skeletal muscle fibre is a syncitium.
Reason: Sarcoplasm of skeletal muscle fibre contain many nuclei.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BLC328

29. **Assertion:** Myasthenia gravis is autoimmune disorder.
Reason: It results due to antibodies are formed that block our own acetylcholine receptor at neuromuscular junction.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BLC329

30. **Assertion:** Biceps and triceps are called antagonistic muscles.
Reason: This is due to the fact that they contract and relax together.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BLC330

Exercise – I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	2	4	2	1	1	1	2	4	3	2	4	4	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	1	1	3	1	2	2	3	2	4	2	1	1	1	3

Exercise - II
Locomotion and Movement

1. Which of the following statements is correct?
(1) Cytoplasmic streaming helps in pseudopodia formation or amoeboid movement
(2) Cytoplasmic streaming is caused by contracting microfilament
(3) Both (1) and (2)
(4) Locomotion is not a voluntary movement

BLC331

2. I. *Paramecium* employs cilia for pushing food in cytopharynx and in locomotion
II. Hydra takes help of tentacles for both food capturing and locomotion
III. All locomotion's are movements and vice-versa
IV. Methods of locomotion very with habitats and the demands of situation
V. Ciliated epithelium is found in respiratory tract, renal tubules and reproductive tracts

Which of the above statements is false?
(1) I and III (2) III
(3) III and V (4) IV and V

BLC332

3. Which of the following statements is false?
(1) Locomotion and many other movements require coordinated muscular activities
(2) Muscle is a specialised tissues of endodermal in origin
(3) There are about 639 muscles which contribute about 40-50% of adult body weight
(4) Muscles show contractibility, excitability and flexibility

BLC333

4. Which of the following statements about the skeletal muscles is correct?
(1) They are striated muscles
(2) They are voluntary muscles
(3) They are primarily involved in locomotory actions
(4) All

BLC334

5. Which of the following statements about visceral muscles is correct?
(1) They are non-striated muscles (smooth muscles)
(2) They are involuntary muscles
(3) They have various functions
(4) All

BLC335

6. Which of the following statements is false?
(1) Smooth muscles are found in urinary bladder, alimentary canal and genital tract.
(2) A striated muscle is syncytium (multinucleate).
(3) The cytoplasm of striated muscle is called endoplasm.
(4) The plasma membrane and ER of striated muscles are called sarcolemma and sarcoplasmic reticulum respectively.

BLC336

7. Select the true statement(s)-
(1) A-band is present in the middle of sarcomere
(2) H-zone is present in the middle of A-band
(3) M-line is present in the middle of H-zone
(4) All of the above

BLC337

8. Match column I with column II-

	Column I		Column II
A.	Structure and functional unit of a myofibril	I.	H-zone
B.	Protein of thin filament	II.	Myosin
C.	Protein of thick filament	III.	Sarcomere
D.	The central part of thick filament not overlapped by thin filament	IV.	Actin

(1) A-I, B-II, C-III, D-IV
(2) A-I, B-III, C-II, D-IV
(3) A-I, B-IV, C-III, D-II
(4) A-III, B-IV, C-II, D-I

BLC338

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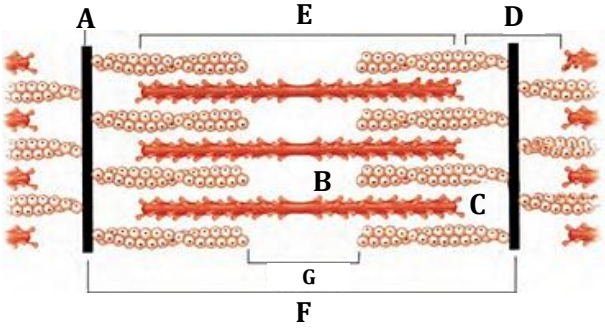
9. Which of the following statements about the striated muscles is false?

I. In the centre of each I-band is an elastic fibre (Z-line) which bisects it.
II. Thin filaments are firmly attached to the Z-line
III. M-line is a fibrous membrane in the middle of A-bands
IV. A sarcomere comprises one full A-bands of 2 half I-bands.

(1) All (2) IV
(3) I and II (4) None

BLC339

10. Choose the letter from the figure that most appropriately corresponds to the structure-



I. A-band
II. I-band
III. Sarcomere
IV. H-zone
V. Myosin
VI. Actin, Troponin, Tropomyosin
VII. Z-line

(1) I-E, II-D, III-F, IV-G, V-B, VI-C, VII-A
(2) I-E, II-D, III-C, IV-G, V-B, VI-A, VII-F
(3) I-E, II-D, III-F, IV-G, V-C, VI-A, VII-B
(4) I-E, II-D, III-F, IV-A, V-B, VI-C, VII-G

BLC340

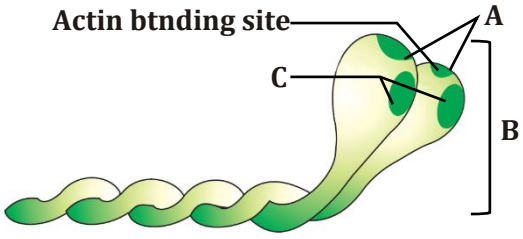
11. Which of the following statements about the molecular arrangement of actin and myosin in myofibrils is false?

I. Each actin (thin filament) is made of 2F (filamentous) actin
II. F-actin is the polymer of G (globular) actin
III. 2F-actins are twisted into a helix
IV. Two strands of tropomyosin (protein) lie in the grooves of F-actin
V. Troponin molecules (complex proteins) are distributed at regular intervals on the tropomyosin.
VI. Troponin forms the head of the myosin molecule.
VII. The myosin is a polymerised protein

(1) I, II, III (2) Only VII
(3) IV, VI (4) Only III

BLC341

12. The below figure is related with myosin monomer (Meromyosin). Identify A to C.



(1) A-head, B-cross arm, C-GTP binding sites
(2) A-head, B-cross arm, C-Ca^{+2} binding sites
(3) A-head, B-cross arm, C-ATP binding sites
(4) A-cross arm, B-head, C-ATP binding sites

BLC342

13. Which of the following statements is false?

(1) Each myosin is a polymerised protein
(2) Many meromyosin constitute one thick filament (myosin)
(3) Each meromyosin's tail is called heavy meromyosin (HMM) and head is called light meromyosin (LMM)
(4) The globular head is an active ATPase enzyme and has binding sites for ATP and active sites for actin

BLC343

14. ATP provides energy for muscle contraction by allowing for-

(1) An action potential formation in the muscle cell
(2) Cross-bridge detachment of myosin from actin
(3) Cross-bridge detachment of myosin to actin
(4) Release of ca^{+2} from sarcoplasmic reticulum

BLC344

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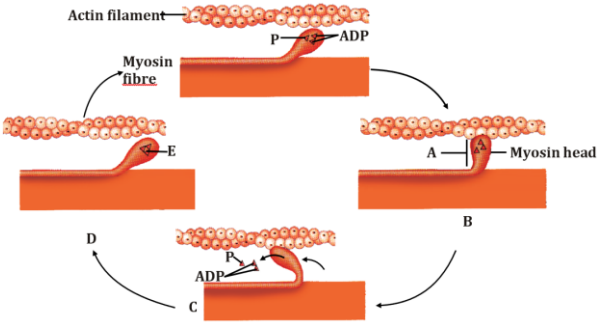
15. Put the following phrases in proper order to describe what occurs at the neuromuscular junction to trigger muscle contraction.

I. Receptor sites on sarcolemma.
II. Nerve impulse.
III. Release of Ca^{+2} from sarcoplasmic reticulum
IV. The neurotransmitter acetylcholine is released
V. Sarcomere shorten
VI. Synaptic cleft
VII. Spread of impulses over sarcolemma on T-tubules

(1) II, I, VI, VII, III, V
(2) II, IV, VI, I, VII, III, V
(3) I, II, III, IV, V, VI, VII
(4) VII, VI, V, IV, III, II, I

BLC345

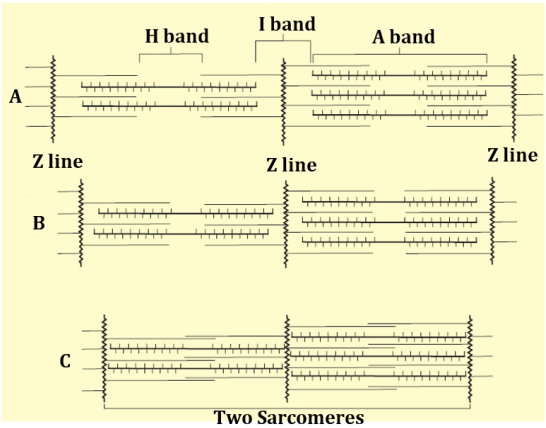
16. Now identify A to E-



(1) A-Cross bridge, B-cross bridge formation, C-Breaking of cross bridge, D-Sliding (rotation), E-ATP
(2) A-Cross bridge, B-Cross bridge formation, C-Sliding/rotation, D-Cross bridge breaking, E-ATP
(3) A-Cross bridge, B-Breaking of cross bridge, C-Sliding/rotation, D-cross bridge formation, E-AMP
(4) A-Cross bridge, B-cross bridge formation, C-sliding /rotation, D-ADP, E-Breaking of cross bridge

BLC346

17. The diagrams given below show 3 different condition of sarcomeres. Identify these conditions-



(1) A-contracting, B-relaxed, C-maximally contracted
(2) A-relaxed, B-contracting, C-maximally contracted
(3) A-maximally contracted, B-contracting, C-relaxed
(4) A-relaxed, B-maximally contracted, C-contracting

BLC347

18. When a skeletal muscle shortens during contraction which of these statements is false?

(1) The I band shortens
(2) The A-band shortens
(3) The H-zone becomes narrow
(4) The sarcomeres shorten

BLC348

19. Which of the following statements is correct?

(1) During muscle contraction chemical energy changes into mechanical energy
(2) Muscle fatigue is due to lactic acid formation due to anaerobic respiration
(3) The reaction time of the fibres can vary in different muscles
(4) All

BLC349

20. Skeletal system consists of-

(1) Only bones
(2) Only cartilage
(3) A framework of bones and a few cartilage
(4) A framework of cartilage and a few bones

BLC350

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21. Match column I with column II-

	Column I		Column II
A.	Cranium/Brain box	I.	29
B.	Skull (cranial and facial bones)	II.	8
C.	Face	III.	14
D.	Hind limb	IV.	12 pairs
E.	Ribs	V.	30

(1) A-I, B-II, C-III, D-V, E-IV
(2) A-II, B-I, C-III, D-V, E-IV
(3) A-I, B-II, C-III, D-IV, E-V
(4) A-V, B-IV, C-III, D-II, A-I

BLC351

22. Which of the following statements about human vertebral column is false?
(1) Vertebral column consists of 26 vertebrae
(2) It is ventrally placed
(3) It extends from the base of skull and constitutes the main framework of the trunk
(4) Neural canal in vertebra is the passage for spinal cord

BLC352

23. Which of the following is not the function of vertebral column?
(1) Protects spinal cord and supports the head
(2) Serves as the point of attachment for ribs and musculature of the back
(3) Both
(4) Supports Tarsals and Metacarpals

BLC353

24. Match column I with column II-

	Column I		Column II
A.	False ribs	I.	1st to 7th pair
B.	True ribs	II.	11th and 12th pair
C.	Floating ribs	III.	8th to 10th pair
D.	Sternum	IV.	One

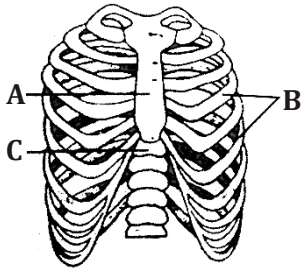
(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-I, C-II, D-IV

BLC354

25. Identify the ribs-
a. Ribs are attached to the sternum ventrally and to the vertebrae dorsally
b. Ribs are attached to sternum through costal cartilage (hyaline) of 7th rib
c. Ribs are not attached to sternum
I. True ribs
II. False ribs
III. Floating ribs
(1) a-I, b-II, c-III (2) a-I, b-II, c-I
(3) a-II, b-I, c-III (4) a-II, b-II, c-I

BLC355

26. The accompanied figure is rib cage. Identify A, B and C respectively-



(1) Coccyx, ribs, vertebral column
(2) Sternum, ribs, vertebral column
(3) Scapula, ribs, vertebral column
(4) Tarsal, ribs, vertebral column

BLC356

27. Which of the following statement is correct?
(1) Pectoral and pelvic girdle bones help in the articulation of the upper and lower limbs respectively with the axial skeleton
(2) Each girdle is formed of 2 halves
(3) Each half of pectoral girdle consists of a clavicle (collar bone) and scapula
(4) All

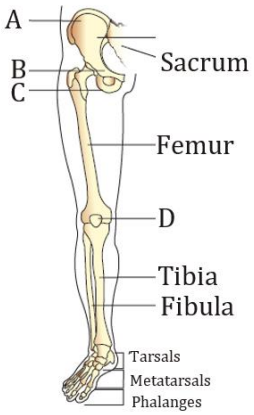
BLC357

28. Which of the following statements is false?
(1) Scapula has the spine which projects as acromion process
(2) Below acromion process is a glenoid cavity
(3) Each clavicle (collar bone) articulates with acromion
(4) Clavicle is long S-shaped bone with 4 curvatures

BLC358

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29. Study the accompanying figure. Identify A, B, C and D-



(1) A-Pubis, B-ilium, C-Ischium, D-Patella
(2) A-Ischium, B-Pubis, C-ileum, D-Patella
(3) A-ileum, B-Pubis, C-Ishium, D-Patella
(4) A-ilium, B-Pubis, C-Ishium, D-Patella

BLC359

30. **Assertion :** Osteoporosis is a age related disorder in which done mass decreases.
Reason : Decreased levels of Estrogen is a common cause.
(1) Both **A** and **R** are correct but **R** is not the correct explanation of **A**
(2) **A** is correct but **R** is not correct
(3) **A** is not correct but **R** is correct
(4) Both **A** and **R** are correct but **R** is the correct explanation of **A**

BLC360

31. **Assertion :** A motor along with muscle fibre connected to it constitute a motor unit.
Reason : The Junction between a motor neuron and the sarcolemma of the muscle fibre is called the motor end plate.
(1) Both **A** and **R** are True and **R** is a correct explanation of the **A**
(2) Both **A** and **R** are True but **R** is not a correct explanation of the **A**
(3) **A** is True but **R** is not False
(4) Both **A** and **R** are False

BLC361

32. **Assertion :** Considerable movement is possible at the Hinge Joint.
Reason : Such joint is also called synovial joint and have almost frictionless movement due to synovial fluid.
(1) Both **A** and **R** are True and **R** is a correct explanation of the **A**
(2) Both **A** and **R** are True but **R** is not a correct explanation of the **A**
(3) **A** is True but **R** is not False
(4) Both **A** and **R** are False

BLC362

33. **Assertion:** Ciliary movement occurs in most of our internal tubular organs which are lined by ciliated epithelium.
Reason: Passage of ova through the female reproductive tract is also facilitated by the ciliary movement.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BLC363

34. **Assertion:** Arthritis or inflammation of a joint makes the joint painful.
Reason: Some toxic substances are deposited at the joint.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BLC364

35. **Assertion:** Each myosin filament is a polymerized carbohydrate.
Reason: Actin filament is made up of meromyosin.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BLC365

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Exercise – II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	2	4	4	3	4	4	4	1	3	3	3	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	2	4	3	2	2	4	4	1	2	4	4	4	4
Question	31	32	33	34	35										
Answer	2	1	2	3	4										

Exercise - I
Excretory Products and Their Elimination

1. Tick mark the correct option ?
(1) Aquatic amphibians → Ammonotelic.
(2) Terrestrial amphibians → Uricotelic
(3) Marine fishes → Ammonotelic
(4) Land snail → Ureotelic.

BEE332

2. Read the following four statements (A-D) and select the correct option ones ?
(A) The medulla is divided into a few conical masses projecting into the calyces.
(B) The cortex extends in between the medullary pyramids as renal columns called columns of Bertini.
(C) Kidneys are blackish brown, bean shaped structures situated between the levels of last thoracic and third lumbar vertebra close to the ventral inner wall of the abdominal cavity.
(D) Each kidney of an adult human measures 10-12 cm in length, 5-7 cm in width, 2-3 cm in thickness with an average weight of 120-170 gm.
(1) Statements A, B & C
(2) Statements B, C & D
(3) Statement A, B & D
(4) Statement B and C

BEE333

3. Which one is the primary function of sweat glands in human?
(1) To facilitate a cooling effect on the body surface.
(2) To help in removal of some of the wastes like NaCl, urea, Lactic acid etc.
(3) It helps in osmoregulation.
(4) It maintains the NaCl in our body fluid.

BEE334

4. Dialysing unit (Artificial kidney) contains a fluid which is almost same as plasma except that it has -
(1) High nutrients
(2) High nitrogenous wastes.
(3) No nitrogenous wastes
(4) High NaCl & urea

BEE335

5. Match the abnormal conditions given in column-A, with their explanations given in column-B and select the correct option ?

	Column-I		Column-II
(A)	Uremia	(i)	Accumulation of urea in urine
(B)	Renal calculi	(ii)	Presence of Glucose in urine
(C)	Glomerulone-phritis	(iii)	Mass of crystallised salt with in kidney
(D)	Glycosuria	(iv)	Inflammation in glomeruli
		(v)	Accumulation of urea in blood

(1) A-ii, B-iii, C-iv, D-i
(2) A-i, B-ii, C-iii, D-iv
(3) A-v, B-iii, C-iv, D-ii
(4) A-v, B-iii, C-iv, D-i

BEE336

6. We have to observe that glucose and ketone bodies pass through urine in a person. Such condition indicates the sign of :-
(1) Diabetes insipidus
(2) Glomerulonephritis
(3) Diabetes mellitus
(4) Renal failure.

BEE337

7. Read the following statements (a-e) and select the correct option which includes all the correct ones?
(a) Micturition is a reflex process
(b) pH of human urine is approximately 6.0.
(c) 25-30 mg of urea is excreted out per day along with urine.
(d) An adult human excretes, 1 to 1.5 litres of urine per day.
(e) During the micturition reflex contraction signal provides to urethral sphincter and simultaneously relaxation signal provides to smooth muscles of wall of urinary bladder.
(1) Statements a, b and c
(2) statements b, c and e
(3) statements a, b and d
(4) statements c, d and e

BEE338

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8. Which of the following statements is correct ?
(1) ADH :- Conversion of angiotensinogen in blood to angiotensin.
(2) Aldosterone :- Reabsorption of Na^+ and water.
(3) ANF :- Enhance Na^+ reabsorption.
(4) Renin :- Causes vasodilation.

BEE339

9. Which one is incorrect about mechanism of concentration of the filtrate ?
(1) NaCl is transported by the ascending limb of 'Loop of Henle' which is the exchanged with the descending limb of vasa-recta.
(2) Help in maintaining an increasing osmolarity towards the inner medullary interstitum.
(3) The proximity between the 'Loop of Henle' and vasa recta as well as counter current in them help in maintaining the osmolarity of medullary interstitium.
(4) Small amount of urea enter in the thick segment of ascending limb of 'Loop of Henle' which is transported back by collecting duct.

BEE340

10. Read the following points with respect to distal convoluted tubule of nephron :-
(a) Its wall is made up by simple cuboidal epithelium.
(b) Conditional reabsorption of H_2O & NaCl takes place in it.
(c) It is also capable of reabsorption of HCO_3^- ions.
(d) It helps in Na^+–K^+ balance in blood.
(e) It also helps in maintainance of pH of body fluid.
How many points are correct ?
(1) Two (2) Three
(3) One (4) All

BEE341

11. Substances like amino acid and glucose in the filtrate are reabsorbed .....(A)..... in the tubular epithelial cells whereas the nitrogenous wastes product are reabsorbed by .....(B)..... What are the A & B and select the correct option ?
(1) A = Actively; B = passively
(2) A = Passively; B = Actively
(3) A = Actively; B = Actively
(4) A = Passively; B = Passively

BEE342

12. Fill in the blanks :-
(a) The glomerular capillary blood pressure cause filteration of blood through 3 layers i.e. the .....(a)..... of glomerular blood vessel, the .....(b)..... of bowman's capsule and .....(c).... between these two layers.
(b) On an average .....(d)..... of blood is filtered by the kidneys per minute which constitute roughly.....(e)..... of the blood pumped-out by each ventricle of the heart in a minute.

	a	b	c	d	e
(1)	Endothelium	Epithelium	Basement membrane	600-800 ml	1/8th
(2)	Epithelium	Endothelium	Basement membrane	1100-1200 ml	1/5th
(3)	Endothelium	Epithelium	Basement membrane	1100-1200 ml	1/5th
(4)	Endothelium	Epithelium	Basement membrane	1500 ml	1/4th

BEE343

13. Match the items of column I with those of column II :

	Column-I		Column-II
(a)	Ammonotelism	(i)	Birds
(b)	Bowman's capsule	(ii)	Water reabsorption
(c)	Micturition	(iii)	Bony fish
(d)	Uricotelism	(iv)	Urinary bladder
(e)	ADH	(v)	Renal tubule

	a	b	c	d	e
(1)	iii	v	i	iv	ii
(2)	iii	v	iv	i	ii
(3)	i	v	iv	iii	ii
(4)	iii	v	iv	ii	i

BEE344

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14. Read the given terms and select the correct option with their appropriate explaination :-

(a) Uremia (b) ANF
(c) micturition (d) glycosuria

(1) (a) accumulation of urea in blood
(b) process of release of urine and it is a involuntary process

(2) (b) release from atria of heart
(c) process by which urine is concentrated

(3) (d) presence of glucose in urine
(b) can cause dilation of blood vessels

(4) (a) accumulation of uric acid in blood
(d) presence of glucose in urine

BEE345

15. JG cells of Juxta glomerular apparatus release to

(1) ANF (2) Rennin
(3) ADH (4) Renin

BEE346

16. Identify the **correct** combination :-

(1) Green glands – crustacean – uricotelic
(2) Malpighian tubule – insect – uricotelic
(3) Nephridia – annelida – uricotelic
(4) Flame cells – Aschelminthes - ammonotelic

BEE347

17. Which of the following is **incorrect** option?

(1) inflammation of glomeruli of kidney is called glomerulonephritis
(2) the epithelial cells of Bowmans capsule are called podocytes
(3) PCT and DCT of nephron are situated in the cortical region of kidney
(4) the ascending limb of loop of henle is permeable to water but allows transport electrolytes only actively

BEE348

18. Select the correct option with the reference of hemodialysis :-

(1) when glucose concentration is high in blood than glucose molecules are removed by hemodialysis
(2) anticoagulant like heparin used during hemodialysis
(3) it is also used for removal of ketone bodies during diabetes mellitus
(4) all of the above

BEE349

19. How many statements are correct ?

(a) vasa recta is absent or highly reduced in cortical nephrons
(b) ammonia is less toxic as compared to urea and requires large amount of water for its elimination
(c) GFR in a healthy individual is approximately 125 ml / minute
(d) PCT also help to maintain the pH and ionic balance of the body fluid

(1) one (2) two
(3) three (4) four

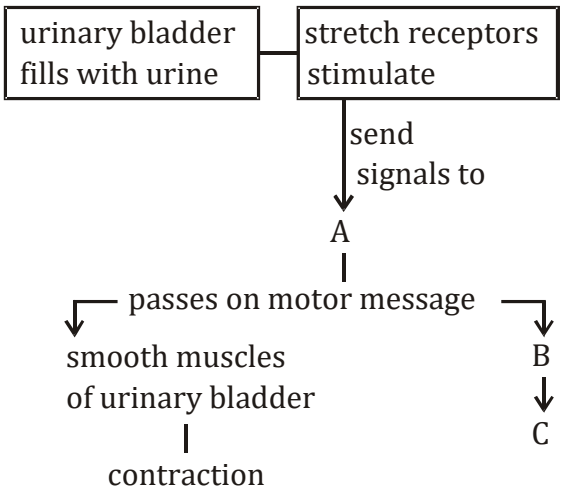
BEE350

20. What will happen when excessive loss of fluid from the body then ?

(1) activates receptors which stimulates the hypothalamus to release ADH from the neuro hypophysis
(2) osmoreceptors switch off and secretion of vasopressin reduces
(3) release of aldosterone from adrenal medulla
(4) angiotensinogen is converted into angiotensin-I

BEE351

21. Study the given chart and choose the correct option with appropriate alphabets



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	A	B	C
(1)	PNS	Urethral Sphincter	relaxation
(2)	CNS	Urethral Sphincter	relaxation
(3)	CNS	Urethral Sphincter	contraction
(4)	PNS	Urethral Sphincter	contraction

BEE352

22. If the kidney fails to function :-
(1) other organs can carry all the functions of excretion for some times
(2) renal failure becomes a life threatening event
(3) they rapidly degenerate their nephrons
(4) cells store their own metabolic wastes until the kidney functions again

BEE353

23. Glucose and 75% water is absorbed in :-
(1) proximal convoluted tubule
(2) distal convoluted tubule
(3) collecting duct
(4) loop of Henle

BEE354

24. Glomerular filtration is possible if :-
(1) afferent arteriole has more diameter than the efferent arteriole
(2) afferent arteriole has less diameter than the efferent arteriole
(3) both have the same diameter
(4) afferent capillaries have less diameter than efferent capillaries

BEE355

25. **Assertion :** Aquatic animals like whales and seals are said to be ureotelic animals.
Reason : It is because of the fact that their main nitrogeneous waste product is urea.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.

BEE356

26. **Assertion :** Kidney maintains the osmotic concentration of the blood.
Reason : Kidney eliminates either hypotonic or hypertonic urine according to the need of the body.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.

BEE357

27. **Assertion :** In the descending limb of loop of Henle, the urine is hypertonic, whereas in ascending limb of loop of Henle, the urine is hypotonic.
Reason : Descending limb is impermeable to sodium, while ascending limb is impermeable to water.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.

BEE358

28. **Assertion :** Secreting hypotonic urine is effective in reducing urinary loss of water.
Reason : Hypotonic urine is more concentrated and higher in osmotic pressure than the blood.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.

BEE359

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29. **Assertion :** Aldosterone is a steroid hormone and is important in the control of sodium and potassium ion concentration in mammals.
Reason : It upgrades sodium ion concentration in the ECF by promoting reabsorption of sodium ions from renal tubules and excretion of potassium ions in urine.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.
BEE360

30. **Assertion** : Main constituent of human urine is ammonia.
Reason : If human urine is allowed to stand for some time, it smells strongly of ammonia.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If Assertion is false but Reason is true.
BEE361

Exercise – I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	3	1	3	3	3	3	2	4	4	1	3	2	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	2	3	1	2	2	1	1	1	1	1	4	1	4

Exercise - II
Excretory Products and Their Elimination

1. The terms "Ammonotelic", "Ureotelic", and "Uricotelic" are used to describe-
(1) Modes of excretory system development
(2) The actions of hormones on the excretory systems
(3) The types of nitrogenous waste produced by various classes of vertebrates
(4) Modification of kidney tubules to enhance excretion

BEE362

2. Which of the following statements is wrong?
(1) Kidney does not play any significant role in the removal of ammonia
(2) Ureotelic animals excrete most of the nitrogenous waste in urea
(3) Ammonia and urea are the waste products derived from the metabolic breakdown of proteins
(4) None of the above is wrong

BEE363

3. Match the column I with column II.

	Column-I		Column-II
A	Nephridia	I	Crustaceans (Prawn)
B	Malpighian tubules	II	Annelids (Earthworm)
C	Antennal gland or green glands	III	Insects (Cockroach)

(1) A-I, B-II, C-III (2) A-III, B-II, C-I
(3) A-II, B-III, C-I (4) A-II, B-I, C-III

BEE364

4. Which of the following statements is false?
I. Outer cortex and inner medulla are the two zones in kidney
II. Medulla is divided into about 8 to 12 medullry pyramids
III. Pyramid projects into calyx
IV. Inwards extension of cortex between the pyramids is called renal column of Bertini
(1) I and IV (2) II and IV
(3) IV (4) None

BEE365

5. Which of the following statements is false?
(1) Renal tubule starts with a double walled cup like structure called Bowman's capsule
(2) In majority of nephrons, the loop of Henle is too short and such nephrons are cortical nephrons
(3) Juxta medullary nephron has long loop of Henle
(4) None

BEE366

6. Which is the correct pathway for passage of urine in humans?
(1) Collecting tubule → ureter → bladder → urethra
(2) Renal vein → renal ureter → bladder → urethra
(3) Pelvis→ Medulla→ bladder→ urethra
(4) Cortex→ Medulla→ bladder→ ureter

BEE367

7. Match the column I with column II.

	Column-I		Column-II
A.	Delivers blood to glomerulus	I	Ascending and descending limb
B.	Carries urine to pelvis, also acts in water reabsorption	II	Renal artery
C.	Collects filtrate from Bowman's capsule	III	Collecting duct
D.	Loop of Henle	IV	PCT

(1) A-II, B-III, C-IV, D-I
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-III, C-II, D-I

BEE368

8. Which of the following is incorrect?
(1) Blood vessel leading to glomerulus is called efferent arteriole
(2) Vasa recta, peritubular capillaries, Glomerulus all have blood
(3) Cortical nephron has no or highly reduced vasa recta
(4) Vasa recta runs parallel to the Henle's loop in juxtamedullary nephrons

BEE369

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9. Which of the following statements is correct?

I. Renal vein take blood away from kidney

II. Loop of Henle conserves water

III. Podocytes occur in inner wall of Bowman's capsule

IV. Ultrafiltrate/nephric filtrate is plasma minus proteins

(1) I and II (2) I and III

(3) III and IV (4) I, II, III, IV

BEE370

10. Juxtaglomerular apparatus, a special sensitive cellular region is formed in-

(1) PCT and DCT

(2) PCT and DCT at the location of their contact

(3) PCT and loop of Henle at the location their contact

(4) DCT and afferent arteriole at the location of their contact

BEE371

11. Which of following statements is false?

(1) The kidney has built in mechanism for regulation of GFR

(2) Tubular secretion does not play any significant role in urine formation

(3) The amount of urine output per day in normal adult is about 1.5 L

(4) During urine formation tubular cells secrete H^+, K^+ and NH_3 in the filtrate

BEE372

12. Which of the following statements about proximal convoluted tubule (PCT) is false?

(1) It is lined by simple cuboidal brush border epithelium which increases the surface area

(2) Nearly all he essential nutrients, 70-80% electrolytes, 70% H_2O are reabsorbed by PCT

(3) PCT is not the site of selective secretion

(4) PCT helps to maintain the pH and ionic balance of body fluids

BEE373

13. PCT helps to maintain the pH and ionic balance of body fluids by-

(1) Selective secretion of H^+ and NH_3 in filtrate

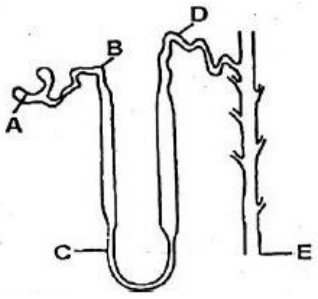
(2) Reabsorption of HCO^-_3 from filtrate

(3) Both (1) and (2)

(4) Secreting regulatory hormone like renin and angiotensinogen

BEE374

14. Use following diagram to complete the statements about the human nephron-



I. The composition of the filtrate would be most like plasma in the tubule.

II. The urine would be most concentrated in the collecting duct.

III. Most of the glomerular filtrate is reabsorbed into peritubular capillary.

IV. Conducting of urine to pelvis of the kidney from the structure.

V. Most water is reabsorbed by the structure.

(1) I-A, II-C, III-B, IV-E, V-D

(2) I-A, II-E, III-B, IV-C, V-D

(3) I-A, II-B, III-E, IV-C, V-D

(4) I-A, II-E, III-B, IV-E, V-B

BEE375

15. I. Reabsorption in this region is minimum

II. This region plays a significant role in the maintenance of high osmolarity of intestinal fluid

III. Its descending limb is permeable to water but almost impermeable of electrolytes

IV. Its ascending limb is impermeable to water but allows transport of electrolyte actively or passively

V. In descending limb filtrate is hypertonic while in ascending limb filtrate is hypotonic

The above characteristics are associated with-

(1) PCT
(2) Loop of Henle
(3) DCT
(4) Bowman's capsule

BEE376

16. Which of the following statements is correct?

I. Reabsorption of water occurs passively in the initial segment of nephron
II. Nitrogenous waste are absorbed by passive transport
III. Conditional reabsorption of Na^+ and water takes place in DCT
IV. DCT reabsorbs HCO_3^-
V. DCT is capable of selective secretion of H^+, K^+ and NH_3 to maintain pH and Na+- K+ balance in blood
VI. Substances like glucose, amino acids, Na^+, etc in the filtrate are reabsorbed actively

(1) I and II (2) II and III
(3) IV and V (4) All

BEE377

17. Which of the following statements is false regarding the collecting duct?

I. Collecting duct is a straight duct
II. It extends from the cortex to medulla
III. Large amount of water could be reabsorbed from it to produce concentrated urine.
IV. Small amount of urea diffuses out from it into the medulla to keep up the osmolarity
V. It plays a role to maintain pH and ionic balance of blood by the selective secretion of H^+ and K^+ ions.

(1) Only I (2) Only III
(3) IV and V (4) None

BEE378

18. Medullary gradient is developed by all the following except-

(1) Reabsorption of Na+ from ascending limb of Henle's loop into medullary interstitium
(2) Reabsorption of Na+ from descending limb of Henle's loop
(3) Diffusion of small amount of urea from collecting duct into medullary interstitium
(4) Proximity between Henle's loop and vasa recta as well as the counter current in them

BEE379

19. The high osmolarity of the renal medulla is maintained by all of the following except-

I. Diffusion of salt from the ascending limb of the loop of Henle
II. Active transport of salt from the upper region of the ascending limb
III. The spatial arrangement of juxtamedullary nephrons
IV. Diffusion of urea from the collecting duct
V. Diffusion of salt from the descending limb of the loop of Henle

(1) Only I (2) Only V
(3) III and IV (4) I and V

BEE380

20. Which of the following sequences is correct for regulation of kidney function?

(1) An excess loss of water from body → stimulates hypothalamus → Osmoreceptors → Neurohypophysis → ADH → Increases water permeability of DCT and CT→ Prevention of diuresis
(2) An excess loss of fluid from body→ Osmoreceptors →Hypothalamus → Neurohypophysis→ ADH → Increases water permeability of DCT and CT → Prevention of diuresis.
(3) An excess loss of fluid from body→ Osmoreceptors →Hypothalamus → Neurohypophysis→ Aldosterone →

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water permeability of DCT and CT Increases → Prevention of diuresis.

(4) An excess loss of fluid from body→ Osmoreceptors →Hypothalamus → Adenohypophysis → ADH → Increases water permeability of DCT and CT → Prevention of diuresis.

BEE381

21. ADH is synthesised by, _____ released by _____ and acts on _____ .

(1) Hypothalamus, Neurohypophysis, DCT and CT

(2) Hypothalamus, Neurohypophysis, Loop of Henle

(3) Hypothalamus, Adenohypophysis, DCT and CT

(4) Hypothalamus, Adenohypophysis, Loop of Henle

BEE382

22. Which of the following sequence is correct?

(1) An increase in body fluid volume→ switch off the osmoreceptors→suppresses the ADH release

(2) ADH-Constricting effect on blood vessel → B.P. high → Glomerular blood flow more → GFR more

(3) Angiotensinogen → Angiotensin I → Angiotensin-II → Adrenal cortex → Aldosterone

(4) All

BEE383

23. Which of the following statements is false?

(1) Angiotensin II, being a powerful vasoconstrictor, increases glomerular pressure and thereby GFR

(2) Angiotensin II activates the adrenal cortex to release aldosterone

(3) Aldosterone promotes reabsorption of Na^+ and water from the DCT and CT leading to an increase in B.P. and GFR

(4) ANF causes vasoconstriction

BEE384

24. RAAS (Renin-Angiotensinogen-Aldosterone system).

(1) Is Triggered when the juxtaglomerular cells of JGA releases renin in response to various stimuli

(2) Is responsible for regulation of kidney function

(3) Are stimulated when ANF is more in blood

(4) (1) and (2) are correct

BEE385

25. Which of the following is true about Atrial Natriuretic factor (ANF)?

(1) An increase in blood volume and B.P. stimulates cardiac atria to release ANF

(2) ANF promotes vasoconstriction and thereby decreases B.P.

(3) ANF acts as a check on RAAS

(4) (1) and (3)

BEE386

26. Following are the steps of dialysis-

A. Blood is passed into a vein.

B. Blood is mixed with heparin.

C. Blood is mixed with anti-heparin.

D. Blood is drained from convenient artery.

E. Blood is passed through a coiled and porous cellophane tube bathing in dialysis fluid.

F. Removal of nitrogenous wastes from blood.

The correct sequence of steps is-

(1) A → B → C → D → E → F

(2) F → C → E → B → A → D

(3) D → B → E → F → C → A

(4) D → C → E → F → B → A

BEE387

27. **Assertion :** Mammal, Terrestrial amphibian mainly excrete urea.

Reason : Adaptation for conservation of water.

(1) Both A & R are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Assertion is False but the Reason is True.

BEE388

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28. **Assertion :** Protonephridia are excretory structures in Cephalochordate.
Reason : Protonephridia are primarily concerned with ionic and fluid volume regulation.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Assertion is False but the Reason is True.
BEE389

29. **Assertion :** Reabsorption of water occurs passively in the final segments of the nephron.
Reason : Juxta medullary nephron are very short and runs deep into the medulla.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Assertion is False but the Reason is False.
BEE390

30. **Assertion:** Angiotensin II increases the glomerular blood pressure thereby GFR.
Reason: To release renin, angiotensin II activates the JG cells.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.
BEE391

31. **Assertion:** Both ammonia and urea are excreted by earthworms.
Reason: Excretion in earthworm depends on the environment.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.
BEE392

32. **Assertion :** During micturition, urine is prevented from flowing back into the ureters.
Reason : Urethral sphincters relax during micturition.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.
BEE393

33. **Assertion:** Glomerular filtration is considered as ultrafilteration.
Reason: Blood is filtered so finely through filtration membrane that almost all the constituent of the plasma except the proteins pass onto the lumen of Bowman 's capsule.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.
BEE394

34. **Assertion :** A fall in GFR activate RAAS which is present in juxta medullary nephrons.
Reason : Macula densa release renin which convert angiotensinogen in blood to angiotensin I and further to angiotensin II.
(1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.

BEE395

35. Which of the following statements is correct?
(1) ADH – prevents conversion of angiotensinogen in blood to angiotensin
(2) Aldosterone – facilitates Na^+ reabsorption
(3) ANF – enhances sodium reabsorption
(4) Renin – causes vasodilation

BEE396

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Exercise – II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	3	4	4	1	1	1	4	4	2	3	3	4	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	4	2	2	2	1	4	4	4	4	3	1	1	3	3
Question	31	32	33	34	35										
Answer	1	2	1	3	2										

Exercise - I

Neural Control and Coordination

1. **Statement I:** When a neuron is not conducting any impulse, the axonal membrane is nearly impermeable to potassium ions.
Statement II: Ionic gradients are maintained by the active transport of ions by sodium potassium pump which transports $3Na^+$ inwards and $2K^+$ outwards into the cell.
(1) Statement-I is true & statement-II is false.
(2) Statement-I is false & statement-II is true.
(3) Both statement-I & II are true.
(4) Both statement-I & II are false.

BNC316

2. The centre which is responsible for the integration of information received from the semi-circular canals of the ear and auditory system are:
(1) Pons
(2) Medulla oblongata
(3) Cerebellum
(4) Hypothalamus

BNC317

3. **Assertion (A):** Neural system has quick coordination.
Reason (R): Neural system provides an organised network of point-to-point connection.
(1) Both A and R are correct and R is the correct explanation of A.
(2) Both A and R are correct but R is not the correct explain of A.
(3) A is correct but R is incorrect.
(4) Both A and R are incorrect.

BNC318

4. Match the following columns.

	Column – I		Column – II
i.	Somatic neural system	a.	Relays impulse from CNS to involuntary organs
ii.	Afferent fibres	b.	CNS to peripheral tissues/organs
iii.	Efferent fibres	c.	Relays impulse from CNS to skeletal muscles
iv.	Autonomic neural system	d.	Tissues/organs to CNS

(1) (i)-(c), (ii)-(b), (iii)-(d), (iv)-(a)
(2) (i)-(a), (ii)-(b), (iii)-(c), (iv)-(d)
(3) (i)-(c), (ii)-(d), (iii)-(b), (iv)-(a)
(4) (i)-(d), (ii)-(b), (iii)-(c), (iv)-(a)

BNC319

5. **Assertion (A):** Association are large region of cerebral cortex which are responsible for complex function like intersensory association, memory etc.
Reason (R): They are clearly sensory and motor in functioning which help them for complex functioning.
(1) Both A and R are correct but R is not the correct explain of A.
(2) Both A and R are correct and R is the correct explanation of A.
(3) A is correct but R is incorrect.
(4) Both A and R are incorrect.

BNC320

6. Which of the following statement is correct ?
(1) Neural system of cockroach consists of network of neurons only.
(2) Afferent nerve fibres transmit impulses from CNS to peripheral tissues
(3) Autonomous nervous system is divided into somatic neural system and peripheral neural system.
(4) The neural system of all animals is composed of highly specialised cells called neurons which can detect, receive & transmit impulse different kinds of stimuli.

BNC321

7. Which of the following statement is not correct ?
(A) During conduction of impulse, the rise in the stimulus induced permeability to Na^+ is extremely short lived.
(B) Resting membrane potential of the membrane at the site of excitation is re-establish by the influx of Na^+.
(C) Na^+–K^+ pump transports, 3 Na^+ to outside and $2K^+$ to inside of axon.
(D) Neuron recognise information of nerve impulse but never transmit it.

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(E) Positive charge on inner axonal membrane is due to outflux and influx of K^+.

(1) A, B & E (2) B, D and E
(3) A, D and E (4) A, B, D and E

BNC322

8. Neurotransmitters in chemical synapses :-
(1) Are released by synaptic vesicle in post synaptic membrane and are received by the receptors in pre synaptic membrane
(2) Are released in synaptic cleft and always have excitatory effect.
(3) Are attached to receptors in post synaptic membrane & transmit inpulses faster than electrical synapse.
(4) Are released by synaptic vesicles through presynaptic membrane in the synaptic cleft & are received by receptors in post synaptic membrane.

BNC323

9. Which of the following option is correctly matched ?
(1) Cerebellar hemisphere – corpus callosum – smallest commissure
(2) Complex regions (cerebral cortex) – Association area – memory & communication
(3) Hindbrain – hypothalamus – intelligence and memory
(4) Midbrain – thalamus – largest choroid plexus

BNC324

10. Brain stem is formed by :-
(1) Hypothalamus, mid brain & cerebellum
(2) Corpora quadrigemina, thalamus & medulla oblongata
(3) Diencephalon, pons and cerebellum
(4) Mid brain, pons & medulla oblongata

BNC325

11. Which of the following statement is correct ?
(1) Electrical synapses are rare in our body.
(2) Neurons are excitable cells because their membranes are in a polarised state.
(3) Piamater is the outermost meninge and is in contact with brain tissues.
(4) Both (1) and (2)

BNC326

12. Corpora quadrigemina, the four round swellings are present :-
(1) On ventral portion of mid brain
(2) On lower region of pons
(3) In front of hypothalamus
(4) On the dorsal portion of midbrain

BNC327

13. The __A__ transmit the impulse via a dorsal nerve root into the CNS. The __B__ then carries signals from CNS to the effector. A & B in the above statements are:-
(1) A-Afferent nerve fibre, B-Afferent nerve fibre
(2) A-Efferent nerve fibre, B-Efferent nerve fibre
(3) A-Efferent nerve fibre, B-Afferent nerve fibre
(4) A-Afferent nerve fibre, B-Efferent nerve fibre

BNC328

14. Structure in brain which is surrounded by cerebrum and is the major coordinating centre for sensory and motor signaling :-
(1) Epithalamus (2) Crura cerebri
(3) Thalamus
(4) Aqueduct sylvius

BNC329

15. Which of the following statement is not correct regarding exciting stage of neuron?
(1) Generation of action potential
(2) More permeability for Na^+ as compared to K^+
(3) Depolarisation of axonal membrane
(4) More negative charge inside

BNC330

16. Consider the given statements
(a) Integrated system in our body, coordinate all activities of the organs
(b) Neural part of integrated system provides chemical integration
(c) Endocrine part of integrated system provides organised connectivity for quick coordination
(1) a and c are true, c is false
(2) a is true , b and c are false
(3) a and c are true, b is false
(4) c is true , a and b are false

BNC331

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17. Which of the following is true for human neural system?
(1) It is ectodermal in origin
(2) Somatic neural system relays impulses from CNS to skeletal muscles.
(3) Neurons show unidirectional conduction of impulse
(4) All of these

BNC332

18. When outer surface of the axonal membrane possesses a positive charge while its inner surface is negatively charged it is called.
(1) Depolarisation (2) Polarisation
(3) Repolarisation (4) Both (2) and (3)

BNC333

19. Choose the correct option with reference to structures with their characteristics and location :

	Structure	Characteristic	Location
(1)	Neuron	Conductivity excitability	Medullated nerve fibres are found in CNS formed by Schwann cells
(2)	Midbrain	Consist of 4 optic lobes	Between forebrain of cerebellum
(3)	Hypothalamus	Secretion of hormones	On ventral surface of cerebellum
(4)	Piamater	Highly vascular outer most meninx	Around brain

BNC334

20. During repolarisation, outer surface of axonal membrane is positively charged due to the :-
(1) Efflux of Na^+ ions
(2) Efflux of Cl^- ions
(3) Efflux of K^+ ions
(4) Influx of K^+ ions

BNC335

21. Read the following
- Conductivity - Excitability
- Contractibility - Elasticity
- All or none

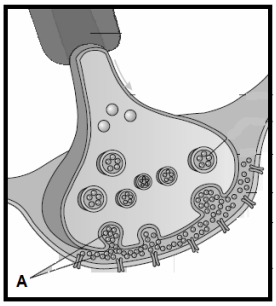
How many properties are shown by a nerve cell?
(1) Two (2) Three
(3) Four (4) Five

BNC336

22. Which of the following is not the function of hypothalamus?
(1) Control body temperature
(2) Synthesis of hormones which are released by neurohypophysis
(3) Storage of memory and seat of intelligence
(4) Control the secretion and inhibition of hormones of the adenohypopysis

BNC337

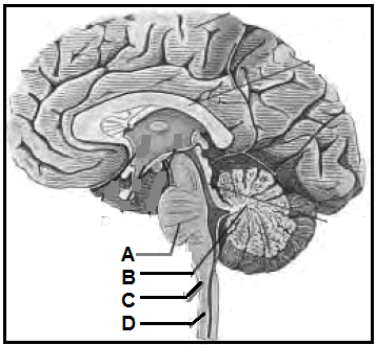
23. Observe the given diagram and select the correct option regarding labeled part A.



(1) It may be either excitatory or inhibitory
(2) It is synthesized by mitochondria
(3) It is stored in synaptic vesicles of synaptic knob
(4) All of these

BNC338

24. Identify A, B, C and D in the given diagram.



	A	B	C	D
(1)	Pons	Cerebellum	Medulla oblongata	Spinal cord
(2)	Cerebellum	Spinal cord	Medulla oblongata	Pons
(3)	Cerebellum	Pons	Spinal cord	Medulla oblongata
(4)	Pons	Cerebellum	Spinal cord	Medulla oblongata

BNC339

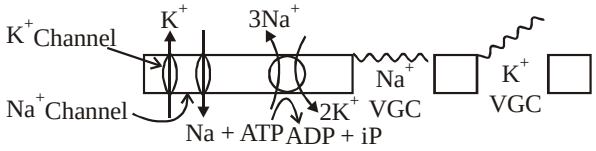
25. Match the column I with column II.

	Column-I		Column-II
(1)	Amygdala	(I)	Largest part of brain
(2)	Cerebrum	(II)	Midbrain
(3)	Cerebellum	(III)	Limbic lobe
(4)	Corpora quadrigemina	(IV)	Hindbrain

	A	B	C	D
(1)	II	I	III	IV
(2)	III	I	IV	II
(3)	III	IV	I	II
(4)	IV	III	II	I

BNC340

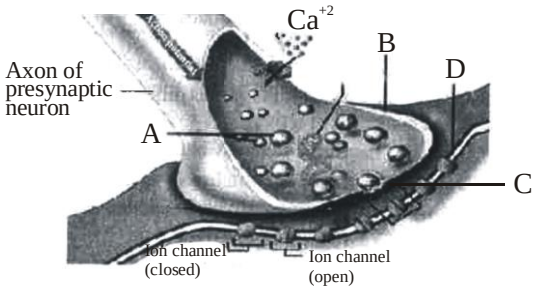
26. In the given diagram, which stage of conduction of nerve impulse through a nerve fibre is observed?



(1) Polarisation (2) Resting stage
(3) Repolarisation (4) Depolarisation

BNC341

27. The given diagram shows the axon terminal and synapse with parts labelled as A, B, C & D. Choose the correct combination of labelling from the given options.



(1) A - Synaptic vesicle, B - Axon terminal, C - Synaptic cleft, D - Postsynaptic membrane
(2) A - Axon terminal, B - Synaptic vesicle, C - Postsynaptic membrane, D - Synaptic cleft.
(3) A - Synaptic vesicle, B - Synaptic cleft, C -Axon terminal, D - Post synaptic membrane
(4) A - Post synaptic membrane, B - Axon terminal, C - Synaptic vesicle, D - Synaptic cleft

BNC342

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	1	3	3	4	2	4	2	4	4	4	4	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27			
Answer	2	4	4	2	3	2	3	4	1	2	3	1			

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Exercise - II
Neural Control and Coordination

1. The neurons in all animals are capable of
i. detecting various stimuli in environment
ii. receiving and transmitting stimuli to CNS
iii. processing various internal and external stimuli
Select the most appropriate option.
(1) I, II and III are correct
(2) Only II is correct
(3) II and III are correct
(4) I and II are correct

BNC343

2. **Assertion-**Neural organisation become complex in vertebrates as compared to invertebrates.
Reason-The ganglion in insects acts as a brain.
(1) Both assertion and reason are true and reason is correct explanation of assertion
(2) Both assertion and reason are true, but Reason is not the correct explanation of assertion
(3) Assertion is true, but reason is false
(4) Assertion is false, but reason is true

BNC344

3. Consider the following statements.
I. Two major division of CNS are somatic neural system and autonomic neural system.
II. Both somatic and visceral neural system are antagonistic in their functions.
Select the correct option.
(1) I is true, III is false
(2) Both I and II are true
(3) I is false, II is true
(4) Both I and II are false

BNC345

4. Visceral nervous system within human body
I. is the division of peripheral nervous system
II. is the division of central nervous system
III. consist of nerve fibres and ganglia
IV. carry impulses from one visceral organ to another.
Which of the following option is the most appropriate?
(1) I and III are correct
(2) II and IV are correct
(3) I, III and IV are correct
(4) II, III, IV are correct

BNC346

5. Match the following columns.

	Column-I		Column-II
(a)	Afferent fibres	(1)	Involuntarily controlled muscles
(b)	Somatic neural system	(2)	Carry impulse away from the CNS
(c)	Autonomic neural system	(3)	Voluntarily controlled muscles
(d)	Efferent fibres	(4)	Carry impulse towards the CNS

Which of the following is the correct option?
(1) a-2, b-1, c-3, d-4
(2) a-2, b-3, c-1, d-4
(3) a-4, b-3, c-1, d-2
(4) a-4, b-1, c-3, d-2

BNC347

6. **Assertion-**A person would not be able perceive hot or cold sensation if afferent fibres of PNS are degenerated in him.
Reason-All the afferent nerve fibres are motor nerves.
(1) Both assertion and reason are true and reason is correct explanation of assertion
(2) Both assertion and reason are true, but reason is not the correct explanation of assertion
(3) Assertion is true, but reason is false
(4) Assertion is false, but reason is true

BNC348

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7. Synaptic knob

I. is terminal bulb like structure of dendrites and axons.

II. contains neurotransmitter-filled vesicles

III. is a protoplasmic extension of cell body

Which of the following option is most appropriate?

(1) I and II are correct

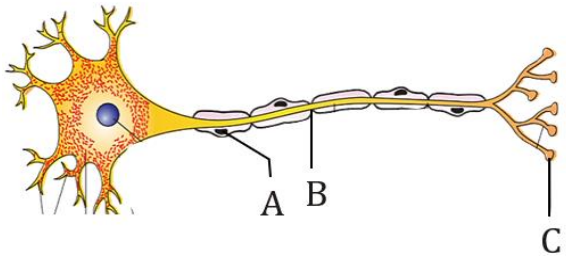
(2) Only III is correct

(3) Only II is correct

(4) I and III are correct

BNC349

8. Refer to the given diagram of the structure of a neuron and identify A, B and C.



Select the correct option.

(1) A-Nissl's granule, B-Axon, C-Schwann cell

(2) A-Schwann cell; B-Nodes of Ranvier, C-Synaptic knob

(3) A-Synaptic knob; B-Dendrite, C-Synaptic knob

(4) A-Nucleus, B-Myelin sheath, C-Nissl's granule

BNC350

9. Match the following columns.

	Column-I		Column-II
(a)	Neurotran smitters	(1)	Ribosomal granules
(b)	Nissl's granules	(2)	Short and branched
(c)	Dendrites	(3)	Contained in synaptic knob
(d)	Axon	(4)	Carry impulse away from cell body

Select the correct option.

(1) a-1, b-3, c-2, d-4

(2) a-3, b-1, c-4, d-2

(3) a-3, b-1, c-2, d-4

(4) a-1, b-3, c-4, d-2

BNC351

10. Match the following columns.

Column-I (Types of neurons)		Column-II (Location)	
(1)	Multipolar neuron	(I)	Embryonic stages
(2)	Bipolar neuron	(II)	Retina of eyes
(3)	Unipolar neuron	(III)	Cerebral cortex

Select the correct option.

(1) a-3, b-2, c-1 (2) a-2, b-1, c-3

(3) a-1, b-3, c-2 (4) a-2, b-3, c-1

BNC352

11. Consider the following statements.

I. The axons of neurons can receive signals from other neurons.

II. Multipolar neuron contains numerous axons.

Select the correct option.

(1) I is true, II is false

(2) Both I and II are true

(3) I is false, II is true

(4) Both I and II are false

BNC353

12. **Assertion-**The speed of nerve impulse along axon would slowdown in the absence of Schwann cells

Reason-Schwann cells are abundantly found in cells body of neurons.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of assertion

(3) Assertion is true, but Reason is false

(4) Assertion is false, but Reason is true

BNC354

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13. The resting axonal membrane is
I. permeable to K^+ ions
II. permeable to Na^+ ions
III. impermeable to negatively charged proteins of axoplasm
Select the correct option.
(1) I and II are correct
(2) I, II and III are correct
(3) II and III are correct
(4) I and III are correct

BNC355

14. Consider the following statements.
(A) At rest, the axoplasm inside the axon contain low concentration of K^+ ions.
(B) The concentration gradient across axonal membrane is generated due to the different concentration of Na+ and K+ ions across it.
Select the correct option.
(1) A is true, B is false
(2) Both A and B are false
(3) A is false, B is true
(4) Both A and B are true

BNC356

15. Na^+ - K^+ pump
I. Needs energy (ATP) to work
II. Expels 3 Na^+ for every 2K^+ ions imported
III. Works against a concentration gradient
IV. Maintains resting potential
(1) All are correct
(2) Only II and III are correct
(3) Only I and III are correct
(4) None is correct

BNC357

16. Match the following columns.

	Column-I		Column-II
A	Synaptic vesicles	I	Bind to neurotransmitters
B	Receptors of post synaptic membrane	II	Contain neurotransmitters
C	Electrical synapse	III	Rare in human system
D	Chemical synapse	IV	Slow conduction of nerve impulse

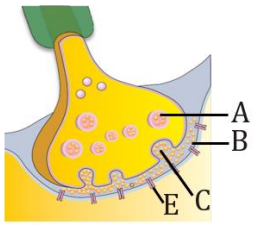
Select the correct option.
(1) A-II, B-I, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-I, B-II, C-III, D-IV
(4) A-I, B-II, C-IV, D-III

BNC358

17. Consider the following statements.
(A) The neurotransmitters help in the transmission of impulses at chemical synapses.
(B) Neurotransmitters have no role in electrical synapses. Select the correct statement.
(1) A is true, B is false
(2) A is false, B is true
(3) Both A and B are false
(4) Both A and B are true

BNC359

18. From the following diagram of axon terminal and synapse, identify at least two correctly labelled structures.



(1) B-Receptor, C-Neurotransmitter
(2) A-Synaptic Vesicles, E-Receptor
(3) C-Post synaptic membrane, D-K^+ ions
(4) D-Na^+ ions, A-Neurotransmitters

BNC360

19. To release the neurotransmitters, synaptic vesicles
(A) get burst open
(B) require stimulation through action potential
(C) Get digested by the lysosomes at axon terminals
Which of the following option is most appropriate?
(1) A and B are correct (2) B is correct
(3) B and C are correct (4) A is correct

BNC361

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20. **Assertion-**Neurons are excitable cells.
Reason-The membrane of neurons remain in polarised
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of assertion
(3) Assertion is true, but Reason is false
(4) Assertion is false, but Reason is true

BNC362

21. Consider the following statements.
I. Synaptic cleft is the point of fusion of pre-synaptic and post synaptic membrane at synapse.
II. Impulse transmission across chemical synapse is faster than that across an electrical synapse.
Select the correct option.
(1) I is true, II is false
(2) Both I and II are true
(3) I is false, II is true
(4) Both I and II are false

BNC363

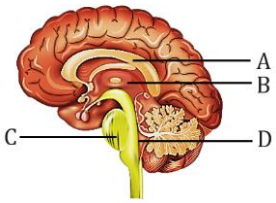
22. Match the following columns.

	Column-I		Column-II
A	Cerebrum	I	Grey matter of brain
B	Cerebral cortex	II	Major part of human brain
C	Corpus callosum	III	Tract of nerve fibres
D	Association areas	IV	Neither sensory nor motor in function

Which of the following is the correct option?
(1) A-IV, B-III, C-II, D-I
(2) A-II, B-I, C-III, D-IV
(3) A-I, B-IV, C-I, D-II
(4) A-III, B-II, C-IV, D-I

BNC364

23. Refer to the diagram showing sagittal section of human brain.



Identify the parts labelled as A -D.
(1) A-Cerebrum; B-Medulla; C-Cerebellum; D-Medulla
(2) A-Hypothalamus; B-Cerebellum; C-Medulla; D-Pons
(3) A-Corpus callosum; B-Thalamus; C-Pons; D-Cerebral aqueduct
(4) A-Thalamus; B-Corpus callosum; C-Medulla; D-Cerebellum

BNC365

24. Thalamus in human brain
(A) is surrounded by cerebrum
(B) acts as a major coordinating centre for sensory and motor signalling
(C) is under the direct control of hypothalamus.
Which of the following option is most appropriate?
(1) A and B are correct
(2) B and C are correct
(3) Only C is correct
(4) A, B and C are correct

BNC366

25. Match the following columns.

	Column-I		Column-II
A	Hypothalamus	I	Inner part of cerebral cortex
B	White matter	II	Neurosecretory cells
C	Amygdala	III	Between thalamus and pons
D	Midbrain	IV	Part of limbic system

Select the correct option.
(1) A-III, B-II, C-I, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-I, B-III, C-II, D-IV
(4) A-IV, B-II, C-III, D-I

BNC367

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26. Match the following columns.

	Column-I		Column-II
A	Brain stem	I	Emotions
B	Cerebellum	II	Brain ventricle containing CSF
C	Limbic	III	Breathing and consciousness
D	Cerebral aqueduct	IV	Balance and coordination

Select the correct option.

(1) A-IV, B-III, C-II, D-I

(2) A-II, B-I, C-IV, D-III

(3) A-III, B-IV, C-I, D-II

(4) A-I, B-II, C-III, D-IV

BNC368

27. **Assertion-**Association areas can carry out complex functions like communication and memory.

Reason-These areas are completely motor in nature.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of assertion

(3) Assertion is true, but Reason is false

(4) Assertion is false, but Reason is true

BNC369

28. **Statement I:** When an impulse arrives at the axon terminal, it stimulates the movement of the synaptic vesicles towards the membrane where they fuse with plasma membrane.

Statement II: The neurotransmitters bind to their specific receptors, present on the pre-synaptic membrane.

(1) Statement-I is correct & statement-II is not correct.

(2) Statement-I is incorrect & statement-II is correct.

(3) Both statement-I & II are correct.

(4) Both statement-I & II are incorrect.

BNC370

29. Which of the following is incorrectly matched.

	List-I	List-II
(1)	Synapse	Junctions between two neurons
(2)	Node of Ranvier	Gaps between two adjacent myelin sheaths
(3)	Unipolar neuron	With one axon and one dendrite
(4)	Synaptic knob	Bulb-like structure which possess synaptic vesicles.

BNC371

30. **Assertion (A):** The membrane of neurons are in a polarised state.

Reason (R): The outer surface of axonal membrane possesses a positive charge while its inner surface become negatively charged and therefore is polarised.

(1) Both A and R are correct and R is the correct explanation of A.

(2) Both A and R are correct but R is not the correct explain of A.

(3) A is correct but R is incorrect.

(4) Both A and R are incorrect.

BNC372

31. **Assertion (A):** Inner part of cerebral hemisphere is called white matter.

Reason (R): Inner part of cerebral hemisphere constitutes of fibres of the tract which are covered by unmyelinated sheath giving white appearance.

(1) Both A and R are correct and R is the correct explanation of A.

(2) Both A and R are correct but R is not the correct explain of A.

(3) A is correct but R is incorrect.

(4) Both A and R are incorrect.

BNC373

32. Choose the incorrect statement.

(1) Neural organisation in insects consists of number of ganglia and neural tissues.

(2) Afferent nerve fibres transmit impulse from tissue/organs to the CNS and efferent fibres transmit regulatory impulses from the CNS to peripheral tissue/organs.

(3) Somatic neural system relays impulses from the CNS to skeletal muscles.

(4) Visceral nervous system is the part of peripheral nervous system.

BNC374

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Exercise – II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	3	4	1	3	3	3	2	3	1	4	3	4	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	2	2	1	4	2	3	1	2	3	3	1	3	1
Question	31	32													
Answer	3	1													

Exercise – I
Chemical Coordination and Integration

1. Consider the given diagrammatic representation of the mechanism of action for 2 categories of hormones. In which of the following option, correct answer for blanks A to I are indicated ?

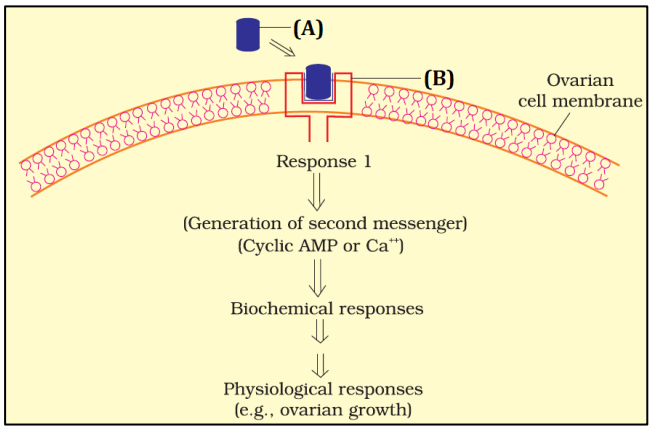


Figure-I

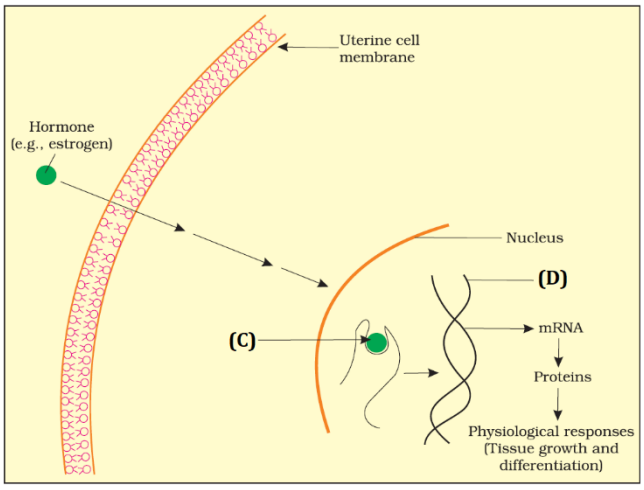


Figure-II

(1) A–Steroid hormone, B–Receptor, C–Protein, D–Genome
(2) A–Non-steroid hormone, B–Protein, C–mRNA, D–Genome
(3) A–Non-steroid hormone, B–Receptor, C–Hormone receptor complex, D–Genome
(4) A–Steroid hormone, B–Hormone receptor complex, C–Genome, D–mRNA

BCI429

2. Which one of the following flows directly into blood from the site of its production to act on an organ away from it ?
(1) Enzyme (2) Hormone (Renin)
(3) Blood (4) Lymph

BCI430

3. Listed below are the hormones of anterior pituitary. Tick the wrong entry
(1) Growth hormone
(2) Follicle stimulating hormone
(3) Oxytocin
(4) Adrenocorticotrophic hormone

BCI431

4. Naina is about to face an interview. But during the first five minutes before the interview she experiences sweating, increased rate of heart beat, respiration etc. Which hormone is responsible for her restlessness ?
(1) Estrogen and progesterone
(2) Oxytocin and vasopressin
(3) Adrenaline and noradrenaline
(4) Insulin and glucagon

BCI432

5. Hormones are called chemical signals that stimulate specific target tissues. Their specificity is due to the presence of signal receiving 'receptors' only in the respective target tissues. Where are these receptors present in case of hormones of protein nature?
(1) Extra cellular matrix
(2) Blood
(3) Plasma membrane
(4) Nucleus

BCI433

6. Choose the correct answer among the following options.

A. Epinephrine	i. Increase in muscle growth
B. Testosterone	ii. Decrease in blood pressure

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C. Glucagon — iii. Decrease in liver glycogen content
D. Atrial natriuretic factor — iv. Increase heart beat

Options:
(1) A-ii, B-i, C-iii, D-iv
(2) A-iv, B-i, C-iii, D-ii
(3) A-i, B-ii, C-iii, D-iv
(4) A-i, B-iv, C-ii, D-iii

BCI434

7. Underproduction of X hormone leads to Addison's disease. Name a hormone that stimulates the secretion of X hormone.
(1) TSH (2) FSH
(3) MSH (4) ACTH

BCI435

8. All the following tissues in mammals except one consists of a central 'medullary' region surrounded by a cortical region.
(1) Ovary (2) Adrenal
(3) Liver (4) Kidney

BCI436

9. The correct set of a single endocrine gland hormone is.
(1) Oxytocin, prolactin, ACTH
(2) Oxytocin, vasopressin, ANF
(3) Thyroxine, secretin, ACTH
(4) Epinephrine, cortisol, ICSH

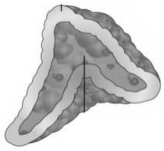
BCI437

10. Antidiuretic hormone's secretion increases when the hypothalamus is stimulated by:
(1) Angiotensin receptors
(2) Glucose receptors
(3) Osmoreceptors
(4) Renin receptors

BCI438

11. Which of the following hormone has no action on gene expression?
(1) Cortisol (2) Aldosterone
(3) Progesterone (4) Glucagon

BCI439

12. 

Which function is not related with above gland?
(1) Sugar metabolism
(2) Fight or flight response
(3) Electrolyte balance
(4) Ca^{2+} regulation

BCI440

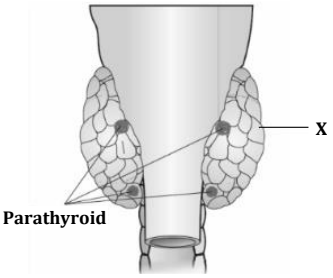
13. Select the correct matching of gland with hormone and function

	Endocrine gland	Hormone	Function
(1)	Parathyroid	Thyrocalcitonin	Ca^{2+} regulation
(2)	Pancreas	Insulin	Hypoglycemia
(3)	Pineal	Melanin	Pigmentation
(4)	Thymus	Thyroxine	Immunity

BCI441

14. Which of the following statement is wrong related to pituitary gland?
(1) Degeneration in neuron of posterior part cause diabetes insipidus.
(2) Oxytocin is released from anterior pituitary
(3) Anterior pituitary is related with body growth
(4) Hypothalamus regulates posterior pituitary directly

BCI442

15. 

The above gland X is not related with the following function :-
(1) Metabolism
(2) Metamorphosis
(3) Erythropoiesis
(4) Immuno suppression

BCI443

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16. Match the hormones given in column-I with the terms given in column-II

	Column-I		Column-II
A	ADH	I	Pituitary
B	ACTH	II	Mineralocorticoid
C	Aldosterone	III	Diabetes mellitus
D	Insulin	IV	Diabetes insipidus
E	Adrenaline	V	Vasodilator in skeletal muscle

(1) A–I, B–IV, C–II, D–III, E–V
(2) A–IV, B–II, C–I, D–III, E–V
(3) A–IV, B–I, C–II, D–III, E–V
(4) A–IV, B–I, C–III, D–II, E–V

BCI444

17. Which one of the following hormones interact with membrane-bound receptors ?
(1) Iodothyronine (2) Estradiol
(3) Glucagon (4) Cortisol

BCI445

18. ADH, responsible for reabsorption of water and reduction of urine secretion, is synthesized by:-
(1) Posterior pituitary gland
(2) Juxtaglomerular apparatus
(3) Hypothalamus
(4) Anterior pituitary gland

BCI446

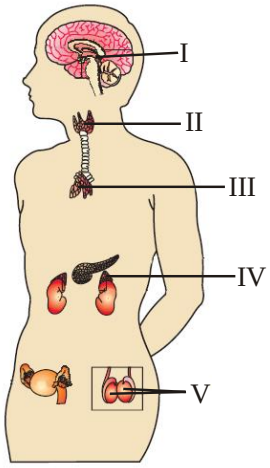
19. Pick the odd one out from each given words and then match it with the correct option.
I. Thyroid, Adrenal, Pituitary, Prostate
II. Cretinism, Goitre, Myxodema, Scurvy
III. Insulin, Blood sugar, Adrenaline, Thyroxine
IV. Estrogen, Progesterone, Testosterone, Prolactin
(1) I-Prostate, II-Scurvy, III-Blood sugar, IV-Prolactin
(2) I-Adrenal, II-Goitre, III-Insulin, IV-Prolactin
(3) I-Thyroid, II-Cretinism, III-Adrenaline, IV-Estrogen
(4) I-Pituitary, II-Myxodema, III-Thyroxine, IV-Progesterone

BCI447

20. The hormones that initiate ejection of milk, stimulates milk production and growth of ovarian follicles, are respectively known as :-
(1) PRL, Oxytocin and LH
(2) Oxytocin, PRL and FSH
(3) LH, PRL and FSH
(4) PRH, Oxytocin and LH

BCI448

21. The given diagram represents the location of human endocrine glands I, II, III, IV and V.



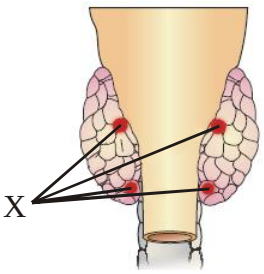
Which of the following is correctly matched with their secretions ?

Glands	Their secretions
I	Melatonin
II	Thymosin
III	Epinephrine
IV	Aldosterone
V	Testosterone

(1) I, II and III only (2) I, IV and V only
(3) II, IV and V only (4) II, III and V only

BCI449

22. The hormone released by label "X" in the given figure helps to restore Y. Identify X and Y.

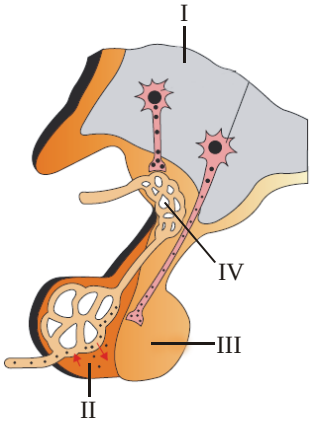


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X	Y
(1) Thyroxine	Too much calcium in the blood.
(2) PTH	Increase levels of calcium in blood.
(3) Thymosin	Decreased level of blood sugar.
(4) Adrenaline	Excessive loss of sodium in extracellular fluid.

BCI450

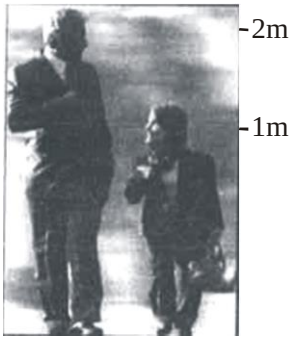
23. The Hypophyseal portal system transports releasing and inhibiting hormones from the hypothalamus into which of the following parts marked as I, II, III and IV ?



(1) I (2) II
(3) III (4) IV

BCI451

24. Which endocrine gland secretes the hormone that causes the abnormalities shown in the below picture ?



(1) Hypothalamus
(2) Pancreas
(3) Adenohypophysis
(4) Neurohypophysis

BCI452

25. **Assertion :** Enlargement of thyroid gland is known as goitre.
Reason: Goitre occurs due to the deficiency of iodine only.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BCI453

26. **Assertion** : Androgen affects male sexual behaviour (libido).
Reason : Estrogen regulates female sexual behaviour.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BCI454

27. **Assertion :** The posterior pituitary is under the direct neural regulation of the hypothalamus.
Reason : Hypothalamic neuron via axon send their secretion to posterior pituitary for release.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BCI455

28. **Assertion :** Glucagon is said to lower down the blood sugar level.
Reason : Glucagon increases the utilization of glucose in the tissue and the synthesis of liver glycogen
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BCI456

29. **Assertion** : PTH increases the Ca^{2+} level in blood.
Reason : PTH stimulates the process of bone resorption.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BCI457

30. **Assertion** : Oxytocin is also known as Antidiuretic Hormone (ADH).
Reason : Oxytocin can cause an increase in the renal reabsorption of water
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BCI458

Exercise – I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	3	3	3	2	4	3	1	3	4	4	2	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	3	1	2	2	2	2	3	3	2	1	4	1	4

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Exercise – II Chemical Coordination and Integration

1. Consider the following statements.
(A) Exocrine glands are the ductless glands.
(B) Sweat glands and gastric glands release their secretions directly into the blood
Select the correct option.
(1) Both (A) and (B) are true
(2) (A) is true but (B) is false
(3) Both (A) and (B) are false
(4) (A) is false but (B) is true

BCI459

2. Which of the following statements is false?
(1) Hormones provide chemical coordination, integration and regulation in the human body
(2) Hormones regulate metabolism growth and development of our organs
(3) Besides hypothalamus, pituitary, pineal, thyroid, adrenal, parathyroid, thymus, etc., GIT heart, kidney, etc also produce hormones.
(4) Hormone can be used again and again like biocatalyst

BCI460

3. The hormones produced by hypothalamic nuclei
(1) regulate the functions of the anterior pituitary
(2) regulate the functions of the posterior pituitary
(3) regulate the functions of both anterior and posterior pituitary
(4) inhibit the secretion of posterior pituitary hormones

BCI461

4. Which of the following hypothalamic hormones is incorrectly matched with its function?
(1) TRH: Stimulates secretion of thyrotropin
(2) PIH: suppresses the secretion of prolactin
(3) GH-RH: stimulates secretion of growth hormone
(4) CRH: stimulates the release of prolactin

BCI462

5. Portal blood vessels connect the ____ to the ___
(1) Hypothalamus, brain
(2) Hypothalamus, posterior pituitary
(3) hypothalamus, anterior pituitary
(4) Anterior pituitary, posterior pituitary

BCI463

6. In adults, adenohypophysis accounts for 75% of the total weight of the gland and consists of
(1) pars distalis
(2) pars tuberalis and pars intermedia
(3) pars nervosa
(4) pars intermedia

BCI464

7. Which of the following statement is incorrect about the posterior pituitary?
(1) The posterior pituitary lobe is not a true endocrine gland
(2) It serves as a hormone storage region.
(3) The posterior pituitary lobe and infundibulum together make neurohypophysis
(4) Oxytocin and vasopressin hormones are synthesized in the posterior pituitary lobe.

BCI465

8. Malfunctioning of endocrine gland deviates the body from homeostasis and causes several disorders. Which of the following disorder is correctly matched with its respective endocrine gland?
(1) Gigantism: Hyposecretion of growth hormone
(2) Dwarfism: Hypersecretion of growth hormone
(3) Acromegaly: Hypersecretion of growth hormone
(4) Gigantism: Hypersecretion of somatostatin

BCI466

9. A person is diagnosed with hypersecretion of growth hormone due to a pituitary tumor. Select the incorrect statement about his medical condition.
(1) Hypersecretion of growth hormone would cause hyperglycaemia.
(2) There will be a reduced secretion of insulin.
(3) Insulin secretion would remain unaffected.
(4) Hypersecretion of growth hormone results in a diabetes

BCI467

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10. Hypersecretion of prolactin hormone in females causes inappropriate lactation and absence of menstrual cycle. Based on the given information, select the pair of correct statements.

A. Prolactin is required for milk ejection from mammary glands.
B. The blood level prolactin is increased just before menstruation
C. During pregnancy, prolactin inhibiting hormone suppresses the release of prolactin.
D. Sucking action of newborn inhibits the release of PIH.

(1) A and B (2) B and C
(3) A and C (4) B and D

BCI468

11. Select the option that correctly matches the hormone with its source endocrine gland and respective function/target organs.

	Hormone	Endocrine gland	Function/target organ
(1)	TSH	Anterior pituitary	All body cells
(2)	ACTH	Posterior pituitary	Stimulates secretion of glucocorticoids from the adrenal cortex
(3)	LH	Anterior pituitary	Triggers secretion of androgens in males
(4)	FSH	Posterior pituitary	Stimulates the growth of ovaria follicles

BCI469

12. Diabetes insipidus is caused by
(1) hyposecretion of insulin by the pancreas
(2) hyposecretion of ADH by the posterior pituitary
(3) increased sensitivity of kidneys for ADH
(4) hypersecretion of ADH by the posterior pituitary

BCI470

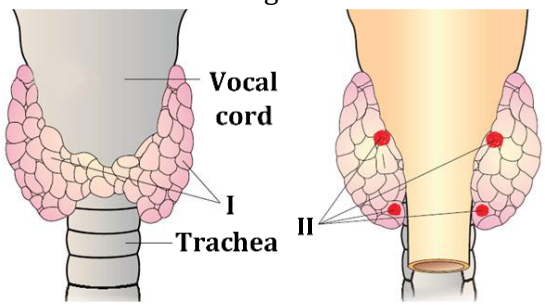
13. **Assertion-**TSH stimulates the thyroid gland to secrete thyroid hormones.
Reason-The hormones of anterior pituitary that regulate the secretions of other endocrine glands are called tropic hormones.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.

BCI471

14. Which of the following set of functions is not regulated by the hormone of the pineal gland?
(1) Diurnal rhythm and body temperature
(2) Metabolism and pigmentation
(3) Growth of bones and defense capability.
(4) Diurnal rhythm and defense capability

BCI472

15. Following is the diagrammatic view of the position of endocrine glands.



Select the option that correctly labels the glands and their respective hormones.
(1) I-Thyroid gland-Thyroxine and TSH
(2) II-Parathyroid gland-Thyroxine and calcitonin
(3) I-Thyroid gland-Thyroxine and calcitonin
(4) I-Parathyroid gland-PTH and calcitonin

BCI473

16. Most of the T_4 is converted into T_3 in the target tissues because-
(1) T_4 is more active
(2) T_3 is more active
(3) T_4 has a shorter half-life
(4) follicular cells cannot synthesize T_3

BCI474

17. Thyroid hormones are required for normal growth and development of humans because
(1) Thyroid hormones increase the basal metabolic rate in most body tissues
(2) thyroid hormones are regulated by negative feedback systems
(3) thyroid hormones contain iodine atoms
(4) thyroid hormones stimulate glycogen formation

BCI475

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18. Which of the following disorders of the endocrine system is incorrectly matched with its description?
(1) Hypothyroidism : Iodine deficiency
(2) Goitre : Enlarged thyroid gland
(3) Hyperthyroidism : Cretinism
(4) Exophthalmic goitre : Hyperthroidism
BCI476

19. Which of the given statements correctly differentiates myxedema from Grave's disease?
(1) Hypothyroidism in adult ages causes Graves' disease
(2) Graves' disease is more common among males than females
(3) Myxedema causes swelling in facial tissues due to the accumulation of interstitial fluid.
(4) Myxedema is an auto-immune disorder
BCI477

20. The physiological role of parathyroid gland does not include.
(1) increased activity of osteoclasts
(2) bone resorption
(3) reduced loss of Ca^{2+} and Mg^{2+} ions into urine
(4) reduced loss of HPO_4^{2-} into the urine
BCI478

21. Which of the following hormones can play a significant role in osteoporosis?
(1) Aldosterone and Prolactin
(2) Progesterone and Aldosterone
(3) Estrogen and Parathyroid hormone
(4) Parathyroid hormones and prolactin
BCI479

22. **Assertion-**Calcitonin and PTH exhibit an antagonistic effect on blood level of calcium ions.
Reason-Insulin and glucagon exhibit antagonistic effects on the blood levels of glucose.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.
BCI480

23. The endocrine gland that functions as a component of the lymphatic system is -
(1) thyroid gland (2) thymus gland
(3) parathyroid gland (4) pineal gland
BCI481

24. Which of the following endocrine gland is responsible for reduced immune responses in old ages?
(1) Thyroid gland (2) Pineal gland
(3) Thymus gland (4) Pituitary gland
BCI482

25. Which of the following categories of hormones is correctly matched with its examples?
(1) Catecholamines : adrenaline and noradrenaline
(2) Emergency hormones : adrenaline and insulin
(3) Glucocorticoids : aldosterone
(4) Mineralocorticoids : cortisol
BCI483

26. Consider the following statements
(a) Some chemicals act as both neurotransmitters and hormones
(b) Norepinephrine is released as a neurotransmitter by sympathetic division
(c) Norepinephrine is released as a hormone by the thyroid gland
(d) Norepinephrine is released as a hormone by the parathyroid gland.
Which of the two statements are correct?
(1) a and b (2) a and c
(3) b and c (4) c and d
BCI484

27. Secretion of hormones from adrenal medulla is controlled by
(1) parasympathetic nervous system
(2) pituitary gland
(3) sympathetic nervous system
(4) peripheral nervous system
BCI485

28. The fight or flight response generated by the release of hormones from adrenal medulla includes
(1) dilation of pupils and increased heart rate and blood pressure
(2) glycogenolysis and lipid synthesis
(3) increased muscular movement of the gastrointestinal tract
(4) reduced blood glucose levels
BCI486

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29. Which of the following sets of physiological functions correctly describes the role of cortisol in the human body?
(1) Anti-inflammatory response and suppression of the immune response
(2) Breakdown of RBCs in spleen
(3) Upregulation of uptake of amino acids
(4) Reabsorption of Na^+ from kidneys

BCI487

30. Which of the following pairs of endocrine glands are composite glands?
(1) Pancreas and thymus gland/Heterocrine gland
(2) Adrenal glands and thymus gland
(3) Pancreas and Ovary
(4) Adrenal gland and pineal gland

BCI488

31. Which of the following hormones is correctly matched with its description/effects?
(1) Hyperglycemic hormone: Glucagon
(2) Hypoplycemia : Glucagon
(3) Hypercalcemic : hormone: calcitonin
(4) Addison's disease: catecholamines

BCI489

32. The water- soluble fuels that are normally exported by the liver but overproduced during fasting or diabetes mellitus when not treated are known as
(1) fatty acids (2) glucose
(3) ketone bodies (4) amino acids

BCI490

33. Match the following hormones with the respective disease

	Column-I		Column-II
A	Insulin	I	Addison's disease
B	Thyroxin	II	Diabetes insipidus
C	Corticoids	III	Acromegaly
D	Growth hormone	IV	Goitre
		V	Diabetes mellitus

Select the correct option.
(1) A-V, B-I, C-II, D-III
(2) A-II, B-IV, C-III, D-I
(3) A-V, B-IV, C-I, D-III
(4) A-II, B-IV, C-I, D-III

BCI491

34. Testes serves as an endocrine gland. Which of the given statement about testicular hormones is correct?
(1) LH from the posterior pituitary stimulates interstitial cells to secrete testosterone.
(2) FSH stimulate secretion of androgens from Sertoli cells.
(3) FSH and androgens regulate spermatogenesis.
(4) LH and FSH are synthesised and secreted by testes in males.

BCI492

35. Androgens are responsible for libido in both human males and females. The source of androgens in human females is
(1) Ovaries
(2) Oviducts
(3) Corpus luteum
(4) Adrenal cortex

BCI493

36. Match the following structures in column I with the correct description in column II.

	Column-I		Column-II
A	Ovarian follicles	I	Progesterone
B	Corpus luteum	II	LH
C	Graafian follicle	III	Prolactin
D	Mammary glands	IV	Estrogen

Select the correct option.
(1) A-IV, B-III, C-II, D-I
(2) A-IV, B-I, C-II, D-III
(3) A-III,B-IV, C-II, D-I
(4) A-III, B-I, C-IV, D-II

BCI494

37. Which of the following statement is incorrect about the effects and regulation of female sex hormones?
(1) Ovarian and uterine cycles are under the regulation of gonadotropin releasing hormone
(2) In females, FSH triggers the development of follicles
(3) Ovarian follicles secrete estrogen under influence of FSH and LH
(4) Estrogen from ovarian follicles is converted into progesterone

BCI495

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38. **Assertion-**Corpus luteum is a temporary endocrine gland.
Reason-Corpus luteum secretes female sex hormones.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.

BCI496

39. Atrial natriuretic hormone I factor (ANF) secreted by atrial wall of our heart has exactly the opposite function of this hormone secreted by zona glomerulus-
(1) ADH (2) Aldosterone
(3) Androgen (4) Calcitonin

BCI497

40. Match the hormones in column I with their functions in column II.

	Column-I		Column-II
A	Progesterone	I	Inhibits uterine contraction
B	Atrial natriuretic factor	II	Formation of RBCs
C	Erythropoietin	III	Formation of alveoli in mammary glands
D	Relaxin	IV	Lowers blood pressure

Select the correct option.
(1) A-IV, B-III, C-II, D-I
(2) A-IV, B-I, C-II, D-III
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-I, C-IV, D-III

BCI498

41. Which of the following hormones of the gastrointestinal tract is wrongly matched with its function?

	Column-I		Column-II
A	Gastrin	I	Inhibition of gastric secretions
B	Gastric inhibitory peptide (GIP)	II	Stimulates bile juice secretion
C	Secretion	III	Secretion of gastric juice
D	Cholecystokinin	IV	Stimulates secretion of pancreatic juice

Select the correct option.
(1) A-III, B-I, C-IV, D-II
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-III, C-IV, D-I

BCI499

42. Identify the hormone with its correct matching of source and function.
(1) Oxytocin-posterior pituitary, growth, and maintenance of mammary glands
(2) Melatonin-pineal gland, regulates the normal rhythm of the sleep wake cycle
(3) Progesterone-corpus luteum, stimulation of growth and activities of female secondary sex organs.
(4) Atrial natriuretic factor-ventricular wall, increases the blood pressure.

BCI500

43. Steroid hormones initiate the production of target cell substances in which manner?
(1) They initiate second messenger activity
(2) They bind with membrane protein
(3) They initiate DNA transcription
(4) They activate enzyme pathways

BCI501

44. Which of the following hormones does not act by a second messenger system?
(1) Glucagon (2) Epinephrine
(3) FSH (4) Testosterone

BCI502

45. Which of the given sets of endocrine gland secrete/release only water soluble hormones?
(1) Pancreas and thyroid gland
(2) Parathyroid gland and pineal gland
(3) Adrenal gland and thyroid gland
(4) Parathyroid gland and pancreas

BCI503

46. **Assertion-**Aldosterone can diffuse freely through the lipid bilayer.
Reason-Steroid hormones are lipid soluble
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.

BCI504

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47. **Assertion:** cAMP serves as the second messenger for protein hormones.
Reason-Insulin is a protein hormone
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.
BCI505

48. **Assertion :** Catecholamine increases the concentration of glucose in blood.
Reason : Catecholamine stimulates glycogenolysis.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BCI506

49. **Assertion**: Failure of secretion of thyroid gland hormones from childhood causes cretinism disease.
Reason: Thyroid stimulating hormone maintains BMR and promotes the growth of body tissues.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BCI507

50. **Assertion**: Thyroid gland regulate blood calcium level.
Reason: Thyroid gland secrete protein hormone called thyrocalcitonin which affects blood calcium level.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BCI508

Exercise – II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	3	4	3	1	4	3	3	4	3	2	1	3	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	3	3	4	3	2	2	3	1	1	3	1	1	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	3	3	3	4	2	4	1	2	3	1	2	3	4	2
Question	46	47	48	49	50										
Answer	1	2	1	3	1										

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Structural Organisation in Animals (Animal Tissue)

1. Which of the following junctions help to stop substances from leaking across a tissue ?
(1) Adhering junction (2) Gap junction
(3) Tight Junction (4) Both (1) & (2)
BSA474

2. Presence of tissue in a multicellular organism ensures :-
(1) Faster development
(2) Division of labour
(3) Higher reproductive potential
(4) Body strength
BSA475

3. Trapped dust particles are pushed out of the respiratory tract by :-
(1) Mast cells (2) Plasma Cells
(3) Cilia (4) Fibroblasts
BSA476

4. Epithelial cells of the intestine are involved in food absorption because :-
(1) Pinocytic vesicles are found
(2) Microvilli are found
(3) Zymogen granules are found
(4) Phagocytic vesicles are found
BSA477

5. The epithelium present in the inner layer of stomach and intestine is :-
(1) Columnar (2) Squamous
(3) Cuboidal (4) Ciliated
BSA478

6. The cells of connective tissue which do not secrete the matrix are :-
(1) Mast cells (2) Fibroblasts
(3) Macrophages (4) All of the above
BSA479

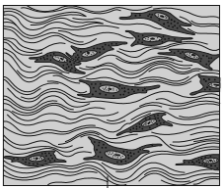
7. The most widely distributed connective tissue in the animal's body is:-
(1) Areolar (2) Adipose
(3) Cartilage
(4) Tendons & ligaments
BSA480

8. Which type of muscle helps in peristaltic movements in the gastro intestinal tract?
(1) Striated (2) Smooth
(3) Cardiac (4) Skeletal
BSA481

9. Which one of the following contains the largest quantity of extracellular material ?
(1) Striated muscle
(2) Areolar tissue
(3) Stratified
(4) Myelinated nerve fibres
BSA482

10. Which types of tissues our heart is made up of?
(A) Epithelial (B) Connective
(C) Muscular (D) Nervous.
(1) A only (2) A and B only
(3) A, B & C (4) A, B, C & D
BSA483

11. Identify the given diagram and choose the structure which represents this condition :-



(1) Cartilage (2) Ligament
(3) Skin (4) Bone
BSA484

12. Choose wrong statement with respect to connective tissue:-
(1) It is the most abundant tissue in the body of complex animals.
(2) In all connective tissues including blood the cell secretes fibers of structural protein called collagen or elastin.
(3) Cartilage, bone and blood are various types of specialised connective tissue.
(4) Loose connective tissue is present beneath the skin.
BSA485

13. Match column-I (Type of epithelium) with column-II (Descriptions) and choose the correct option.

	Column-I (Types of epithelium)		Column-II (Descriptions)
A	Squamous epithelium	I	It is composed of a single-layer of cube-like cells
B	Cuboidal epithelium	II	Have cilia on their free surface

C	Columnar epithelium	III	It is composed of a single layer of tall and slender cells
D	Ciliated epithelium	IV	It is made up of a single thin layer of flattened cells with irregular boundaries

(1) A – IV; B – I; C – III, D – II
(2) A – I; B – IV; C – III, D – II
(3) A – IV; B – I; C – II, D – III
(4) A – IV; B – III; C – I, D – II

BSA486

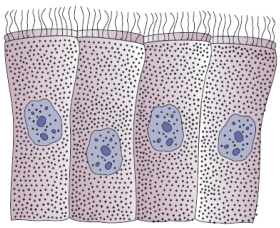
14. Match the types of connective tissue given in column-I with their examples given in column-II and choose the correct option.

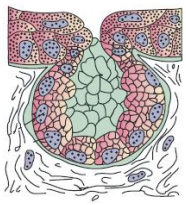
Column-I (Types of connective tissue)			**Column-II (Examples)**
A	Loose connective tissue	I	Tendons and ligaments
B	Dense regular tissue	II	Skin
C	Dense irregular tissue	III	Cartilage, bones, blood
D	Specialised connective tissue	IV	Fibroblasts, macrophages and mast cells

(1) A – I; B – IV; C – II, D – III
(2) A – I; B – IV; C – III, D – II
(3) A – IV; B – I; C – II, D – III
(4) A – IV; B – II; C – I, D – III

BSA487

15. A student was given a sample of two tissues. He observes the tissues under the microscope and draws their figures (1 and 2) as shown below.

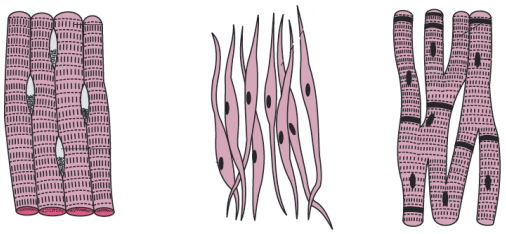

1


2

Identify the tissues (1 and 2)
(1) 1 = Columnar cells bearing cilia;
2 = Unicellular glandular epithelium
(2) 1 = Cuboidal cells bearing cilia;
2 = Multicellular glandular epithelium
(3) 1 = Compound cells bearing cilia;
2 = Unicellular glandular epithelium
(4) 1 = Columnar cells bearing cilia;
2 = Multicellular glandular epithelium

BSA488

16. The following figures A, B and C are types of muscle tissue. Identify A, B and C :-


A B C

(1) A-Smooth muscle, B-Cardiac muscle, C-Skeletal muscle
(2) A-Skeletal muscle, B-Smooth muscle, C-Cardiac muscle
(3) A-Cardiac muscle, B-Smooth muscle, C-Skeletal muscle
(4) A-Smooth muscle, B-Skeletal muscle, C-Cardiac muscle

BSA489

17. Consider the following statements:
(a) A tissue is composed of similar cells which perform specific functions.
(b) Epithelial tissues are characterized by a free surface toward body fluid or outside environment.
Select the correct option.
(1) a is true, b is false
(2) Both a and b are true
(3) a is false, b is true
(4) Both a and b are false

BSA490

18. Match the following columns.

Column-I			**Column-II**
A	Squamous epithelium	I	Stomach and intestine
B	Cuboidal epithelium	II	Lungs and blood vessels
C	Columnar epithelium	III	Tubular parts of nephrons

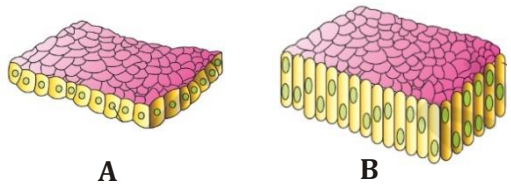
Select the correct option
(1) A-III, B-I, C-II (2) A-I, B-II, C-III
(3) A-II, B-III, C-I (4) A-III, B-II, C-I

BSA491

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19. Identify the tissues A and B shown in the following diagram.



A B

Select the correct option.

	A	B
(1)	Cuboidal epithelium	Columnar epithelium
(2)	Cuboidal epithelium	Squamous epithelium
(3)	Columnar epithelium	Cuboidal epithelium
(4)	Compound epithelium	Pseudostratified

BSA492

20. Match the following columns.

Column-I			Column-II
A	Globlet cells	I	Multicellular glandular epithelium
B	Salivary glands	II	Unicellular glandular epithelium
C	Buccal cavity	III	Compound epithelium
D	PCT	IV	Cuboidal epithelium

Select the most appropriate option.

(1) A-II, B-I, C-III, D-IV
(2) A-III, B-II, C-IV, D-I
(3) A-IV, B-III, C-I, D-II
(4) A-I, B-IV, C-II, D-III

BSA493

21. Select the incorrect statement.

(1) Multicellular glandular epithelium is formed of clusters of cells.
(2) Compound epithelium is actively involved in secretion and absorption of substances
(3) Pancreatic and salivary ducts are internally lined by compound of epithelium
(4) None of these

BSA494

22. Cell junctions :-

(a) Are formed in epithelial tissues
(b) Provide structural and functional link between adjacent cells of tissues.
(c) Are alternatively called gap junctions

Select the most appropriate option.

(1) a, b, c are correct
(2) Only b is correct
(3) b and C are correct
(4) a and b are correct

BSA495

23. Match the following cell structure with its characteristic feature.

Column-I			Column-II
A	Tight junctions	I	Cement neighbouring cells together to form sheet
B	Adhering junctions	II	Transmit information through chemical to another cells
C	Gap junctions	III	Establish a barrier to prevent leakage of fluid across epithelial cells
D	Synaptic junctions	IV	Cytoplasmic channels to facilitate communications between adjacent cells

Select the most appropriate option.

(1) A-IV, B-III, C-I, D-II
(2) A-II, B-IV, C-I, D-III
(3) A-IV, B-II, C-I, D-III
(4) A-III, B-I, C-IV, D-II

BSA496

24. **Assertion:** Squamous epithelium helps in the diffusion of gases in lungs.

Reason: Squamous epithelium bear microvilli.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true

BSA497

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25. **Assertion :** Compound epithelium is composed of two or more layers of cells.
Reason : Compound epithelium has protective functions.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true
BSA498

26. Select the incorrect statement regarding connective tissues.
(1) It helps to connect and support other tissues of the body
(2) Blood is a fluid connective tissue.
(3) It is composed of structural protein fibres, collagen or elastin.
(4) Their ground substance is composed of polypeptides and its derivatives
BSA499

27. Myofibrils are :-
(1) Contracted muscle fibres
(2) Structural components of all muscle fibres
(3) Striated muscle fibres
(4) Skeletal muscle fibres
BSA500

28. Consider the following statements.
(a) All involuntary muscles are smooth in appearance.
(b) All straited muscles are voluntary muscles.
Select the correct option.
(1) a is true b is false
(2) Both a and b are true
(3) a is false b is true
(4) Both a and b are false
BSA501

29. **Assertion :** Both skeletal muscles and cardiac muscles are striated appearance.
Reason : Cardiac muscles are involuntary in nature.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true
BSA502

30. **Assertion :** Neurons are excitable cells.
Reason : Neurons are found abundantly throughout the body.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true
BSA503

31. **Assertion:** Muscle fibres are said to be contractile in nature.
Reason: Cells of muscle tissues can shorten forcefully and again return to the relaxed state.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BSA504

32. **Assertion:** Smooth muscle fibres appear to be striated.
Reason: This is due to regular alternate arrangement of thick and thin filaments in Skeletal muscle fibres.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is false but the reason is true.
(4) If both the assertion and reason are false.

BSA505

33. **Assertion:** Epithelial tissues protect the under lying and over lying tissues.
Reason: Materials are exchanged at the surfaces across the epithelial tissues.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BSA506

34. **Assertion:** The cells of columnar epithelium in absorptive surfaces often bears microvilli on their free ends.
Reason: Microvilli on their free ends for increase surface area.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BSA507

35. **Assertion:** Blood is connective tissue
Reason: In blood, cells use to secrete fibres of structural proteins called collagen or elastin.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false

BSA508

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	3	2	1	3	1	2	2	4	2	2	1	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	3	1	1	2	2	4	3	1	4	2	2	2	2
Question	31	32	33	34	35										
Answer	1	3	2	1	3										

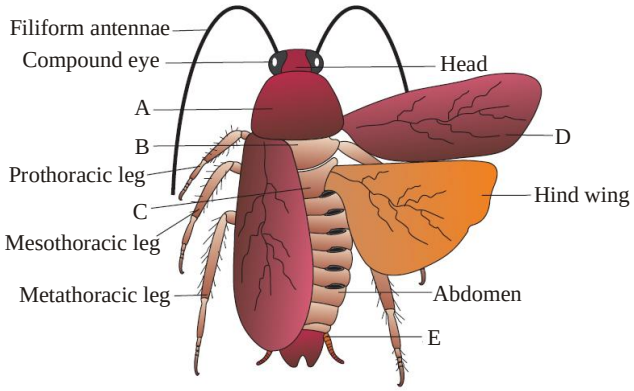
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Structural Organisation in Animals (Cockroach)

1. Which is not a character of *Periplaneta*?
(1) Segmented body (2) Jointed legs
(3) Compound eyes (4) 1- pair wings

BSA509

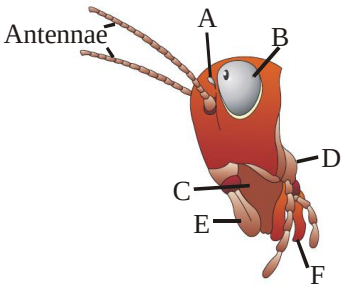
2. The following figure shows the external features of cockroach with few structures labelled as A, B, C, D & E.



Identify A to E :-
(1) A-Mesothorax, B-Pronotum, C-Metathorax, D-Tegmina, E-Anal style
(2) A-Pronotum, B-Metathorax, C-Mesothorax, D-Tegmina, E-Sterna
(3) A-Pronotum, B-Mesothorax, C-Metathorax, D-Tegmina, E-Anal cerci
(4) A-Pronotum, B-Mesothorax, C-Metathorax, D-Tegmina, E-Anal style

BSA510

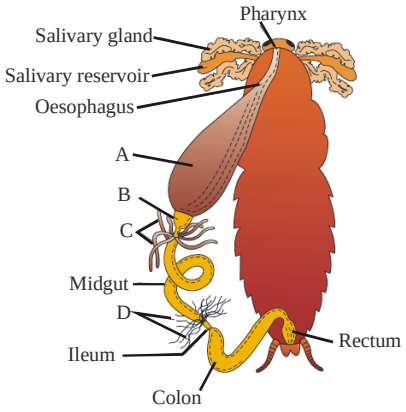
3. The figure given below shows the head region of cockroach. Identify A to F.



(1) A = Compound eye, B-Ocellus, C-Maxilla, D = Mandible, E-Labrum, F-Labium
(2) A = Ocellus, B-Compound eye, C = Mandible, D-Maxilla, E-Labrum, F = Labium
(3) A-Ocellus, B-Compound eye, C-Mandible, D-Maxilla, E-Labium, F-Labrum
(4) A-Ocellus, B-Compound eye, C-Maxilla, D-Mandible, E-Labrum, F-Labium

BSA511

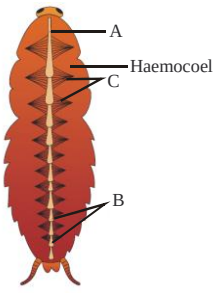
4. The given figure shows the digestive system of cockroach with few structures marked as A, B, C and D.



Identify structures A to D :-
(1) A-Gizzard, B-Crop, C-Hepatic caecae, D-Malpighian tubules
(2) A-Crop, B-Gizzard, C-Hepatic caecae, D-Malpighian tubules
(3) A-Crop, B-Gizzard, C-Malpighian tubules, D-Hepatic caecae
(4) A-Gizzard, B-Crop, C-Malpighian tubules, D-Hepatic caecae

BSA512

5. The given figure shows open circulatory system of cockroach with structures marked as A, B and C. Which structure is a 12 pair of wing shaped involuntary muscles and maintain blood circulation ?



(1) A (2) B
(3) C (4) Both A and B

BSA513

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6. Which of the following statements are correct about the fore wings in cockroach?
(1) They are mesothoracic
(2) They are opaque, dark and leathery
(3) They are not used in flight
(4) All

BSA514

7. In cockroach, hepatic caecae present at the junction of of alimentary canal–
(1) Hind gut and foregut
(2) Midgut and hind gut
(3) Foregut and mid gut
(4) Pharynx and Oesophagus

BSA515

8. In addition to malpighian tubules which type of structures also help in excretion in cockroach–
(1) Nephrocytes (2) Uricose gland
(3) Urate cells (4) All

BSA516

9. A pair of jointed filamentous structure called anal cerci present in cockroach–
(1) Between 9th–10th segment
(2) 9th segment
(3) 10th segment
(4) 10th segment of male only

BSA517

10. Which is/are correct statement/s?
(A) Gizzard or proventriculus helps in grinding of food
(B) Six chitinous teeth like structures present in gizzard
(C) 6–8 blind tubules called hepatic or gastric caeca is present at the junction of foregut and mid gut, which secrete digestive juice
(D) 100–150 malpighian tubules present between midgut and hindgut
(1) A and B
(2) A, B and C
(3) B, C and D
(4) A, B, C and D

BSA518

11. In which of the following statement is correct for arthrodial membrane :
(1) It is joined to wings with segment
(2) Join sclerites to each other
(3) Only present between mesothorax & metathorax
(4) Found around genitalia

BSA519

12. Which of the following type of development found in *Periplaneta americana*–
(1) Hemimetabolous
(2) Paurometabolous
(3) Ametabolous
(4) Holometabolous

BSA520

13. How many structures are not related to cockroach?
a –Haemocoel
b –Hexapoda
c – Nocturnal
e – 10 abdominal segments
f – Hypopharynx act as lower lip
(1) 1 (2) 2 (3) 3 (4) 4

BSA521

14. Testis of cockroach located in -
(1) 1-3 abdominal segments
(2) 2-5 abdominal segments
(3) 4-6 abdominal segments
(4) 2-4 abdominal segments

BSA522

15. How many statement/s is/are correct for cockroach?
(i) Size of cockroach may be 0.6–7.6 mm.
(ii) Only dorsally tergites are joined to each other by arthrodial membrane.
(iii) A ring of 6–8 pairs blind tubules called hepatic caecae is present at the junction of fore gut and mid gut.
(iv) Three pair ganglia lie in the throax and 6 pair in the abdomen.
(1) ii and iii are correct
(2) Only iii and iv are correct
(3) ii, iii, iv are correct
(4) All are incorrect

BSA523

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16. How many statements are correct for circulatory system of cockroach?
(i) Blood vascular system of cockroach is an open type.
(ii) Visceral organs located in the haemocoel are bathed in blood.
(iii) The haemolymph is composed of colourless plasma and haemocytes.
(iv) Blood pigment absent in haemolymph.
(1) 2 (2) 1 (3) 4 (4) 3

BSA524

17. Read following and pick wrong matches–
(a) Collaterial gland → Female Periplaneta only
(b) Anal cerci → Male Periplaneta only
(c) Phallic gland → Female Periplaneta only
(d) Anal style → Male Periplaneta only
(1) a and c (2) a, b, c and d
(3) b and c (4) d only

BSA525

18. Which is correct statement for cockroach :
(A) Cockroach is serious pests and vectors of several disease
(B) Cockroach is nocturnal and omnivorous, live in dump place
(C) Body of cockroach divided in to three part head thorax and abdomen
(D) Exoskeleton of cockroach in called sclerites
(1) A, B, C (2) B, C, D
(3) A, B (4) A, B, C, D

BSA526

19. Which is wrong statement for cockroach–
(1) In cockroach tergites and sternite both are attach by articular membrane
(2) Head made up of six segment during embryonic stage
(3) One pair antenna present
(4) Mouth part is bitting and sucking type

BSA527

20. Which is correct statement–
(A) The mouth part consisting of a labrum (upperlip), a pair of mandible, a pair of maxillae, and a labium (lower lip)
(B) Hypopharynx act as tongue
(C) Thorax divided in to three parts and each thorax segment bears a pair of walking legs
(1) A and B (2) B and C
(3) A and C (4) A, B and C

BSA528

21. Which is correct statement for cockroach –
(1) The first pair of fore wing arise from prothorax and called tegmina or elytra, it is opaque dark and leathery and not for flight
(2) The second pair of wing arise from mesothorax, use for flight, it is transparent and membranous
(3) The abdoman in both male and females consists of 12 segment
(4) In females, the 7th sternum is boat shaped and together with the 8th and 9th sterna form a brood or genital pouch

BSA529

22. Which is correct statement for cockroach–
(1) Males bear a pair of short thread like anal styles which are absent in females
(2) In both sexes, the 10th segment bears a pair of jointed filamentous structure called anal cerci
(3) The alimentary canal divided in to three regions foregut, midgut, hindgut
(4) All of these

BSA530

23. Which is correct statement :
(A) Gizzard or proventriculus help in grinding of food
(B) Six chitinous teeth like structure present in gizzard
(C) 6-8 blind tubules called hepatic or gastric caeca is present at the junction of foregut and mid gut, which secrete digestive juice
(D) 100-150 malpighian tubules present between midgut and hindgut
(1) A and B (2) A, B and C
(3) B, C and D (4) A, B, C and D

BSA531

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24. In cockroach the wings that help in flying are-
(1) Fore wings
(2) Mesothoracic wings
(3) the hind wings which are transparent and membranous
(4) Tegmina

BSA532

25. Tick mark the false statement regarding the heart of cockroach–
(1) Heart is present along the mid dorsal line of thorax and abdomen
(2) It is 13–chambered
(3) It is neurogenic
(4) It is myogenic

BSA533

26. The nerve cord of Cockroach is–
(1) Double, Ventral, Solid
(2) Double, Dorsal, Hollow
(3) Single, Dorsal, Solid
(4) Depends on species

BSA534

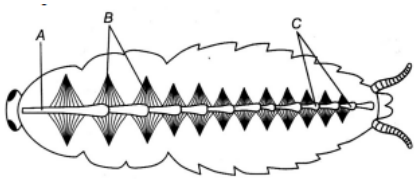
27. Read following statements and pick those who are not incorrect–
(a) Cockroach is devoid of endoskeleton
(b) Ootheca of Cockroach has unfertilized eggs and number is 16.
(c) Number of chambers in the heart of Cockroach is 13.
(1) a, b and c (2) b and c
(3) a and b (4) a and c

BSA535

28. Alary muscles are associated with–
(1) Circulation (2) Nervous control
(3) Excretion (4) Chewing

BSA536

29. Given below is the figure of open circulatory system of cockroach. Identify A, B and C and choose the correct options.



(1) A-Lateral aorta, B-Ciliary muscles, C-Chambers of heart
(2) A-Internal aorta, B-Alary muscles, C-Chambers of heart
(3) A-Anterior aorta, B-Alary muscles, C-Chambers of heart
(4) A-Posterior aorta, B-Fibrous muscles, C-Chambers of heart

BSA537

30. Main sound receptor organ of Cockroach is :–
(1) Anal cerci (2) Anal styles
(3) Wings (4) Legs

BSA538

31. **Assertion :** Filiform antennae are found in cockroach.
Reason : Antennae are thin filamentous, slender and club shaped at the end.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true

BSA539

32. **Assertion:** Head of cockroach shows great mobility in all direction
Reason: Cockroach possess flexible neck
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true

BSA540

33. **Assertion:** Cockroach is uricotelic
Reason: Fat bodies, nephrocyte and urecose glands help in excretion in cockroach.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true
(4) Both Assertion and Reason are false.

BSA541

34. **Assertion:** Cockroach possesses mosaic vision.
Reason: Eye of cockroach contains several ommatidia.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BSA542

35. **Assertion:** Development in *P. americana* is paurometabolous type.
Reason: Development in cockroach occurs through nymphal stage.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BSA543

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	2	2	3	4	3	4	3	4	2	2	1	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	4	4	4	4	4	4	3	4	1	4	1	3	1
Question	31	32	33	34	35										
Answer	3	1	2	1	1										

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Structural Organisation in Animals (Frog)

1. Which type of movement is not performed by frog?
(1) Burrowing
(2) Climbing & Running
(3) Leaping & Walking
(4) Leaping & Burrowing

BSA544

2. What is correct about the most common species of Frog in India?
(1) Colour of dorsal side is olive green
(2) 5 digits in forelimbs & 4 digits in hindlimb
(3) Male Frog lacks vocal cords
(4) Performs hibernation, but not aestivation

BSA545

3. The length of alimentary canal of an adult frog is :-
(1) Long, as it is carnivore
(2) Long, as it is herbivore
(3) Short, as it is carnivore
(4) Short, as it is herbivore

BSA546

4. Respiration in tadpole of frog takes place by :-
(1) Lungs (2) Gills
(3) Buccal cavity (4) Skin

BSA547

5. Vasa-efferentia in male frog opens into :
(1) Vas-deferens (2) Bidder's canal
(3) Cloaca (4) Urino-genital duct

BSA548

6. Frog has the ability to change the colour to hide them from their enemies is called–
(1) Camouflage (2) Hibernation
(3) Aestivation (4) Both (2) and (3)

BSA549

7. Which statement is incorrect for frog–
(1) Special venous connection between liver and intestine as well as the kidney and lower parts of the body are present.
(2) The excretory system consists of a pair of kidney, ureters, cloaca and urinary bladder
(3) In female frog the ureters and oviduct open combinedly in the cloaca
(4) Elimination of nitrogen wastes is carried out by a well developed excretory system.

BSA550

8. Select the incorrect statement w.r.t. Frog–
(1) Protective colouration in frog is called mimicry.
(2) Ventral side of skin is generally olive green.
(3) Final digestion takes place in the intestine.
(4) Only male possess sound producing vocal sacs

BSA551

9. The ureter act as urinogenital duct which opens into cloaca in–
(1) Only male frog
(2) Only female frog
(3) Both male and female frog
(4) Neither male nor female frog

BSA552

10. RBC's of Frog contain–
(1) Nucleus but not haemoglobin
(2) Haemoglobin but not nucleus
(3) Both nucleus and haemoglobin
(4) Plasma proteins

BSA553

11. Fore brain of frog does not include which of the following structures–
(1) Olfactory lobe
(2) Optic lobe
(3) Cerebral hemisphere
(4) Diencephalon

BSA554

12. What is incorrect about Frog–
(1) Ear is the organ of equilibrium and hearing
(2) Frog contains simple eyes
(3) Vasa efferentia arise from kidneys
(4) There is no functional connection of ovaries with kidneys

BSA555

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13. Bidder's canal is present in _____of frog and is meant for the passage of_____.
Fill the blanks with correct options respectively.
(1) Testes, Sperms (2) Kidney, Ova
(3) Testes, Urine (4) Kidney, Sperms
BSA556

14. Which of the following structures in frog receives blood through vena cava–
(1) Right atrium
(2) Left atrium
(3) Sinus venosus
(4) Conus arteriosus
BSA557

15. In frog, emulsification of fat is carried out by–
(1) Bile (2) Pancreatic Juice
(3) HCl (4) Both (1) and (2)
BSA558

16. In forg, mesorchium is a thin fold of membrane extending between–
(1) Two testes
(2) Ovary and kidneys
(3) Two kidneys
(4) Kidneys and testes
BSA559

17. Bidder's canal is found in–
(1) Testes of frog
(2) Ovary of mammal
(3) Kidney of frog
(4) Kidney of mammal
BSA560

18. Find the incorrect statement about the digestive system of Frog–
(1) The alimentary canal is long because frogs are carnivorous
(2) Food is captured by the bilobed tongue
(3) Final digestion takes place in intestine
(4) Digested food is absorbed by the numerous finger like folds in the inner wall of intestine called villi and microvilli.
BSA561

19. Where is bile stored in the body of frog?
(1) Pancreas (2) Liver
(3) Gall bladder (4) Bidder's canal
BSA562

20. How many chambers are present in a frog's heart?
(1) 2 (2) 4 (3) 3 (4) 5
BSA563

21. How many pairs of cranial nerves arise from the brain of frog?
(1) 12 (2) 8 (3) 10 (4) 6
BSA564

22. In the brain of frogs, optic lobes are present in the _____.
(1) midbrain
(2) hindbrain
(3) forebrain
(4) medulla oblongata
BSA565

23. Which of these statements is true with respect to frog?
(1) They are pests for agricultural crop
(2) Legs of frogs can be eaten
(3) They do not play a part in ecological balance
(4) They are herbivores
BSA566

24. What is the shape of testes in male frogs?
(1) Triangular (2) Irregular
(3) Spherical (4) Ovoid
BSA567

25. Which of these characteristics of frog is the reason for its short alimentary canal?
(1) Carnivorous
(2) Sexual dimorphism
(3) Amphibian
(4) Chordate
BSA568

26. **Assertion :** Bidder's canal are present in the kidney of frog for passage of sperm.
Reason : Vasa efferentia from the testes open into the anterior part of bidder's canal.
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.
BSA569

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27. **Assertion :** Frog exhibits sexual dimorphism.
Reason : Male frog can be distinguished by the presence of a copulatory pad on first digit of hind limbs.
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.
BSA570

28. **Assertion :** Frog has organs of touch called sensory papillae.
Reason : External ear is absent in frog.
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.
BSA571

29. **Assertion :** Blood of frog is red coloured.
Reason : Their blood possess haemoglobin as a respiratory pigment which is red in colour.
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.
BSA572

30. **Assertion :** Mimicry observed in frog as protactive coloration.
Reason : Frogs have ability to change the colour to hide from their enemies.
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.
BSA573

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	3	2	2	1	3	2	1	3	2	3	4	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	1	3	3	3	1	2	4	1	1	3	2	1	1

Exercise – I — Animal Kingdom

1. Pneumatic bones are present in :-
(1) *Corvus* (2) *Psittacula*
(3) *Hyla* (4) Both (1) and (2)
BAD552

2. Notochord is :-
(1) Ectodermal in origin
(2) Endodermal in origin
(3) Mesodermal in origin
(4) Some part is ectodermal and some part is of endodermal origin
BAD553

3. Collar cells or choanocytes are found in :-
(1) Lining of spongocoel
(2) Lining of canals
(3) Both (1) and (2)
(4) Outer layer of body wall
BAD554

4. Fertilisation and development in Sponges is :-
(1) External & direct
(2) External & Indirect
(3) Internal & direct
(4) Internal & indirect
BAD555

5. Features of Aschelminthes are :-
(1) Tissue level of organisation & acoelomate
(2) Organ Level of organisation & acoelomate
(3) Organ system level of organisation and coelomate
(4) Organ system level of organisation and pseudocoelomate
BAD556

6. Locomotory structure in Annelida are :-
(1) Parapodia
(2) Setae
(3) Longitudinal and Circular muscles
(4) All of these
BAD557

7. Larvae and adults of Echinoderms have respectively :-
(1) Radial and bilateral symmetry
(2) Bilateral and radial symmetry
(3) Both have radial symmetry
(4) Both have bilateral symmetry
BAD558

8. Excretory organ of Hemichordates are known as:-
(1) Nephridia (2) Green gland
(3) Proboscis gland (4) Organ of Bojanus
BAD559

9. Worm like, fossorial marine animals are found in:
(1) Arthropoda (2) Echinodermata
(3) Mollusca (4) Hemichordata
BAD560

10. Four pairs of gills with an operculum is present in :-
(1) *Trygon* (2) *Myxine*
(3) *Clarias* (4) *Pristis*
BAD561

11. Body of amphibians is generally divided into:-
(1) Head and tail
(2) Head, neck and tail
(3) Head and trunk
(4) Head, neck, trunk and tail
BAD562

12. Epidermal scales are found in :-
(1) *Hemidactylus* (2) *Ichthyophis*
(3) *Scoliodon* (4) *Betta*
BAD563

13. Match the column-I and column-II and select the correct answer :-

	Column-I		Column-II
(1)	*Chameleon*	i	Garden Lizard
(2)	*Calotes*	ii	Wall lizard
(3)	*Hemidactylus*	iii	Tree lizard
(4)	*Bangarus*	iv	Krait
		v	Viper

(1) 1-ii, 2-i, 3-ii, 4-v (2) 1-iii, 2-ii, 3-i, 4-iv
(3) 1-iii, 2-i, 3-ii, 4-iv (4) 1-i, 2-ii, 3-iii, 4-v
BAD564

14. Air sacs of birds helps in :-
(1) Gaseous exchange
(2) Internal perspiration
(3) Supplement respiration
(4) Both (2) and (3)
BAD565

15. Reptilian origin of birds are marked by :-
(1) Presence of pneumatic bones
(2) Presence of feathers
(3) Presence of oil gland at the base of tail
(4) Presence of scales on hind limbs
BAD566

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16. Features used for animal classifications are :-
(1) Arrangement of cells and body symmetry
(2) Nature of Coelom and pattern of alimentary canal
(3) Patterns of digestive, circulatory and reproductive systems.
(4) All of the above

BAD567

17. Select the correct option with the reference of pseudocoelomate animals:
(1) These animals are diploblastic
(2) They may be free living or parasitic in plants and animals
(3) They have blind-sac type of body plan
(4) These animals are gregarious in habit

BAD568

18. Which of the following is the most distinctive feature of Echinodermata?
(1) Presence of larva
(2) Organ system level of body organisation
(3) Presence of water vascular system
(4) All of these

BAD569

19. Match the column I with column II:

	Column-I		Column-II
(A)	Saw fish	i	*Magur*
(B)	Fresh water fish	ii	Limbless amphibian
(C)	Cuttle fish	iii	*Pristis*
(D)	*Ichthyophis*	iv	*Trygon*
		v	*Sepia*

(1) A-iv B-i C-v D-ii
(2) A-i B-iii C-iv D-v
(3) A-iii B-i C-v D-ii
(4) A-iii B-iv C-v D-i

BAD570

20. Identify the correct option for Sea horse:
(1) It is a fresh water fish
(2) Mouth is located ventrally and skin is covered with cycloid scales
(3) It has bony endoskeleton and air bladder is present
(4) Gill slits are separate and without operculum

BAD571

21. Identify the character of class- Aves:
(1) Skin is dry and generally without glands
(2) They are able to maintain a constant body temperature
(3) Fertilisation is internal
(4) All of these

BAD572

22. Read the following characters:
- They have two pairs of limbs
- Homoiothermous and heart is four-chambered
- Fertilisation is internal
- Different types of teeth are present in jaw

Above characters belong to-
(1) Aves (2) Reptilia
(3) Mammalia (4) Amphibia

BAD573

23. Study the given table and select the correct option:

	Animal	Character	Taxa
(1)	Lancelet	Exclusively fresh water animal	Phylum - Cephalochordata
(2)	Hagfish	Close type of circulatory system	Division - Gnathostomata
(3)	*Hyla*	Skin is moist, Oviparous	Class - Amphibia
(4)	*Struthio*	Skin is moist, fertilisation is internal	Class - Aves

BAD574

24. Identify the correct match:
(1) *Ophiura* – Brittle star – Arthropoda
(2) *Dentalium* – Chiton – Mollusca
(3) *Limulus* – King crab – Echinodermata
(4) *Ancylostoma* – Hookworm – Aschelminthes

BAD575

25. Study the given animal and select the correct option for this animal :-

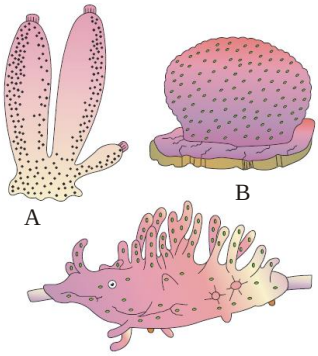


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(1) It respires by tracheal system
(2) Presence of three pairs of legs
(3) Bilateral symmetry and triploblastic
(4) All of these

BAD576

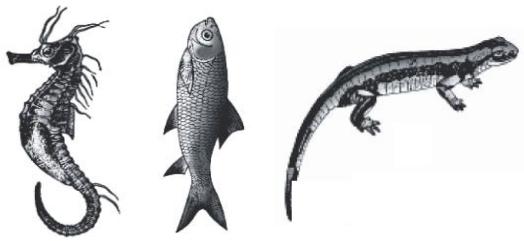
26. Examine the figures A, B and C. In which one of the four options all the items A, B and C are correctly identified?



	A	B	C
(1)	*Sycon*	*Euspongia*	*Spongilla*
(2)	*Euspongia*	*Spongilla*	*Sycon*
(3)	*Spongilla*	*Sycon*	*Euspongia*
(4)	*Euspongia*	*Sycon*	*Spongilla*

BAD577

27. Refer the figures A, B and C and choose the correct option which shows animals that regulate buoyancy with the help of air bladder:-



(1) A and B
(2) A and C
(3) B and C
(4) All of the above

BAD578

28. **Assertion:** A closed circulatory system is found in annelids.
Reason: Annelids posses true coelom.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BAD579

29. **Assertion:** Bats and whales are classified as mammals.
Reason: Bats and whales have four-chambered heart.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BAD580

Exercise – I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	3	4	4	4	2	3	4	3	3	1	3	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Answer	4	2	3	3	3	4	3	3	4	3	1	1	2	2	

Exercise – II Animal Kingdom

1. Choose the correct option :-

(1) Ctenophores and platyhelminthes possess complete digestive system

(2) Aschelminthes to chordates, all possess organ system level of organization along with complete digestive system.

(3) Coelenterates and aschelminthes possess organ system level of organization along with complete digestive system.

(4) Poriferan may possess complete digestive system.

BAD581

2. Which of the following is incorrect?

(1) Some division of labour (activities) occur among the cells in the members of phylum porifera.

(2) Division of labour (activities) is completely absent among the cell in poriferans

(3) Open circulatory system is found in Tunicates, hemichordates and mollusc.

(4) All of these

BAD582

3. Choose the incorrect statement.

(I) Notochord is ectodermally derived rod-like structure.

(II) Notochord is formed on the dorsal side during embryonic development.

(III) The animals from porifera to Echinoderms are without notochord.

(IV) In some chordates, notochord is replaced by the vertebral column and these chordates are called vertebrates.

(1) I and II (2) I, II and III

(3) II, III and IV (4) I only

BAD583

4. Identify the correct labels A and B.

Notochord	Nerve Cord
A	Part of nervous system
Found in chordates only	Found in chordates as well as non-chordates
Dorsal side in chordates	**B**

(1) A = Exoskeleton, B = Dorsal in chordates as well as in non-chordates

(2) A = Endoskeleton, B = Ventral in chordates as well as in non-chordates

(3) A = Exoskeleton, B = Ventral in chordates and dorsal in non-chordates

(4) A = Endoskeleton, B = Dorsal in chordates and ventral in non-chordates

BAD584

5. Match the following columns:-

	Column-I		Column-II
(A)	Porifera	(1)	Canal system
(B)	Aschelminthes	(2)	Water vascular system
(C)	Annelida	(3)	Muscular pharynx
(D)	Arthropoda	(4)	Joined appendages
(E)	Echinodermata	(5)	Metameres

Select the correct option :-

	A	B	C	D	E
(1)	1	3	5	4	2
(2)	1	2	3	4	5
(3)	5	4	3	2	1
(4)	4	3	1	2	5

BAD585

6. **Assertion :** The primary character of chordates is the presence of dorsal hollow nerve cord.

Reason : Vertebral column is derived from the notochord.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Assertion is false, but Reason is true.

BAD586

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7. Which of the following is not a characteristic of class Porifera?

(I) Development is indirect (larval stage is present).

(II) Mostly asymmetrical and usually marine.

(III) Primitive multicellular animals with cellular level of organization.

(IV) Choanocytes line the spongocoel and the canals.

(V) Sexes are separate

(1) I and IV (2) II only

(3) V only (4) III and IV

BAD587

8. Consider the following statement about cnidarians :-

(I) They have tissue level of organization and triploblastic.

(II) Digestion is extracellular and intracellular.

(III) Corals have a skeleton of calcium sulphate.

(IV) They possess a central gastrovascular cavity with a single opening mouth in hypostome.

(1) Statements I and III are correct

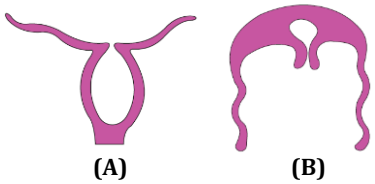
(2) Statements II correct only

(3) Statements I, II and III are correct

(4) Statements III and IV are correct

BAD588

9. Here two basic body forms of cnidarians are given.



(A) **(B)**

(1) A and B are free swimming forms

(2) A and B are sessile form

(3) A produce B asexually and B form the 'A' sexually

(4) B produce A sexually and A form the 'B' sexually

BAD589

10. Match the columns.

	Column-I		**Column-II**
(A)	*Gorgonia*	(1)	Sea fan
(B)	*Adamsia*	(2)	Sea pen
(C)	*Physalia*	(3)	Portuguese man of war
(D)	*Pennatula*	(4)	Sea anemone

Select the correct option :-

	A	**B**	**C**	**D**
(1)	1	4	3	2
(2)	1	2	3	4
(3)	4	3	2	1
(4)	3	4	1	2

BAD590

11. **Assertion :** Choanocytes or collar cells line the spongocoel and the canals in poriferans.

Reason : Poriferans possess spicules or spongin fibers.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Assertion is false, but Reason is true.

BAD591

12. Which of the option is correct for the statements given below.

(I) Commonly called sea walnuts or comb jellies.

(II) Bioluminescence is well marked.

(III) Body bear eight external rows of ciliated comb plates.

(IV) They have flame cells for osmoregulation and excretion.

(V) Alimentary canal is complete with a well-developed muscular pharynx.

	Ctenophores	**Platyhelminthes**	**Aschelminthes**
(1)	I, II, III	IV	V
(2)	IV	I, II	III, V
(3)	I, II	III, IV	V
(4)	IV, V	II, III	I

BAD592

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13. **Assertion :** The organisms of Platyhelminthes are usually hermaphrodite.
Reason : These organisms possess internal as well as external fertilization.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Both assertion and Reason are false.

BAD593

14. Match the columns.

	Column-I		Column-II
(A)	Gills	(1)	King crab
(B)	Tracheal system	(2)	Crab, prawn
(C)	Book gills	(3)	Butterfly, cockroach
(D)	Book lungs	(4)	Scorpion, spider

Select the correct option :-

	A	B	C	D
(1)	1	2	3	4
(2)	2	3	1	4
(3)	4	3	2	1
(4)	3	1	4	2

BAD594

15. Choose the incorrect statement.
(1) In cockroaches and prawns, excretion of waste material occurs through malphigian tubules.
(2) In ctenophores, locomotion is mediated by comb plates.
(3) In fasciola, flame cells take part in excretion.
(4) Earthworms are hermaphrodites and yet cross fertilization takes place among them.

BAD595

16. Consider the following statements about Arthropods.
(I) Open circulatory system is found in most arthropods.
(II) Arthropods contain Haemolymph in which internal tissues and organs directly bathes in it.
(1) I is true but II is false
(2) I is false but II is true
(3) Both I and II are true
(4) Both I and II are false

BAD596

17. Choose the incorrect statement for phylum Mollusca.
(1) Body is covered by a calcareous shell and unsegmented.
(2) Feather like gills present for excretion and respiration.
(3) The anterior head region has sensory tentacles.
(4) Mostly terrestrial, triploblastic and acoelomated.

BAD597

18. **Assertion :** In molluscs, feathers like gills are present in the mantle cavity.
Reason : These gills have respiratory and excretory function.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Both assertion and Reason are false.

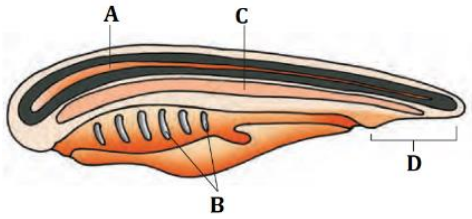
BAD598

19. Choose the correct statement for star fish.
(I) Sexes are separate and reproduction is sexual.
(II) Development is indirect with free swimming larva.
(III) Mouth is present on the upper (dorsal) side and anus on the lower (ventral) side.
(IV) Adults are radially symmetrical.
(1) I and III
(2) I, II and IV
(3) I, II and III
(4) III and IV

BAD599

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20. Animals belonging to phylum-chordata are fundamentally characterized by the presence of structure noted as A, B, C and D. Identify A, B, C and D.



(1) A – Notochord, B – Nerve cord, C – Gill slits, D – Post anal part
(2) A – Nerve cord, B – Notochord, C – Gill slits, D – Post anal part
(3) A – Nerve cord, B – Notochord, C – Post anal part, D – Gill slits
(4) A–Nerve cord, B – Gill slits, C – Notochord, D – Post anal part

BAD600

21. Select the correct difference on the basis the notochord in the following :

	Urochordata	**Cephalochordata**
(1)	Present only in larval tail	Extend from head to tail throughout life
(2)	Present only in adult	Present only in larval tail
(3)	Persistent throughout their life	Present only in adult
(4)	Extend from head to tail throughout life	Present only in larval tail

BAD601

22. Which of the following represents the correct combination without any exception?

	Characteristic	**Class**
(1)	Mammary gland; hair on body; pinnae; two pairs of limbs	Mammalia
(2)	Mouth ventral; gills without operculum, skin with placoid scales; persistent notochord	Chondrichthyes
(3)	Sucking and circular mouth, jaws absent, integument without scales; paired appendages	Cyclostomata
(4)	Body covered with feathers; skin moist and glandular; lungs with air sacs, forelimbs form wing	Aves

BAD602

23. Which of the following is not a characteristic of class Chondrichthyes?
(1) Gill slits are separated and without operculum.
(2) Predaceous with powerful jaws.
(3) Notochord is persistent throughout life.
(4) Air bladder present.

BAD603

24. Choose the incorrect option for the following animal.



(1) Cloaca present
(2) Dioecious, external fertilization, oviparous, indirect development
(3) Body divisible into head and trunk
(4) Eyes are without eyelids.

BAD604

25. Match the following columns.

	Column-I		**Column-II**
(A)	Cyclostomes	(1)	Hemichordata
(B)	Aves	(2)	Urochordata
(C)	Tunicates	(3)	Agnatha
(D)	*Balanoglossus*	(4)	Pisces
(E)	Osteichthyes	(5)	Tetrapod

Select the correct option :-

	A	**B**	**C**	**D**	**E**
(1)	1	2	3	4	5
(2)	2	3	4	1	5
(3)	3	5	2	1	4
(4)	3	1	5	2	4

BAD605

26. Which of the following is incorrect for Petromyzon?
(1) Cranium and vertebral column are cartilaginous
(2) They are freshwater organisms but migrate for spawning to sea water
(3) After spawning within few days, they die
(4) Their larvae, after metamorphosis, return to ocean

BAD606

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27. Identify the correctly matched option.

	Column-I	Column-II	Column-III
(1)	*Petromyzon*	Ectoparasite	Cyclostomata
(2)	*Ichthyophis*	Terrestrial	Reptilia
(3)	*Limulus*	Body covered by chitinous exoskeleton	Pisces
(4)	*Adamsia*	Radially symmetrical	Porifera

BAD607

28. Match the following columns.

	Column-I		Column-II
(A)	*Chameleon*	(1)	Tortoise
(B)	*Testudo*	(2)	Tree lizard
(C)	*Calotes*	(3)	Garden lizard
(D)	*Chelone*	(4)	Turtle

Select the correct option :-

	A	B	C	D
(1)	1	2	3	4
(2)	4	3	2	1
(3)	2	1	3	4
(4)	3	1	4	2

BAD608

29. Which of the following is incorrect about aves?

(1) Endoskeleton is ossified.

(2) Aves are partially homeotherms.

(3) Aves bones are hollow with air cavities.

(4) Aves have sexes separate, fertilization is internal, oviparous with direct development.

BAD609

30. **Assertion :** All vertebrates are chordates.

Reason : Vertebrates possess notochord during embryonic period.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both assertion and Reason are false.

BAD610

31. **Assertion :** Duck bill platypus is not a true mammal.

Reason : All true mammals are viviparous while platypus are egg laying.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both assertion and Reason are false.

BAD611

32. **Assertion :** Reptiles are referred to as poikilotherms.

Reason : Reptiles possess eggs with shells which help them to adapt in land environment.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BAD612

33. **Assertion:** Coelenterates are triploblastic.

Reason: Coelenterates contain mesoderm in between ectoderm and endoderm.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) If the assertion is true but the reason is false.

(4) If both the assertion and reason are false.

BAD613

34. **Assertion:** Annelids shows metameric segmentation.
Reason: Body of annelids divided externally as well as internally.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BAD614

35. **Assertion:** Bioluminescence is well marked in ctenophores.
Reason: All ctenophores are exclusively marine.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BAD615

Exercise – II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	4	4	1	2	3	2	3	1	2	1	3	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	2	2	4	1	2	4	4	3	2	1	3	2	1
Question	31	32	33	34	35										
Answer	1	2	4	1	2										

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Exercise – I
Human Reproduction

1. Which one of the following statement is not correct about testis ?
(1) About 250 testicular lobules in each testes
(2) Spermatogonia are the immature male germ cells of testes.
(3) Width of testes is about 4 cm to 5 cm.
(4) Testes are extra abdominal in human males.

BHR452

2. Find out the option which represents incorrect match :-
(1) Sertoli cells - Nutrition to the germ cells
(2) Male germ cells - by meiotic division form sperms
(3) Interstitial cells - Secrete androgens
(4) Immunologically competent cells present in seminiferous tubules

BHR453

3. How many structures are intratesticular:
Rete testis, vasa efferentia, epididymis, vas deferens, ejaculatory duct. :-
(1) 2 (2) 3 (3) 1 (4) 5

BHR454

4. Which one of the following gland release lubricant at the time of coitus ?
(1) Seminal visicles
(2) Bulbourethral glands
(3) Prostate gland
(4) Both (1) and (3)

BHR455

5. In human females, ovaries are primary sex organs because :-
(1) Produce female gametes
(2) Secrete Female sex hormones
(3) Both (1) and (2)
(4) Have role in lactation

BHR456

6. Find out incorrect match :-
(1) Ampulla - widest part of fallopian tube
(2) Cervix and vagina - Birth canal
(3) Uterine endometrium - non glandular
(4) Uterus - Inverted pear shape

BHR457

7. Which one of the following structure is not a part of female external genitalia :-
(1) Labia majora (2) Labia minora
(3) Hymen (4) Cervical canal

BHR458

8. Which hormone regulates spermatogenesis and also binds to receptors of leydig cells :-
(1) Luteinizing hormone
(2) Testosterone
(3) Follicle stimulating hormone
(4) Progesterone

BHR459

9. Match the hormone in column I with their function in column II :-

	Column-I		Column-II
(a)	FSH	(p)	Increases secretory nature of endometrium
(b)	LH	(q)	Develops female secondary sexual characters
(c)	Progesterone	(r)	Contraction of uterine wall
(d)	Estrogen	(s)	Development of corpus luteum
		(t)	Follicular development

(1) a-t, b-s, c-p, d-q
(2) a-s, b-t, c-q, d-p
(3) a-s, b-r, c-q, d-t
(4) a-t, b-p, c-q, d-s

BHR460

10. Which one of the following is incorrect matching pair :-
(1) Development of corpus luteum- In secretory phase
(2) Menstrual flow - Break down of myometrium
(3) Ovulation - LH and FSH attain peak level
(4) Proliferative phase - maturation of graffian follicle

BHR461

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11. Which one of the following acrosomal enzyme dissolve the zona pellucida of ovum.
(1) Hyaluronidase
(2) Acrosin
(3) Corona penetrating enzyme
(4) Erepsin

BHR462

12. Placenta acts as an endocrine structure and produce several harmones like :-
(a) Human chorionics gonadotropin (hCG)
(b) Human placental lactogen (hPL)
(c) Estrogen
(d) Progesterone
(e) LH
Select the correct option
(1) a and b only (2) b only
(3) a, b, c and e (4) a, b, c and d

BHR463

13. Which one of the following hormone mainly helps in parturition . :-
(1) Progesterone
(2) Thyroid hormone
(3) Oxytocin
(4) Luteinizing hormone

BHR464

14. Blood flowing in the umbilical cord of mammalian embryo is :-
(1) 100% foetal
(2) 50% maternal and 50% foetal
(3) 100% maternal
(4) 75% foetal and 25% maternal

BHR465

15. Secretions from which one of the following are rich in fructose, calcium and some enzymes :-
(1) Liver
(2) Pancreas
(3) Salivary glands
(4) Male accessory glands

BHR466

16. Which hormones is not involved in progress of induction of parturition?
(1) Cortisol (2) Estrogen
(3) Oxytocin (4) Prolactin

BHR467

17. Which event does not occur if fertilization not occur in menstrual cycle.
(1) Progesterone is decreased after 22nd day.
(2) Corpus luteum degenerates.
(3) Degeneration and shedding of endometrium.
(4) Degeneration of ovary

BHR468

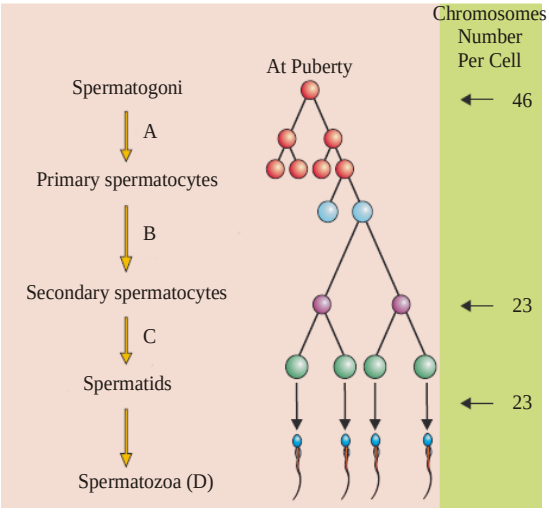
18. The male sex accessory ducts include
(1) Rete testis, vasa efferentia, epididymis, bulbourethral, prostate
(2) vasa efferentia, epididymis, seminal vesicle
(3) vas deference, rete testis, vasa efferentia, epididymis
(4) epididymis, seminal vesicle, vas deference, vasa efferentia

BHR469

19. Find the **incorrect** statement?
(1) hCG, hPL and relaxin are produced in women only during pregnancy
(2) Blastocyst is embedded in the endometrium of the uterus and it leads to pregnancy
(3) The first sign of growing fetus may be noticed by listening to the heart sound carefully through the stethoscope.
(4) The first movement of fetus & appearance of hair in the head are observed during 6th month.

BHR470

20. The figure given below shows a flowchart on spermatogenesis. Identify the correct label marked as A, B, C and D.



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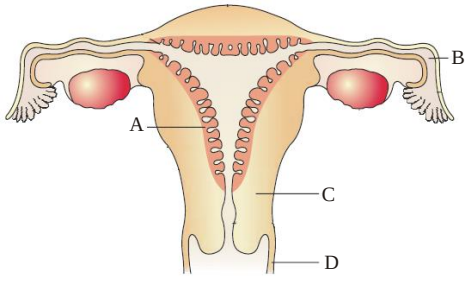
(1) A : first meiotic division;
B : second meiotic division;
C : Differentiation; D : 23.
(2) A : second meiotic division;
B : Differentiation;
C : first meiotic division; D : 46.
(3) A : Differentiation;
B : second meiotic division;
C : first meiotic division; D : 46.
(4) A : Mitosis differentiation;
B : first meiotic division;
C : second meiotic division; D : 23.

BHR471

21. Study the given statement and answer the question. If pregnancy doesn't happen, "During 'P' phase of the menstrual cycle, the 'Q' starts to degenerate usually around day 22 in a 28-day cycle. The drop in 'R' levels causes the lining of the uterus to fall away. This is known as 's'.
Identify P, Q, R and S.
(1) P → Menstrual, Q → Graafian follicle,
R → Estrogen, S → Menarche
(2) P → Luteal, Q → Corpus luteum,
R → Progesterone, S → Menstruation
(3) P → Ovulatory, Q → Endometrium,
R → Follicle stimulating hormone,
S → Menopause
(4) P → Follicular, Q → Secondary oocyte,
R → Luteinizing hormone,
S → Menstruation

BHR472

22. The given diagram shows a human female's reproductive system. Select the part (labelled as A, B, C & D) where semen is released into the female by the penis during copulation.



(1) A (2) B (3) C (4) D

BHR473

23. The placenta is formed from the _____ of the embryo and the _____ of the mother :-
(1) uterus, trophoblast
(2) chorion, endometrium
(3) endometrium, chorion
(4) inner cell mass, endometrium

BHR474

24. _____ is a sticky white or yellow fluid secreted by the breast during the second half of pregnancy and for a few days after birth, before breast milk comes in :-
(1) Placenta (2) Colostrum
(3) Egg yolk (4) Blood cells

BHR475

25. Which of the following is not a paired structure in male ?
(1) Urethra
(2) Vas deferens
(3) Epididymis
(4) Ejaculatory duct

BHR476

26. Study the statement given below and answer the question. "Vigorous contraction of the 'X' at the end of the 'Y' causes expulsion of the foetus." Identify X and Y :-
(1) X - Vagina ; Y - Fertilisation
(2) X - Uterus ; Y - Pregnancy
(3) X - Placenta ; Y - Implantation
(4) X - Embryo ; Y - Ovulation

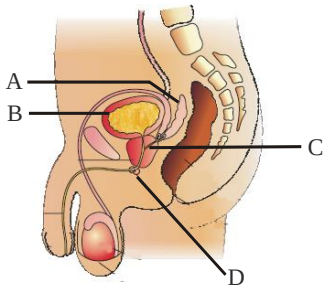
BHR477

27. Fertilization can only occurs if :-
(1) Sperm reaches to the ampullary – isthmic junction before the ovum.
(2) Ovum reaches to the ampullary – isthmic junction before sperm.
(3) Sperms are transported to the uterus and ovum to the fallopian tube simultaneously.
(4) Sperms and ovum are transported simultaneously to the ampulla

BHR478

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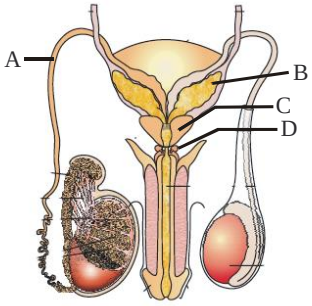
28. The given figure shows the male reproductive system. Some structures are marked as A, B, C and D. Identify the structure whose removal will cause lack of lubrication during coitus.



(1) A (2) B (3) C (4) D

BHR479

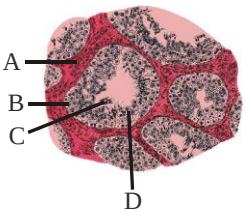
29. Given below is a diagrammatic sketch of a portion of human male reproductive system. Select the correct set of the names of the parts marked as A, B, C and D.



	A	B	C	D
(1)	Ureter	Seminal vesicle	Prostate	Bulbourethral
(2)	Ureter		Seminal vesicle	Bulbourethral gland
(3)	Vas deferens	Seminal vesicle	Prostate	Bulbourethral gland
(4)	Vas deferens	Seminal vesicle	Bulbourethral gland	Prostate

BHR480

30. Given below is the diagrammatic sectional view of seminfiferous tubule with their parts marked as A, B, C and D. Select the option which shows the correct identification of the structure with its characteristics.



(1) A : Spermatozoa, secretes testicular hormones that control spermatogenesis

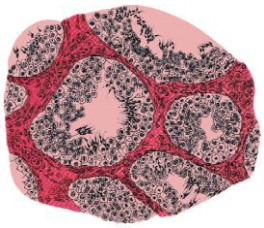
(2) B : Spermatogonium, it is also called germ call which undergo meiotic divisions finally leading to sperm formation.

(3) C : Interstitial cells, present in the interstitial spaces and store and transport the sperms from the testis to the outside through the urethra.

(4) D : Sertoli cells, it maintains low temperature of the testis.

BHR481

31. Identify the figure(A) show sectional view below and match with its characteristics (B) and its location (C).



	A	B	C
(1)	Graafian follicle	Involved in the formation of ovum	Ovary
(2)	Seminiferous tubule	Involved in the formation of sperm	Tesis
(3)	Ovum surrounded by sperm	Process of fertilization	Graafian follicle
(4)	Mammary gland	Involved in milk secretion	Female reproductive system

BHR482

32. Which of the following hormone/hormones is/are released during pregnancy ?

(1) hCG

(2) hPL

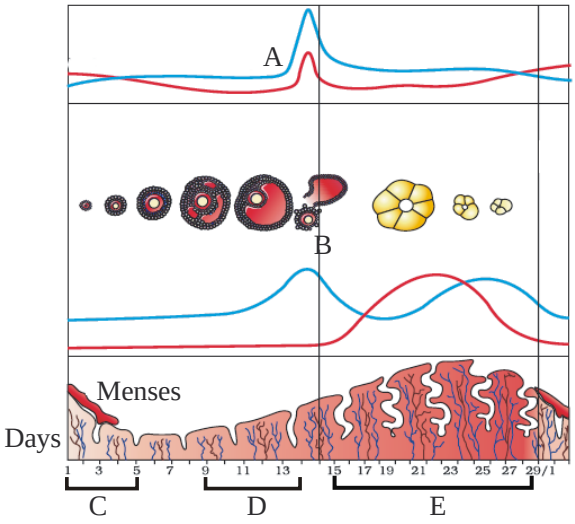
(3) Relaxin

(4) All of the above

BHR483

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33. The figure given below shows the various events occurring during a menstrual cycle with few structures marked as A, B, C and D. Which of the following options shows the correct labelling ?



(1) A→LH, B→Ovulation, C→Menstruation, D→Proliferative phase, E→Luteal phase

(2) A→FSH, B→Implantation, C→Follicular phase, D→Menstruation phase, E→Luteal phase

(3) A→Estrogen, B→Parturition, C→Luteal phase, D→Follicular phase, E-Follicular phase

(4) A→Progesterone, B→Fertilization, C→Menstruation phase, D→Secretory phase, E→Follicular phase

BHR484

34. **Assertion:** All copulations not lead to fertilization.
Reason: Fertilization can only if the ovum & sperm are transported simultaneously to the ampullary-isthmic junction.

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both the assertion and reason are false.

BHR485

35. **Assertion:** Ovulation generally occur during the mid of cycle
Reason: LH surge occur in mid of cycle.

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both the assertion and reason are false.

BHR486

36. **Assertion:** Follicular phase is also referred as proliferative phase.
Reason: During follicular phase, the endometrium of uterus regenerates through proliferation.

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both the assertion and reason are false.

BHR487

Exercise – I — ANSWER KEY

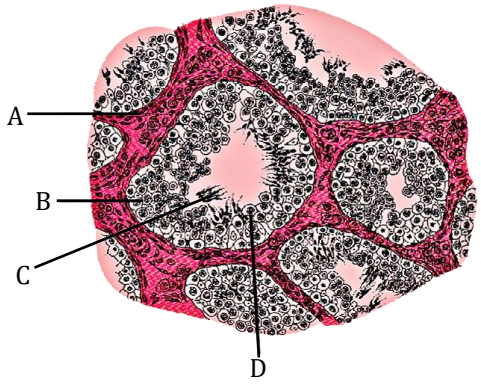
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	1	2	3	3	4	1	1	2	2	4	3	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	4	3	4	4	2	4	2	2	1	2	4	4	3	2
Question	31	32	33	34	35	36									
Answer	2	4	1	1	1	1									

Exercise – II
Human Reproduction

1. Sertoli cells which line the seminiferous tubules from inside
(1) undergo meiotic division to produce sperms
(2) provide nutrition the germ cells
(3) synthesise and secrete testicular hormones
(4) All of these

BHR488

2. Refer to the given figure showing sectional view of seminiferous tubule. In the figure, some parts are labelled as A, B, C and D. identify the part which provides nutrition to the developing sperms.



(1) A (2) C (3) D (4) B

BHR489

3. Match the parts given in Column–I to their characteristic features in Column–II and choose the correct option from the correct option from the codes given below.

	Column I		Column II
(a)	Penis	(1)	Loose fold of skin
(b)	Glans penis	(2)	Male external genitalia
(c)	Foreskin	(3)	External opening urethra
(d)	Urethral meatus	(4)	Enlarged end of penis

Codes-

	a	b	c	d
(1)	2	4	1	3
(2)	3	4	1	2
(3)	2	4	3	1
(4)	4	3	2	1

BHR490

4. Among the following which one is not a male accessory gland? (NCERT Exemplar)
(1) Seminal vesicle
(2) Ampulla
(3) Prostate
(4) Bulbourerthral gland

BHR491

5. Match the Column – I (parts) to Column- II (feature) and choose the correct option from the codes given below.

	Column I		Column II
(a)	Sertoli cells	(1)	Testicular hormones
(b)	Leydig cells	(2)	External opening of urethra
(c)	Epididymis	(3)	Nutrition to the germ cells
(d)	Urethral meatus	(4)	Male sex accessory duct

Codes –

	a	b	c	d
(1)	4	3	2	1
(2)	3	1	4	2
(3)	1	2	3	4
(4)	2	4	1	3

BHR492

6. Seminal plasma is contributed by (NCERT Exemplar)
(I) Seminal vesicle (II)Prostate
(III) Urethra
(Iv) Bulbourethral gland
(1) I and II (2) I, II, and IV
(3) II, III and IV (4) I and IV

BHR493

7. Read the following statements about seminal plasma and choose the correct statement(s) from the given options.
(I) Seminal plasma is secreted by seminal vesicles, prostate, and bulbourethral glands.
(II) It is rich in fructose and calcium.
(III) It contains certain enzymes also.
(1) I and II (2) II, and III
(3) I and III (4) All of these

BHR494

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8. Read the following statements about male reproductive system and choose the incorrect statements from the given options.

(I) It is located in the pelvis region.

(II) The testes are situated outside the abdominal cavity within a pouch called scrotum.

(III) Each testis has about 350 testicular tubules.

(IV) Penis, the male external genitalia is made up of special tissues to facilitate insemination.

(1) I and III (2) III and IV

(3) I and IV (4) Only III

BHR495

9. Match the parts of female reproductive system given in Column –I with their function in Column-II and choose the correct option from the codes given below.

	Column I		Column II
(a)	Ovary	(1)	Delivery of baby
(b)	Fimbriae	(2)	Steroid hormone
(c)	Myometrium	(3)	Secretion of milk
(d)	Cells of alveoli	(4)	Collection of ovum

Codes –

	a	b	c	d
(1)	2	4	1	3
(2)	4	3	2	1
(3)	3	4	1	2
(4)	1	4	3	2

BHR496

10. Match the layers of uterus given in Column-I with their characteristic features given in Column–II and choose the correct option from the codes given below.

	Column I		Column II
(A)	Perimetrium	(1)	Thick layer of smooth muscles
(B)	Myometrium	(2)	Thick membranous layer
(C)	Endometrium	(3)	Glandular layer
		(4)	Thin membranous layer

Codes –

	a	b	c
(1)	2	4	3
(2)	4	1	3
(3)	1	2	3
(4)	3	1	2

BHR497

11. Match the parts of female external genitalia in Column-I with their characteristic features in Column – II. Choose the correct option from the codes given below.

	Column I		Column II
(a)	Mons pubis	(1)	Fleshy folds of tissue
(b)	Labia majora	(2)	Cushion of fatty tissue
(C)	Hymen	(3)	Tiny finger like structure
(D)	Clitoris	(4)	Covers opening of vagina

Codes –

	a	b	c	d
(1)	2	1	4	3
(2)	4	3	2	1
(3)	1	4	3	2
(4)	2	1	3	4

BHR498

12. Read the following statements about uterus and choose the correct option from the codes given below.

(I) The shape of the uterus is like inverted pear.

(II) The uterus opens into vagina through a narrow cervix.

(III) The uterus along with cervix forms the birth canal.

Codes -

(1) I and III (2) II and III

(3) I and II (4) All of these

BHR499

13. Read the following statements about mammary glands and choose the incorrect statement.

(I) The mammary glands contain glandular tissue and fat.

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(II) The mammary lobes of breasts contain alveoli which secrete milk.
(III) The milk secreted by alveoli is stored in lactiferous duct.
(1) Only II (2) Only III
(3) Only I (4) I and III

BHR500

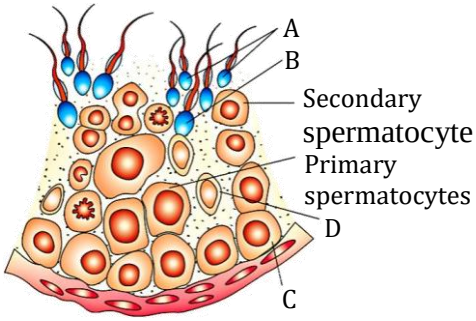
14. Spermatogenesis is the process in which immature male germ cells undergo division to produce sperms. Choose the correct one with reference to above.
(1) Spermatogonia have 46 chromosomes and always undergo meiotic cell division.
(2) Primary spermatocytes divide by mitotic cell division.
(3) Secondary spermatocytes have 23 chromosomes and undergo second meiotic division.
(4) Spermatozoa are transformed into spermatids.

BHR501

15. The difference between spermiogenesis and spermiation is -
(1) In spermiogenesis spermatids are formed, while in spermiation spermatozoa are formed.
(2) In spermiogenesis spermatozoa are formed, while in spermiation spermatids are formed
(3) in spermiogenesis spermatozoa from Sertoli cells are released into the cavity of seminiferous tubules, while in spermiation spermatozoa are formed
(4) in spermiogenesis spermatozoa are formed, while in spermiation spermatozoa are released from Sertoli cells into the cavity of seminiferous tubules.

BHR502

16. Refer to the given figure showing diagrammatic sectional view of a seminiferous tubule. In the figure, some parts are labelled as A, B, C and D. Identify the part which gets activated by FSH.

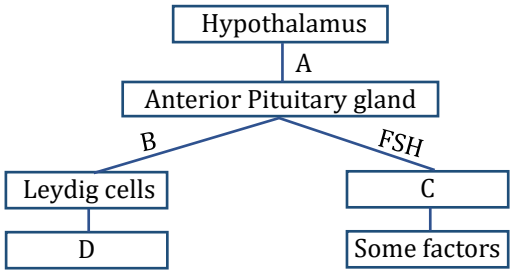


(1) A (2) B
(3) D (4) C

BHR503

17. Refer to the given flowchart. In it, some spaces are mentioned as A, B, C and D. Identify the correct option for them from the codes given below.

Codes-



(1) A-GnRH; B-LH; C-Sertoli cells; D-Androgens
(2) A-LH; B-GnRH; C-Androgens; D-Sertoli cells
(3) A-Androgens; B-LH; C-Interstitial cells; D-FSH
(4) A-FSH; B-Sertoli cells; C-GnRH; D-LH

BHR504

18. Match the Column- I representing parts of the sperm to Column- II showing their functions and choose the correct option.

	Column I		**Column II**
(a)	Head	(1)	Enzymes
(b)	Middle piece	(2)	Sperm motility
(c)	Acrosome	(3)	Energy
(d)	Tail	(4)	Genetic

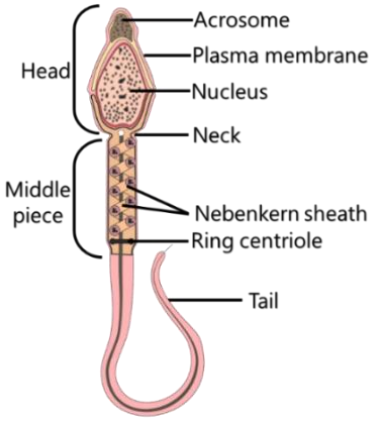
Codes –

	a	**b**	**c**	**d**
(1)	2	4	1	3
(2)	4	3	1	2
(3)	4	1	2	3
(4)	2	1	3	4

BHR505

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19. Refer to the given figure showing structure of a sperm. The figure is followed by four (I-IV) statements. Choose the incorrect statement (s) about it.



(I) The acrosome is filled with enzymes that help in fertilizing the ovum.
(II) The neck possesses numerous mitochondria.
(III) Tail is responsible for sperm motility.
(IV) The human male ejaculates 50–100 million sperms during a coitus.

(1) II and IV (2) I and III
(3) I and II (4) III and IV

BHR506

20. Match Column – I with Column – II and choose the correct option from the code given below.

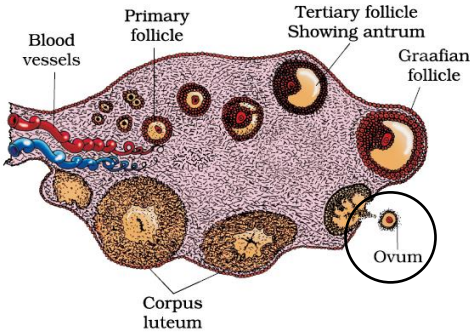
	Column I		Column II
(a)	Oogonia	(1)	Antrum
(b)	Tertiary follicle	(2)	Gamete mother cells
(c)	Secondary follicle	(3)	Haploid
(d)	Secondary oocyte	(4)	More layers of granulosa

Codes –

	a	b	c	d
(1)	4	3	2	1
(2)	1	2	3	4
(3)	2	1	4	3
(4)	3	4	2	1

BHR507

21. Refer to the given figure showing diagrammatic section view of ovary. The encircled part of figure is showing a process of oogenesis. Identify it as well as the follicle which is involved in this process.



(1) Spermiation, Secondary follicle
(2) Menstruation, Primary follicle
(3) Ovulation, Graafian follicle
(4) Ovulation, Tertiary follicle

BHR508

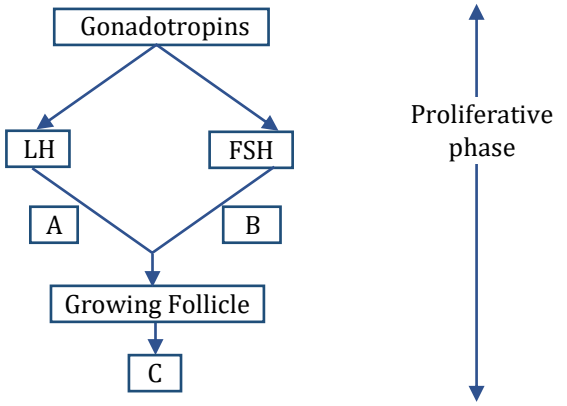
22. **Assertion :** Spermatogenesis starts at the age of puberty.
Reason : There is a significant increase in the secretion of gonadotropin releasing hormone at puberty.
(1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true but reason is not the correct explanation of assertion.
(3) Assertion is true, but reason is false.
(4) Both assertion and reason are false.

BHR509

23. **Assertion :** A large haploid secondary oocyte is formed due to unequal division.
Reason : A tiny second polar body is formed during this division.
(1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason true but reason is not the correct explanation of assertion
(3) Assertion is true but reason is false
(4) Both assertion and reason are false.

BHR510

24. Refer to the given flowchart, in which three parts are labelled as A, B and C. Identify them and the correct option from the codes given below.



	A	B	C
(1)	Increase	Increase	Estrogens
(2)	Decrease	Increase	Estrogens
(3)	Decrease	Decrease	Testosterone
(4)	Increase	Decrease	TSH

BHR511

25. Match the items given in Column-I with those in Column-II and select correct option from the codes given below -

	Column I		Column II
(a)	Proliferative phase	(1)	Breakdown of endometrial lining
(b)	Secretary phase	(2)	Follicular phase
(c)	Menstruation	(3)	Luteal phase

Codes :

	a	b	c
(1)	3	2	1
(2)	1	3	2
(3)	2	3	1
(4)	3	1	2

BHR512

26. No new follicles develop in the luteal phase of the menstrual cycle because -

(1) Both FSH and LH levels are low in the luteal phase

(2) Follicles do not remain in the ovary after ovulation

(3) FSH levels are high in the luteal phase.

(4) LH levels are high in the luteal phase.

BHR513

27. Match the phase of menstrual cycle given in Column-I with the hormones secreted during that phase in Column-II. Choose the correct option from the codes given below.

	Column I		Column II
(a)	Follicular phase	(1)	Progesterone
(b)	Ovulatory phase	(2)	Gonadotropins
(c)	Luteal	(3)	LH surge

Codes –

	a	b	c
(1)	3	1	2
(2)	2	3	1
(3)	3	2	3
(4)	2	1	3

BHR514

28. Read the following settlements about corpus luteum and choose the correct ones from the following options

(I) It is formed during ovulatory phase of menstrual cycle.

(II) It secretes large amounts of progesterone.

(III) In the absence of ovulation, the corpus luteum degenerates.

(IV) The degeneration of corpus luteum causes disintegration of endometrium.

(1) I and III (2) II and III

(3) II and IV (4) I and IV

BHR515

29. In human beings, permanent cessation of menstrual cycle is called:

(1) Menopause (2) Menarche

(3) Ovulation (4) None of these

BHR516

30. Choose the incorrect statement from the following.

(I) High levels of estrogen triggers the ovulatory surge.

(II) Oogonial cells start to proliferate and give rise to functional ova in regular cycles from puberty onwards.

(III) Sperms released from seminiferous tubules are poorly motile/ non – motile.

(IV) Progesterone level is high during the post ovulatory phase of menstrual cycle.

(1) I and III (2) II and IV

(3) I and IV (4) I and II

BHR517

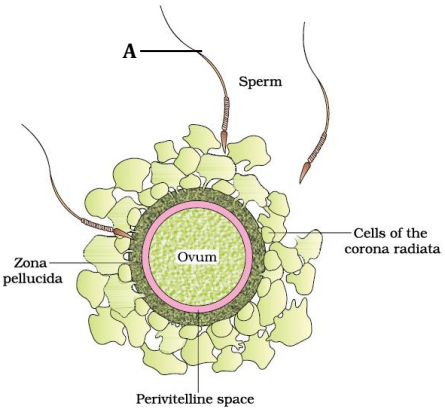
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31. Consider the following features.
(i) Transformation of graafian follicle into corpus luteum.
(ii) Secretion of large amount of progesterone from corpus luteum.
(iii) Maintenance of endometrial lining of the uterus.
Select the correct phase of menstrual cycle that possesses all the above characteristics.
(1) Follicular phase
(2) Secretory phase
(3) Proliferative phase
(4) Ovulatory phase

BHR518

32. Refer to the given figure showing an ovum surrounded by few sperms. Sperm 'A' in the figure is trying to fertilise the ovum. How will this sperm 'A' ensure that no other sperm can fertilise the ovum?



(1) By inducing changes in the cells of corona radiata.
(2) By inducing changes in the pellucida layer of the ovum.
(3) By inducing changes in the perivitelline space of the ovum
(4) By releasing some chemicals to kill other sperms.

BHR519

33. During acrosomal reaction the sperm -
(1) comes in contact with zona pellucida of the ova.
(2) undergoes reactions within the uterine environment of female.
(3) undergoes reaction within the epididymal environment of the male.
(4) produces androgens in the uterus.

BHR520

34. Match the events given in Column-I with their characteristic features in Column-II and choose the correct option from the codes given below.

	Column I		Column II
(a)	Fertilisation	(1)	Female reproductive tract
(b)	Capacitation	(2)	Contact of sperm with zona pellucida
(c)	Acrosomal reaction	(3)	Before fertilisation and after ovulation
(d)	Second polar body	(4)	Ampullary isthmic junction

Codes –

	a	b	c	d
(1)	1	4	2	3
(2)	4	1	2	3
(3)	3	2	4	1
(4)	4	1	3	2

BHR521

35. Morula is known as a developmental stage -
(1) between the zygote and blastocyst
(2) between the blastocyst and gastrula
(3) after the implantation
(4) between implantation and parturition

BHR522

36. Match the following and choose the correct option from the codes given below.

	Column I		Column II
(a)	Trophoblast	(1)	Embedding of Blastocyst in the endometrium
(b)	Cleavage	(2)	Group of cells that would differentiate as embryo
(c)	Inner cell mass	(3)	Outer layer of blastocyst attached to the endometrium
(d)	Implantation	(4)	Mitotic division of zygote

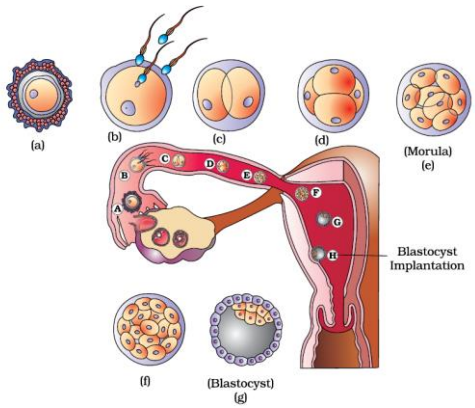
Codes –

	a	b	c	d
(1)	2	1	3	4
(2)	3	4	2	1
(3)	3	1	2	4
(4)	2	4	3	1

BHR523

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37. Refer to the given figure showing transport of ovum, fertilisation and passage of growing embryo through. Fallopian tube. The figure is followed by four statements Choose the incorrect statement about it.



(1) The second meiotic division of secondary oocyte occurs after fertilisation.
(2) The embryo with 8 –16 blastomeres is called morula.
(3) The trophoblast layer of blastocyst gets embedded in the endometrium.
(4) The cells of inner cell mass differentiate to form embryo.

BHR524

38. **Assertion :** All copulations do not lead to fertilisation and pregnancy.
Reason : Fertilisation can only occur if the ovum and sperms are transported Simultaneously to the ampullary – isthmic junction
(1) Both assertion and reason are ture and reason ins the correct explanation of assertion.
(2) Both assertion and reason are true but reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.

BHR525

39. Choose the incorrect statement about placenta.
(1) The placenta facilitates the supply of oxygen and nutrients to the embryo
(2) It helps in the removal of CO_2 and excretory/ waste materials produced by the embryo.
(3) The placenta is connected to the embryo through umbilical cord.
(4) Placenta acts like an endocrine tissue and produces several enzymes also.

BHR526

40. The levels of estrogens, progestogens, cortisol, prolactin, etc., increase many fold in blood during pregnancy. It is necessary for
(1) supporting the fetal growth
(2) metabolic changes in the mother
(3) the maintenance of pregnancy
(4) all of these

BHR527

41. Match Column-I with Column-II and choose the correct option from the codes given below.

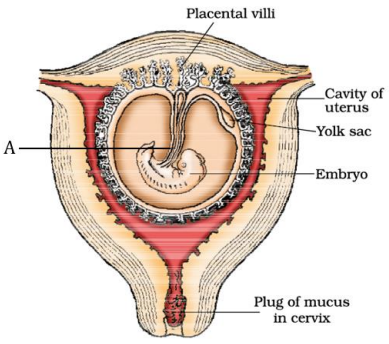
	Column I		**Column II**
(a)	Chorionic villi	(1)	Secretes relaxin hormone
(b)	Placenta	(2)	Finger-like projection on the trophoblast
(c)	Umbilical cord	(3)	Structural and functional units between foetus and mother
(d)	Ovary	(4)	Connects embryo to placenta

Codes –

	a	**b**	**c**	**d**
(1)	2	3	4	1
(2)	3	2	4	1
(3)	2	3	1	4
(4)	4	1	3	2

BHR528

42. Refer to the given figure showing human foetus within the uterus. How will the removal of 'A' in the figure affect the growth of foetus?



(1) Secretion of hCG hormone will stop
(2) Secretion of relaxin hormone will stop
(3) Transportation of substances to and from the embryo will stop
(4) All of these

BHR529

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43. Match Column-I with Column-II and select the correct option using the codes given below.

	Column I		Column II
(a)	Mons pubis	(1)	Embryo formation
(b)	Antrum	(2)	Sperm
(c)	Trophectoderm	(3)	Female external genitalia
(d)	Nebenkern	(4)	Graafian follicle

Codes –

	a	b	c	d
(1)	3	1	4	2
(2)	1	4	3	2
(3)	3	4	2	1
(4)	3	4	1	2

BHR530

44. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column I (Organ)	Column II (Month of development during pregnancy)	
(a)	Heart	(1)	Second month
(b)	Limbs and digits	(2)	First month
(c)	External genitalia	(3)	Fifth month
(d)	Appearance of hair on head	(4)	Third month

Codes –

	a	b	c	d
(1)	1	2	4	3
(2)	2	1	3	4
(3)	2	1	4	3
(4)	3	2	4	1

BHR531

45. Read the following statements about major features of embryonic development at various months of pregnancy. Choose the incorrect statement (s) about it.

(I) The embryo's heart is the first formed organ.

(II) Most of the major organ systems are formed by the end of 2nd month of pregnancy.

(III) The first movement of foetus is observed during 7th months of pregnancy.

(IV) The eyelashes are formed by the end of second trimester.

(1) II and III (2) I and II

(3) III and IV (4) Only IV

BHR532

46. Maths Column-I with Column-II and choose the correct option from the codes given below.

	Column I		Column II
(a)	Gestation period	(1)	Mild uterine contractions
(b)	Parturition	(2)	Duration of pregnancy
(c)	Foetal ejection reflex	(3)	Process of delivery of the foetus
(d)	After birth	(4)	Placental expulsion

Codes –

	a	b	c	d
(1)	2	3	1	4
(2)	2	4	1	3
(3)	4	3	2	1
(4)	1	2	3	4

BHR533

47. Choose the incorrect statement from the following:

(1) Parturition is induced by a complex neuroendocrine mechanism.

(2) The foetal ejection reflex triggers release of oxytocin from the hypothalamus.

(3) Oxytocin acts on the uterine muscles.

(4) The placenta is also expelled out of the uterus after the delivery on infant.

BHR534

48. Choose the incorrect statement from the following.

(1) Internal fertilisation takes place, in birds and mammals.

(2) Colostrum contains antibodies and nutrients.

(3) Polyspermy is prevented by chemical changes on the egg surface.

(4) In the human female implantation occurs almost seven days after fertilisation.

BHR535

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49. **Assertion** : Parturition is a complex neuroendocrine mechanism.
Reason : The signals of parturition originate from the fully developed fetus and placenta.
(1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true, but reason is not the correct explanation of assertion.
(3) Assertion is true, but reason is false.
(4) Both assertion and reason are false.
BHR536

50. **Assertion :** Breast feeding during the initial period of infant growth is recommended
Reason : During initial few days after delivery, colostrum is produced.
(1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true, but reason is not the correct explanation of assertion.
(3) Assertion is true, but reason is false.
(4) Both assertion and reason are false.
BHR537

51. **Assertion:** Lack of menstruation may be indicative of pregnancy in reproductive period of female.
Reason: Menstruation usually occurs if the released ovum is not fertilised.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BHR538

52. **Assertion:** Only one sperm can fertilize ovum.
Reason: A sperm when comes in contact with zona pellucida layer of the ovum, some changes occur in membrane that blocks the entry of other sperm.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BHR539

Exercise – II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	1	2	2	2	4	4	1	2	1	3	2	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	2	1	3	3	1	1	1	3	1	2	3	1	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	3	1	2	1	2	1	1	4	4	1	3	4	3	1
Question	46	47	48	49	50	51	52								
Answer	1	2	3	1	1	1	1								

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Exercise – I
Reproductive Health

1. In which year Indian government had legalized the MTP ?
(1) 1973 (2) 1970 (3) 1971 (4) 1917
BRH150

2. MTP is considered safe up to how many weeks of pregnancy :-
(1) Six weeks (2) Eight weeks
(3) Twelve weeks (4) Eighteen week
BRH151

3. Which one of the following is not a method of contraception :-
(1) Multiload -375
(2) Condoms
(3) Pills of combination of oxytocin and vasopressin
(4) Lippes loop
BRH152

4. Which of the following is a wrong match between the type of contraception and its function?

	Contraceptive Method	Function
(1)	Contraceptive pills	Suppress the secretions of gonadotropins
(2)	IUDs	Inhibit the implantation
(3)	Natural methods	Avoid chances of ovum & sperm meeting
(4)	Sterilisation	Inhibits gamete formation

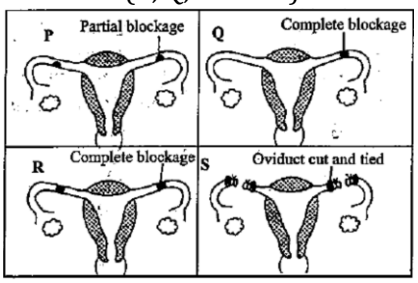
BRH153

5. In which of the following method of contraception no medicines or devices are used?
(1) Lactational amenorrhea
(2) Coitus withdrawal
(3) Periodic abstinence
(4) All of the above
BRH154

6. Which contraceptive method inhibit ovulation and implantation as well as after the quality of cervical mucus to prevent/ retard entry of sperm?
(1) Copper T (2) Oral contraceptives
(3) Condoms (4) cervical cap
BRH155

7. Oral contraceptive pills act-
(1) By preventing sperm motility
(2) By preventing ovulation
(3) By inhibiting motility of cilia of fallopian tube
(4) All of the above
BRH156

8. The given diagram shows the uterine tubes of four women (P, Q, R and S).



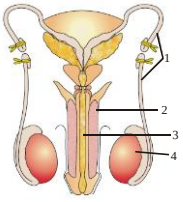
In which two women is fertilization impossible at present?
(1) P and Q (2) Q and R
(3) R and S (4) S and P
BRH157

9. The given figure shows one of the elements releasing intrauterine device. Select the option which shows the correct identification of the device and its feature :-



(1) CuT; suppress sperm motility and its fertilizing capacity.
(2) CuT; make uterus unsuitable for the attachment of blastocysts.
(3) Lippes loop; protect the users from contracting AIDS and STDs.
(4) LNG-20; acts as spermicidal means and decrease the contraceptive efficiency.
BRH158

10. In the given figure which marked number (1, 2, 3, 4) is operated for terminal contraception.



(1) 1 (2) 2
(3) 3 (4) 4
BRH159

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11. **Assertion:** Statutory ban on amniocentesis for sex determination is done.
Reason: There is increase in female foeticides.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BRH160

12. **Assertion:** Couple avoid coitus from day 10 to 17 of the menstrual cycle and it is one of the easy method of contraception's.
Reason: Ovulation occurs during these days (fertile period) so there is no chance of conception by abstaining from coitus during this period
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BRH161

13. **Assertion:** Barrier method is one of the contraceptive methods.
Reason: Barrier method prevents physical meeting of ovum and sperm.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BRH162

14. **Assertion:** Copper releasing IUDs are used as contraceptives.
Reason: Cu ions released form IUDs suppresses the sperm motility and the fertilizing capacity of sperms.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BRH163

15. **Assertion:** As long as mother breast feeds the child fully, chances of conception are almost nil.
Reason: Menstrual cycle and ovulation do not occur during the period of intense lactation.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BRH164

Exercise – I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	3	4	4	2	2	3	1	1	1	1	1	1	1

Exercise – II
Reproductive Health

1. Consider the following statements :
(a) In India, family planning program was initiated in 1982.
(b) Reproductive and Child Health Care (RCH) program is an improved version of family planning programs.
Select the correct option.
(1) (a) is true, (b) are false
(2) Both (a) and (b) are false
(3) (a) is false, (b) is true
(4) Both (a) and (b) are true

BRH165

2. In what way awareness can be created among people about reproduction related aspects?
(a) Through audio-visual and print media advertisements.
(b) Creating fear among adolescents about sexual practices.
(c) Introduction of sex education at school.
Select the most appropriate option:
(1) (a), (b) and (c) are correct
(2) (b) and (c) are correct
(3) (a) and (b) are correct
(4) (a) and (c) are correct

BRH166

3. Consider the following statements:
(a) In amniocentesis, small sample is taken from the foetus blood.
(b) Foetal sex can be determined through amniocentesis by studying the chromosomal pattern in amniotic fluid.
Select the correct option:
(1) (a) is true, (b) are false
(2) Both (a) and (b) are false
(3) (a) is false, (b) is true
(4) Both (a) and (b) are true

BRH167

4. In the context of amniocentesis, which of the following statement is incorrect?
(1) It is used for prenatal sex determination.
(2) It can be used for detection of Down syndrome.
(3) It can be used for detection of cleft palate.
(4) It is usually done when a woman is 14-16 weeks pregnant.

BRH168

5. **Assertion :** Amniocentesis helps to detect foetal disorders during early pregnancy.
Reason : Jaundice can be prevented in foetus by amniocentesis.
(1) Both assertion and reason are true and the reason is the correct explanation of assertion.
(2) Both assertion and reason are true but the reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.

BRH169

6. Under what circumstances, the growth of population declines?
(a) Increased number of people in reproducible age.
(b) Decreased infant mortality rate.
(c) Increased maternal mortality rate.
Select the most appropriate option.
(1) (a), (b) and (c) are correct
(2) (b) and (b) are correct
(3) Only (c) is correct
(4) Only (b) is correct

BRH170

7. Consider the following statements.
(a) 'One child norm' is a threat to population growth rate of India.
(b) Statutory raising of marriageable age of males and females check the population growth rate.
Select the correct option.
(1) a is true, b is false
(2) Both a and b are true
(3) a is false, b is true
(4) Both a and b are false

BRH171

8. Consider the following statements :
(a) Natural method of contraception is a traditional method to prevent pregnancy.
(b) The basic principle of natural method of contraception is to avoid the contact between ovum and sperms.
Select the correct option.
(1) a is true, b is false
(2) a is false, b is true
(3) Both a and b are true
(4) Both a and b are false

BRH172

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9. Match the following columns.

Column-I (Natural methods of contraception)		Column-II (Characteristics)	
(a)	Coitus interruptus	(1)	Avoid intercourse during ovulation
(b)	Lactational amenorrhea	(2)	Avoid insemination into vagina
(c)	Periodic abstinence	(3)	Ovum is not available for fertilisation

Select the correct option.

	a	b	c
(1)	3	1	2
(2)	1	3	2
(3)	3	2	1
(4)	2	3	1

BRH173

10. Select the incorrect statement.

(1) Natural methods of contraception are riskier but reliable.

(2) The act of breastfeeding interrupt ovulation during menstrual cycle.

(3) The chances of conception are higher during the ovulation period.

(4) Natural methods of contraception do not interfere with the sexual drive of an individual.

BRH174

11. The barrier method of contraception.

(a) Can be used by males only

(b) Prevents physical meeting of sperm and ovum

(c) Are used at the time of coitus

Select the most appropriate option.

(1) (a) and (c) are correct

(2) (a), (b) and (c) are correct

(3) (b) and (c) are correct

(4) Only (b) is correct

BRH175

12. Consider the following statements :

(a) Cervical caps and vaults are not cost effective as they cannot be reused.

(b) Contraceptive efficiency of vaults and cervical caps can be increased by is using spermicidal creams along with these barriers.

Select the correct option.

(1) a is true, b is false

(2) Both a and b are true

(3) a is false, b is true

(4) Both a and b are false

BRH176

13. Match the following columns.

Column-I		Column-II	
(a)	LNG 20	(1)	Hormone-releasing IUD
(b)	Lippes loop	(2)	Copper releasing IUD
(c)	Multiload 375	(3)	Non medicated IUD

Select the correct option.

	a	b	c
(1)	2	1	3
(2)	2	3	1
(3)	1	2	3
(4)	1	3	2

BRH177

14. Consider the following statements :

(a) Oral contraceptives inhibit ovulation and implantation to induce contraception.

(b) Menstrual flow does not occur in females after taking oral contraceptives.

Select the correct option.

(1) a is true, b is false

(2) a is false, b is true

(3) Both a and b are false

(4) Both a and b are true

BRH178

15. Which of the following statement is incorrect regarding implants?

(1) They are used by both males and females.

(2) These are placed under the skin.

(3) They contain progestogens or combinations of pills.

(4) They are effective for longer period as compared to pills.

BRH179

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16. Consider the following statements :
(a) Emergency contraceptives are taken within 72 hours of coitus to avoid pregnancy.
(b) Sterilisation is the terminal method to prevent any pregnancy as it is irreversible.
Select the correct option.
(1) a is true, b is false
(2) Both a and b are true
(3) a is false, b is true
(4) Both a and b are false

BRH180

17. **Assertion :** After vasectomy, sperms are not found in the semen or ejaculate.
Reason : Vas deferens is cut and tied in vasectomy.
(1) Both assertion and reason are true and the reason is the correct explanation of assertion.
(2) Both assertion and reason are true but the reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.

BRH181

18. Consider the following statements :
(a) AIDS and Hepatitis B are STIs which are caused by virus.
(b) Gonorrhoea and syphilis are caused by bacteria.
Select the correct option.
(1) a is true, b is false
(2) a is false, b is true
(3) Both a and b are false
(4) Both a and b are true

BRH182

19. Consider the following statements :
(a) Sexually transmitted diseases can be cured completely if detected early and treated properly.
(b) Early symptoms of STIs include itching, fluid discharge and swelling in genital areas.
Select the correct option.
(1) a is true, b is false
(2) Both a and b are true
(3) a is false, b is true
(4) Both a and b are false

BRH183

20. The consequences of delayed treatment of STIs include :
(a) ectopic pregnancy
(b) Still births
(c) Pelvic inflammatory diseases
(d) Uncontrolled growth of cells
Select the most appropriate option.
(1) (a) and (b) are correct
(2) (c) and (d) are correct
(3) (a), (b), (c) and (d) are correct
(4) (a), (b) and (c) are correct

BRH184

21. Consider the following statements :
(a) An infertile couple is unable to produce children due to immunological disorders only.
(b) In human population, males are always fertile while females are either fertile or infertile.
Select the correct option.
(1) a is true, b is false
(2) Both a and b are true
(3) a is false, b is true
(4) Both a and b are false

BRH185

22. Consider the following statements :
(a) Embryo formed by in-vivo fertilisation could be used in intra uterine transfer.
(b) Fully formed embryo is transferred into the fundus region of uterus in ZIFT.
Select the correct option.
(1) a is true, b is false
(2) a is false, b is true
(3) Both a and b are false
(4) Both a and b are true

BRH186

23. Match the following columns.

Column-I		Column-II	
(a)	ZIFT	(1)	Transfer of ovum into fallopian tube
(b)	GIFT	(2)	Transfer of embryo into the uterus Transfer of embryo into the uterus
(c)	IUT	(3)	Transfer of embryo into fallopian tube

Select the correct option.

	a	b	c
(1)	3	2	1
(2)	1	2	3
(3)	2	3	1
(4)	3	1	2

BRH187

24. **Assertion :** In all ART, fertilisation occurs outside the female's body.

Reason : Infertile couples cannot produce gametes.

(1) Both assertion and reason are true and the reason is the correct explanation of assertion.

(2) Both assertion and reason are true but the reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

BRH188

25. **Assertion:** Coitus interruptus is method of contraception.

Reason: It prevents insemination.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) If the assertion is true but the reason is false.

(4) If both the assertion and reason are false.

BRH189

Exercise – II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	3	3	3	3	3	3	4	1	3	3	4	1	1
Question	16	17	18	19	20	21	22	23	24	25					
Answer	2	1	4	2	4	4	1	4	4	1					

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Exercise - I Biotechnology Principles and Processes

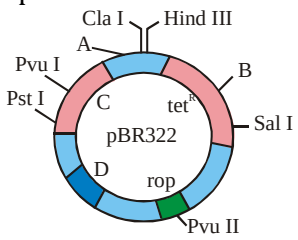
1. The separated fragments resulted from treatment with a restriction enzyme have an overhanging piece of single-stranded DNA at ends, called as :-
(1) Satellite DNA (2) Sticky ends
(3) Plasmid (4) Selectable marker
BBP261

2. The method in which alien DNA is directly injected into the nucleus of an animal cell is known as :-
(1) Biolistics
(2) Micro-injection
(3) Electroporation
(4) Heat-shock treatment
BBP262

3. The DNA polymerase (*Taq* polymerase) used in polymerase chain reaction (PCR) has been isolated from :-
(1) Bacteria (2) Plant
(3) Nematode (4) Fungus
BBP263

4. Which of the following is not a feature of the cloning vector :-
(1) Origin of replication
(2) Many restriction sites of one restriction enzyme
(3) Selectable marker
(4) Restriction sites of more than one restriction enzymes
BBP264

5. Identify A, B, C and D in the given figure of *E.coil* cloning vector pBR322 and select the correct option :-



	A	B	C	D
(1)	Hind I	Eco R I	amp^R	ori
(2)	Hind I	Bam H I	kan^R	amp^R
(3)	Bam H I	Pst I	ori	amp^R
(4)	Eco R I	Bam H I	amp^R	ori

BBP265

6. Gel electrophoresis is used for :-
(1) Construction of recombinant DNA by joining with cloning vectors
(2) Purification of product
(3) Cutting of DNA into fragments
(4) Separation of DNA fragments according to their size
BBP266

7. Which vector can clone only a small fragment of DNA?
(1) BAC (2) YAC
(3) Plasmid (4) Cosmid
BBP267

8. Insertional inactivation is used for :-
(1) Selection of transformed cells
(2) Selection of non-transformed cells
(3) Selection of recombinants
(4) Selection of vector
BBP268

9. After the formation of products in the bioreactors, it undergoes through some processes, before a finished product to be ready for marketing is called :-
(1) Elution
(2) Upstream processing
(3) Downstream processing
(4) Transformation
BBP269

10. The construction of first r-DNA emerged from the possibility of linking a gene encoding antibiotic resistance with native plasmid of :-
(1) Salmonella typhimurium
(2) E. coli
(3) Yeast
(4) Thermus aquaticus
BBP270

11. What is the source of the Ti (Tumor inducing plasmid) which is modified and used as a cloning vector to deliver the desirable genes into plant cells?
(1) Agrobacterium tumifaciens
(2) Thermophilus aquaticus
(3) Pyrococcus furiosus
(4) Aedes aegypti
BBP271

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12. For a DNA to function as a cloning vector the most essential requirement is :-
(1) multiple restriction sites
(2) several selectable markers
(3) circular nature
(4) 'ori' sequence

BBP272

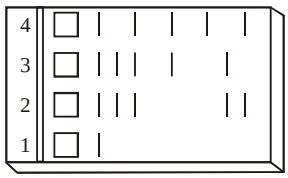
13. Purified DNA ultimately precipitate out after the addition of chilled ethanol. This DNA that separates out can be removed by :-
(1) Electrophoresis
(2) Downstream processing
(3) PCR
(4) Spooling

BBP273

14. Among the following select which is not the tool of rDNA technology :-
(1) Restriction (2) Vectors enzymes
(3) Host (4) Promoter

BBP274

15. In the following figure



Which DNA is not digested ?
(1) Lane 1 (2) Lane 2
(3) Lane 3 (4) Lane 4

BBP275

16. Genetically modified organism are also called:
(1) Transgenic organisms
(2) Clones
(3) Daughter cells
(4) Somaclones

BBP276

17. Given below are two statements:
Statement-I : Restriction enzyme belong to a larger class of enzymes called as nucleases.
Statement-II : Each restriction endonuclease recognise a specific palindromic nucleotide sequence in the DNA.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and II are correct.

BBP277

18. Given below are two statements:
Statement-I : DNA is a positively charged molecule move towards the negative electrode.
Statement-II : Amplification of DNA in PCR is achieved by use of a thermostable DNA polymerase.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BBP278

19. Given below are two statements:
Statement-I : *Agrobacterium tumefaciens* a pathogen of several monocot plants.
Statement-II : *Retroviruses* in animals have the ability to transform cancerous cells into normal cells.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BBP279

20. EFB Stands for :
(1) English Federation of Biology
(2) European Federation of Biology
(3) European Federation of Biotechnology
(4) European Federation of Biosphere

BBP280

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21. The core techniques that enable birth of modern biotechnology are :-
(1) Genetic engineering
(2) Bioprocess engineering
(3) Breeding
(4) Both (1) & (2)

BBP281

22. For alien DNA to replicate, it needs to be a part of:
(1) chromosome without origin of replication site
(2) mitochondrial DNA with origin of replication site
(3) Vector with origin of replication site
(4) cytoplasmic DNA with origin of replication site

BBP282

23. The plasmid joined with required DNA of interest can be transferred into …… .
(1) *Escherichia coli* (2) Plant cell
(3) Animal cell (4) Fungal cell

BBP283

24. The first restriction endonuclease was
(1) Hind-III (2) Hind-II
(3) Hind-I (4) Hind-IV

BBP284

25. Exonuclease cuts DNA from -
(1) specific position within DNA
(2) ends of DNA
(3) Both (1) and (2)
(4) None of these

BBP285

26. Restriction enzyme cuts DNA
(1) between same two bases on opposite strands, randomly DNA.
(2) between same two bases on opposite strands, a little away from centre of DNA sequence recognized.
(3) between two different bases on opposite strands, in centre of DNA sequence recognized.
(4) between two different bases on opposite strands, little away from centre of DNA sequence recognized.

BBP286

27. Which of the following is correct?
(1) Any piece of DNA linked to ori site will be replicated inside host.
(2) Number of replication copies is under control of recognition site
(3) Vector should not be chosen based on number of copies supported by it
(4) More than one option.

BBP287

28. Transformants include -
(1) Cells which have picked vector with foreign DNA ligated to it.
(2) Cells which have picked up vector without foreign DNA ligated to it
(3) Cells which have not picked up vector
(4) Both (1) & (2)

BBP288

29. Normal *E.coli* cell -
(1) Carries resistance against antibiotics ampicillin, tetracycline and kanamycin
(2) Does not carry resistance against antibiotics ampicillin, tetracycline and kanamycin
(3) Carries resistance against ampicillin but not tetracycline and kanamycin
(4) Carries resistance against tetracycline but not ampicillin and kanamycin

BBP289

30. If a foreign gene is ligated at Bam HI site of vector pBR322, then the resistance for _____.
(1) tetracycline is lost
(2) ampicillin is lost
(3) tetracycline is not lost
(4) more than one option

BBP290

31. The figure shows DNA separated out, removed by :



(1) spooning (2) spooling
(3) spilling (4) peeling

BBP291

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32. The fourth letter in the name of restriction endonuclease comes from :-
(1) Species
(2) Genus
(3) Order in which strain is isolated
(4) Strain

BBP292

33. Match column I with column II

	Column-I		Column-II
(a)	Spooling	(i)	Cutting out & extraction of DNA from gel
(b)	Elution	(ii)	Restriction endonuclease
(c)	Transformation	(iii)	Piece of DNA introduced into host bacterium
(d)	Molecular scissors	(iv)	Collection of fine threads of DNA

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
(2) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)
(3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
(4) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

BBP293

34. **Statement-I :** The normal *E. coli* cells carry resistance against the antibiotics.
Statement-II : rop sequence is responsible for controlling the copy number. of the linked DNA.
(1) Both statement-I and statement-II are correct.
(2) Both statement-I and statement-II are incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Statement-I is correct but statement-II is incorrect.

BBP294

35. The construction of first recombinant DNA was done by :-
(1) Meischer and Frederick Griffith
(2) Hershey and Chase
(3) Taylor and Colleagues
(4) Herbert Boyer and Stanley Cohen

BBP295

36. **Statement-I :** When cut by the different restriction enzymes, the resultant DNA fragments have the same kind of sticky ends.
Statement-II: Each restriction exonuclease recognises a specific palindromic nucleotide sequence in the DNA.
(1) Both statement-I and statement-II are correct.
(2) Both statement-I and statement-II are incorrect.
(3) Statement-I is correct but statement-II is incorrect.
(4) Statement-I is incorrect but statement-II is correct.

BBP296

37. Match column-I with column-II.

Column-I		Column-II	
(a)	Ribonuclease	(i)	Bacterial cell wall
(b)	Protease	(ii)	RNA
(c)	Chilled ethanol	(iii)	Plant cell wall
(d)	Lysozyme	(iv)	Protein
(e)	Cellulase	(v)	Fungal cell wall
(f)	Chitinase	(vi)	Precipitation of DNA

(1) a-(i), b-(ii), c-(iii), d-(iv), e-(v), f-(vi)
(2) a-(vi), b-(v), c-(iv), d-(iii), e-(ii), f-(i)
(3) a-(ii), b-(iv), c-(i), d-(vi), e-(iii), f-(v)
(4) a-(ii), b-(iv), c-(vi), d-(i), e-(iii), f-(v)

BBP297

38. **Assertion (A) :** Amplification of DNA is achieved by the use of DNA polymerase isolated from a bacterium *Thermus aquaticus*.
Reason (R) : This DNA polymerase remains active during the high temperature induced denaturation of ds DNA.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true and Reason is false.
(4) Both Assertion and Reason are false.

BBP298

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39. **Statement-I :** DNA fragments separate according to their size through sieving effect provided by agarose gel.
Statement-II : The larger the fragment size, the farther it moves.
(1) Both statement I and statement II are correct.
(2) Both statement I and statement II are incorrect.
(3) Statement I is correct but statement II is incorrect.
(4) Statement I is incorrect but statement II is correct.

BBP299

40. **Assertion (A) :** DNA fragments move towards anode under an electric field.
Reason (R) : DNA fragments are negatively charged molecules.
(1) Both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
(2) Assertion is correct and Reason is not correct
(3) Assertion is not correct but Reason is correct
(4) Both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

BBP300

41. **Assertion (A) :** Restriction enzymes are called molecular scissors.
Reason (R) : They cut DNA molecules at a particular point by recognizing a specific base sequence known as ori sequence.
(1) Both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
(2) Assertion is correct and Reason is not correct.
(3) Assertion is not correct but Reason is correct.
(4) Both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

BBP301

Exercise - I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	1	2	4	4	3	3	3	1	1	4	4	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	3	1	3	4	3	1	2	2	2	1	4	2	1
Question	31	32	33	34	35	36	37	38	39	40	41				
Answer	2	4	2	2	4	2	4	1	3	4	2				

Exercise - II
Biotechnology Principles and Processes

1. Plasmid is :
(1) autonomously replicating extra chromosomal DNA
(2) non-autonomously replicating, extra chromosomal DNA
(3) autonomously replicating chromosomal DNA
(4) non-autonomously replicating, extra chromosomal DNA

BBP302

2. The key tools for recombinant DNA technology are :
(1) Restriction enzyme, polymerase, isomerase, vectors.
(2) Recognition enzyme, polymerase, ligase, vector.
(3) Restriction endonuclease, polymerase, ligase, vector.
(4) Restriction enzyme, polymerase, dehydrogenase, vector.

BBP303

3. In 1963, two restriction endonucleases were isolated in *E. coli* that restricted growth of bacteriophage by -
(1) cutting DNA
(2) adding methyl group to DNA
(3) removing methyl group to DNA
(4) more than one option is correct

BBP304

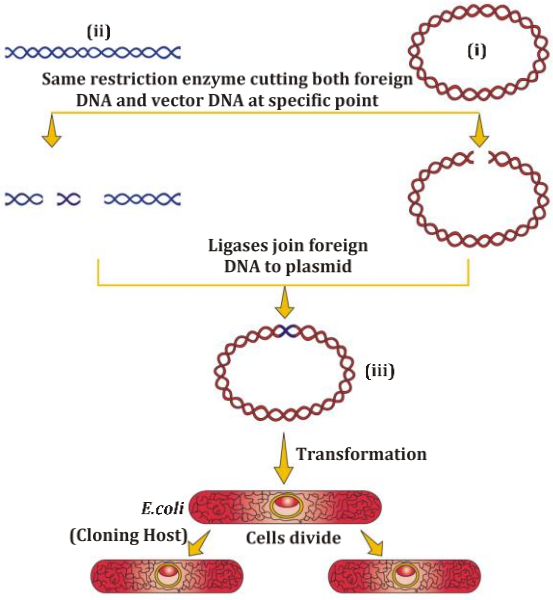
4. The convention for naming restriction endonucleases is :
(1) First two letters come from genus & third from species of prokaryotic cell from which they were isolated.
(2) First two letters come from species & third from genus of prokaryotic cell from which they were isolated.
(3) First letter come from genus & second from species of prokaryotic cell from which they were isolated.
(4) First letter come from species & second two from genus of prokaryotic cell from which they were isolated.

BBP305

5. EcoRI cuts DNA at -
(1) 5' G|AATTC 3'
3' CTTAA|G 5'
(2) 3' GAAT|TC 5'
5' CTTA|AG 3'
(3) 3' G|AATTC 5'
5' CTTAA|G 3'
(4) All of these

BBP306

6. Identify correct labelling.



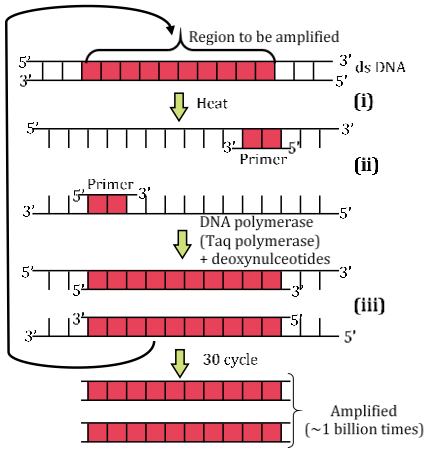
	(i)	(ii)	(iii)
(1)	Vector plasmid	Recombinant DNA	Foreign DNA
(2)	Foreign DNA	Vector plasmid	Recombinant DNA
(3)	Recombinant DNA	Vector plasmid	Foreign DNA
(4)	Vector plasmid	Foreign DNA	Recombinant DNA

BBP307

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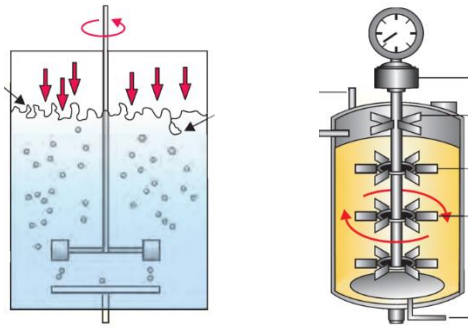
7. Identify correct labelling of sequence :



	(i)	(ii)	(iii)
(1)	Annealing	Denaturation	Extension
(2)	Denaturation	Extension	Annealing
(3)	Denaturation	Annealing	Extension
(4)	Extension	Annealing	Denaturation

BBP308

8. Identify types of stirred-tank bioreactor :-

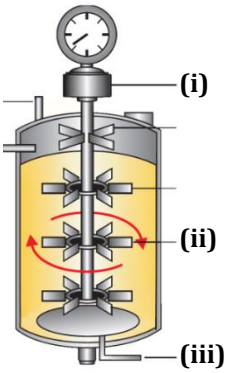


(i) **(ii)**

	(i)	(ii)
(1)	Simple stirred-tank bioreactor	Complex stirred-tank bioreactor
(2)	Complex stirred-tank bioreactor	Simple stirred-tank bioreactor
(3)	Simple stirred-tank	Sparged stirred-tank
(4)	Sparged stirred-tank	Simple stirred-tank

BBP309

9. Identify the correct labels :-



	(i)	(ii)	(iii)
(1)	Motor	Culture broth	Sterile Air
(2)	Culture broth	Motor	Sterile Air
(3)	Motor	Sterile Air	Culture broth
(4)	Sterile Air	Culture broth	Motor

BBP310

10. Identify the correct match from column I, II and III :-

	Column-I	Column-II	Column-III
(1)	Selectable marker	amp^R gene	Identification of recombinants
(2)	Restriction endonuclease	Enzyme	Random cutting of DNA
(3)	pBR322	Cloning vector	Linear DNA
(4)	BAC	Host cell	Isolation of DNA

BBP311

11. Given below are two statement

Statement-1 : The uptake of DNA during transformation is an active process.

Statement-2 : In microinjection, desired gene is directly injected into the nucleus of an animal cell.

In light of above statement, choose the correct answer from the option given below.

(1) Both statement I and statement II are incorrect

(2) Statement I is incorrect but statement II is correct.

(3) Statement I is correct but statement II is incorrect.

(4) Both statement I and statement II are correct.

BBP312

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12. Given below are two statement

Statement-1 : Each restriction endonuclease recognises a specific palindromic nucleotide sequences in the DNA.

Statement-2 : The stickness of the ends facilitates the action of the enzyme DNA ligase.

In the light of the above statements choose the most appropriate answer from the option given below :-

(1) Statement I is correct but statement II is incorrect.

(2) Both statement I and statement II are incorrect

(3) Both statement I and statement II are correct.

(4) Statement I is incorrect but statement II is correct.

BBP313

13. Given below are two statement

Statement-1 : The vector requires a selectable marker, which helps in identifying and eliminating non-transformants and selectively permitting the growth of the transformants.

Statement-2 : Normally the genes encoding resistance to antibiotics are considered useful selectable markers for *E. coli*.

In light of above statements, choose the most appropriate answer from the option given below :-

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect.

(3) Statement I is incorrect but statement II is correct.

(4) Statement I is correct but statement II is incorrect.

BBP314

14. **Assertion (A) :** DNA fragments can be separated by technique known as gel electrophoresis.

Reason (R) : DNA fragments separate according to their size through sieving effect provided by agarose gel.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is not correct but (R) is correct.

(3) (A) is correct but (R) is not correct.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BBP315

15. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : RNAi is a cellular defense mechanism in eukaryotes.

Reason (R) : RNAi is silencing of specific tRNA.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BBP316

16. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : In gel electrophoresis DNA fragments are separated.

Reason (R) : DNA is negatively charged, so it moves towards anode under electric field.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BBP317

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17. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : The first restriction endonuclease is Hind-II and cut specific sequence of six base pairs.

Reason (R) : Exonuclease remove nucleotides from the ends of DNA.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BBP318

18. Identify the correct set of statements :

(a) Origin of replication is responsible for initiating replication.

(b) DNA ligase acts on cut DNA molecules and joins their ends.

(c) Plasmid is autonomously replicating circular extra chromosomal DNA.

(d) More than 900 restriction enzymes that have been isolated from over 230 strains of bacteria.

Choose the correct answer from options given below :

(1) a and c only (2) b, c and d only

(3) a, c and d only (4) a, b, c and d all

BBP319

19. Identify the correct set of statements :

(a) In gel electrophoresis, the most commonly used matrix is agarose.

(b) Plasmids and bacteriophages do not have the ability to replicate within bacterial cells.

(c) Transformation is a procedure through which a piece of DNA is introduced in a host bacterium.

Choose the correct answer from options given below :

(1) a and b only (2) a and c only

(3) b and c only (4) a, b and c

BBP320

20. Identify the correct set of statements :

(a) When different DNAs are cut by same restriction enzyme, the resultant DNA fragments have the same kind of sticky ends.

(b) The seperated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution.

(c) The RNA can be removed by treatment with protease.

(d) If any protein encoding gene is expressed in a heterologous host it is called a recombinant protein.

Choose the correct answer from options given below :

(1) a and b only (2) b, c and d only

(3) a, b and d only (4) a and d only

BBP321

21. **Statement-I :** Ori sequence is a specific sequence which is responsible for initiating replication.

Statement-II : This sequence is not responsible for controlling the copy no. of the linked DNA.

(1) Both statement-I and II are incorrect

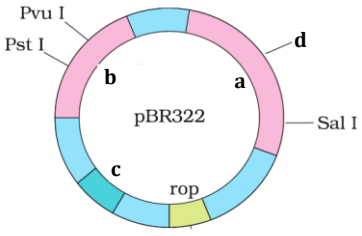
(2) Statement-I is correct but statement-II is incorrect

(3) Statement-I incorrect but statement-II is correct

(4) Both statement-I and II are correct

BBP322

22. The figure below is diagrammatic representation of pBR322 which of the given options correctly identifies its certain components.



(1) a-tet^R, b-amp^R, c-ori, d-BamHI

(2) a-amp^R, b-tet^R, c-ori, d-BamHI

(3) a-tet^R, b-amp^R, c-BamHI, d-ori

(4) a-amp^R, b-tet^R, c-PvuII, d-EcoRI

BBP323

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23. Match column-I with column-II.

Column-I		Column-II	
(a)	Ribonuclease	(i)	Bacterial cell wall
(b)	Protease	(ii)	RNA
(c)	Chilled Ethanol	(iii)	Plant cell wall
(d)	Lysozyme	(iv)	Protein
(e)	Cellulase	(v)	Fungal cell wall
(f)	Chitinase	(vi)	Precipitation of DNA

(1) a-(i), b-(ii), c-(iii), d-(iv), e-(v), f-(vi)
(2) a-(vi), b-(v), c-(iv), d-(iii), e-(ii), f-(i)
(3) a-(ii), b-(iv), c-(i), d-(vi), e-(iii), f-(v)
(4) a-(ii), b-(iv), c-(vi), d-(i), e-(iii), f-(v)

BBP324

24. **Assertion (A) :** Presence of insert results into insertional inactivation of the β-galactosidase gene.
Reason (R) : The colonies produce blue color, these are identified as recombinant colonies.
(1) Both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
(2) Assertion is correct and Reason is not correct
(3) Both Assertion is not correct but Reason is correct
(4) Both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

BBP325

25. Match column-I with column-II.

Column-I		Column-II	
(a)	Microinjection	(i)	Competent cells
(b)	Biolistics	(ii)	Plant cells
(c)	Heat shock	(iii)	Bacterial cells
(d)	Treatment with Ca^{2+}	(iv)	Animal cells

(1) a-(iv), b-(ii), c-(iii), d-(i)
(2) a-(ii), b-(iv), c-(i), d-(iii)
(3) a-(iv), b-(i), c-(ii), d-(iii)
(4) a-(i), b-(ii), c-(iii), d-(iv)

BBP326

26. **Statement-I :** DNA fragments separate according to their size through sieving effect provided by agarose gel.
Statement-II : The larger the fragment size, the farther it moves.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct and statement II is incorrect
(4) Statement I is incorrect and statement II is correct

BBP327

27. Which of the following statements in incorrect?
(1) The first restriction endonuclease isolated was Hind III.
(2) Restriction enzymes cut at specific base sequence is known as recognition sequence.
(3) From to 230 strains of bacteria, 900 restriction enzymes have been isolated.
(4) Restriction enzymes belong to a larger class of enzymes called nuclease.

BBP328

28. Which of the following statement is incorrect w.r.t. Plasmids.
(1) Stanley Cohen and Herbert Boyer in 1972 constructed the first recombinant DNA using a native plasmid of *E. coli*.
(2) It is an autonomously replicating circular extra-chromosomal DNA.
(3) Plasmid DNA act as vectors to transfer the piece of DNA attached to it.
(4) The linking of antibiotic resistance gene with the plasmid vector become possible with help of DNA ligase.

BBP329

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29. Match column-I with column-II.

Column-I		Column-II	
(a)	DNA ligase	(i)	Molecular scissors
(b)	DNA polymerase	(ii)	Initiation of replication
(c)	Restriction enzymes	(iii)	Molecular glue
(d)	Vectors	(iv)	DNA synthesis
(e)	Origin of replication	(v)	Plasmid DNA

(1) a-(i), b-(ii), c-(iii), d-(iv), e-(v)
(2) a-(iii), b-(ii), c-(i), d-(iv), e-(v)
(3) a-(iii), b-(iv), c-(i), d-(v), e-(ii)
(4) a-(iii), b-(ii), c-(i), d-(v), e-(iv)

BBP330

30. Which of the following statements is/are incorrect.
(a) The digested DNA fragments can be separated by a technique known as gel electrophoresis.
(b) The separated bands of DNA are cut out from the agarose gel and extracted from the gel is known as spooling.
(c) The DNA fragments purified are used in constructing recombinant DNA by joining them with cloning vectors.
(d) The most commonly used matrix is agarose gel which is an artificial polymer extracted from sea weeds.
(1) a & b (2) b & d (3) b & c (4) a & d

BBP331

31. **Assertion (A) :** When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of sticky ends.
Reason (R) : These can be joined together (end to end) using DNA ligases.
(1) Both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
(2) Assertion is correct and Reason is not correct.
(3) Assertion is not correct but Reason is correct.
(4) Both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

BBP332

32. Which of the following statements are incorrect?
(a) Plasmids and bacteriophages have the ability to replicate within bacterial cells independent of the control of chromosomal DNA.
(b) The normal *E. coli* cells carry resistance against antibiotics.
(c) Agarose is an artificial polymer extracted from seaweeds.
(d) The smaller the DNA fragment size, the farther it moves in gel electrophoresis.
(1) (a) and (b) (2) (b) and (c)
(3) (c) and (d) (4) None of the above

BBP333

33. A thermostable DNA polymerase is isolated from a _____, _____.
(1) Bacterium, Thermus arciformis
(2) Fungus, Thermus hydrothermalis
(3) Bacterium, Thermus aquaticus
(4) Fungus, Thermus aquaticus

BBP334

34. **Assertion:** Bacterial cells are treated with a specific concentration of a divalent cation.
Reason: It increases the efficiency with which DNA enters the bacterium through pores in its cell membrane.
(1) Assertion is true but Reason is false.
(2) Both assertion and reason are false.
(3) Both assertion and reason are true and reason is the correct explanation of assertion.
(4) Both assertion and reason are true but reason is not the correct explanation of assertion.

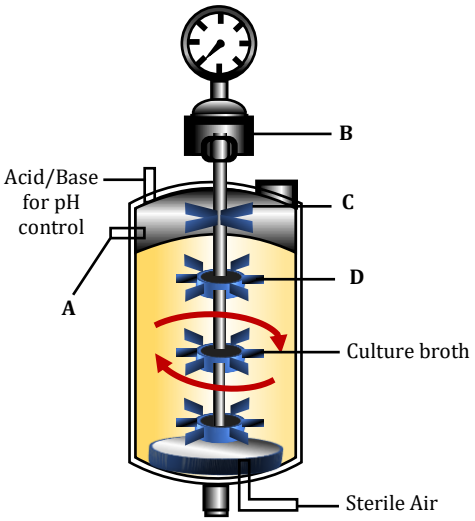
BBP335

35. Which of the following are the steps of down streaming.
(a) Quality control
(b) Clinical trial
(c) Purification
(d) Maintenance of optimal growth conditions
(e) Separation
(f) Production of recombinant protein
(1) a, b, c, d, f (2) a, b, c, e, f
(3) a, b, c, e (4) a, b, c, d, e

BBP336

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36. The given diagram represents simple stirred tank bioreactor. Which of the following represents the correct labelling.

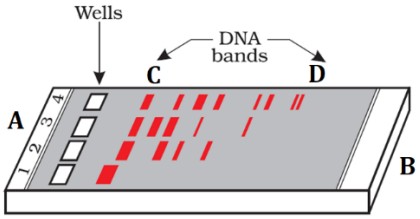


Simple stirred-tank bioreactor

	A	B	C	D
(1)	Acid/ base for pH control	Motor	Flat bladed impeller	Foam breaker
(2)	Motor	Acid/ base for pH control	Foam Breaker	Flat bladed impeller
(3)	Steam for sterilisation	Motor	Foam Breaker	Flat bladed impeller
(4)	Acid/ base for pH control	Motor	Culture broth	Foam breaker

BBP337

37.



	A	B	C	D
(1)	Anode	Cathode	Largest	Smallest
(2)	Cathode	Anode	Smallest	Largest
(3)	Anode	Cathode	Smallest	Largest
(4)	Cathode	Anode	Smallest	Largest

BBP338

38. Which of the following is/are not the characteristic feature of pBR322 vector.

(i) Ori
(ii) Selectable markers amp^R and tet^R
(iii) Restriction sites for Pvu II, Pst I, Pvu I, Hind II
(iv) Presence of more than one recognition sites for a restriction enzyme
(v) rop

(1) (ii) and (iii) (2) (i) and (v)
(3) (iii) and (iv) (4) (ii) and (iv)

BBP339

39. Which of the following statement is incorrect about selectable marker gene for β-galactosidase enzyme.

(i) The presence of a chromogenic substrate gives blue colored colonies if the plasmid does not have insert.
(ii) Selection of recombinants due to inactivation of β-galactosidase is cumbersome process.
(iii) Recombinant colonies do not produce any colour.
(iv) Absence of insert results into insertional inactivation of β galactosidase gene.

(1) (i) and (ii) (2) (ii) and (iii)
(3) (ii) and (iv) (4) (iii) and (iv)

BBP340

Exercise – II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	3	4	3	1	4	3	4	1	1	4	3	1	4	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	1	4	2	3	2	1	4	2	1	3	1	1	3	2
Question	31	32	33	34	35	36	37	38	39						
Answer	1	2	3	1	3	3	4	3	3						

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Exercise - I

Biotechnology and Its Applications

1. Find out incorrect match :-
(1) SCID – ADA deficiency
(2) α-1 antitrypsin - used to treat emphysema
(3) ELISA - it is based on the antisense RNA reaction only (without antigen)
(4) alpha-lactalbumin protein - Rosie

BBN217

2. Identify the mismatched pair :-
(i) Gene transfer - micropropagation
(ii) Early detection & diagnosis - hybridisation technique using probes
(iii) Flavr savr tomato - Gene silencing
(iv) International committee - GEAC
(v) Golden rice - Vitamin-A rich
(1) (i) & (ii) (2) (ii) & (iii)
(3) (iii) & (v) (4) (i) & (iv)

BBN218

3. For which variety of Indian rice, the patent was filed by a USA company :-
(1) Golden rice (2) Brown rice
(3) Basmati rice (4) Fe-fortified rice

BBN219

4. Why has the Indian parliament cleared the second amendment of the Indian patents bill:
(1) To prevent biopiracy
(2) To prevent biopatent
(3) To increase biodiversity
(4) All of these

BBN220

5. Bt-cotton is a GM crop when released for cultivation it will help in :-
(1) Pest resistance
(2) Herbicide tolerance
(3) Increase soil fertility
(4) Reduce post-harvest losses

BBN221

6. Using *Agrobacterium* vectors, nematode specific genes were introduced into host plant and these produced :-
(1) Sense - DNA
(2) Antisense - DNA
(3) ds RNA
(4) All of these

BBN222

7. The protein products of the following Bt toxin genes *cry I Ac* are responsible for controlling:
(1) Boll worm (2) Round worm
(3) Moth (4) Fruitfly

BBN223

8. Eli-Lilly, an American company is famous for:
(1) Producing vitamin-A enriched rice
(2) Producing pest resistant plants, by RNA interference
(3) Producing GH (Growth hormone) synthesised by recombinant DNA technology
(4) Preparing two DNA sequences corresponding to A and B chain of human insulin and introduced them in the plasmids of *E. coli* to produce insulin chain.

BBN224

9. Given below are two statements:
Statement-I : Bt toxin is produced by a bacterium called *Bacillus thuringiensis*.
Statement-II : Bt toxin exists in active form in bacteria.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BBN225

10. Given below are two statements:
Statement-I : About 30 recombinant therapeutics have been approved for human use over the world over.
Statement-II : In India, 12 of these are presently being marketed.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BBN226

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11. Given below are two statements:
Statement-I : The first clinical gene therapy was given in 1990 to a 4 year old girl with Adenosine deaminase (ADA) deficiency.
Statement-II : ELISA is based on the principle of antigen antibody interaction.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BBN227

12. Biotechnology mainly deals with -
(1) Industrial scale production of biopharmaceutical
(2) Biological use of genetically modified microbes, fungi, plants and animals
(3) Both (1) and (2)
(4) None of these

BBN228

13. Golden rice is genetically modified crop plant with incorporated gene meant for biosynthesis of -
(1) Vitamin-E (2) Vitamin-K
(3) Omega-3 (4) Vitamin-A

BBN229

14. The genes from bacterium *Bacillus thuringiensis* are used to produce ___ plants which reduces the amount of __used.
(1) disease resistant, insecticide
(2) insect resistant, fertilizers
(3) disease resistant, industrial enzyme
(4) insect resistant, insecticide

BBN230

15. Which of the following crops are modified using *Bacillus thuringiensis*?
(1) Corn and cotton
(2) Tomato and rice
(3) Potato and soyabean
(4) All of the above

BBN231

16. Bt toxin kills insect by -
(1) Inhibiting protein synthesis
(2) Generating excessive heat
(3) Creating pores leading to cell swelling and lysis in the mid gut epithelial cells
(4) None of these

BBN232

17. Cotton bollworm controlled by -
(1) Cry I Ac, Cry II Ab
(2) Cry I Ac, Cry II Ac, Cry I Ab
(3) Cry II Ac, Cry I Ab
(4) Cry I Ab

BBN233

18. Which of the following nematodes infects the root of the tobacco plants which reduces the production of tobacco?
(1) Meloidogyne incognitia
(2) Ascaris
(3) Wuchereria
(4) Enterobius

BBN234

19. RNAi take place in all _____ organisms as method of _____.
(1) prokaryotes, insect resistant
(2) eukaryotes, insect resistant
(3) eukaryotes, cellular defence
(4) prokaryotes, cellular defence

BBN235

20. The two polypeptides of human insulin are linked together by -
(1) Phosphodiester bonds
(2) Disulphide bridge
(3) Hydrogen bonds
(4) None of the above

BBN236

21. ____ is removed during the maturation of proinsulin to insulin.
(1) A-chain
(2) B-chain
(3) C-chain
(4) Both (1) and (2)

BBN237

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22. Treatment of genetic disorder by manipulating gene is called -
(1) Gene therapy
(2) rDNA technology
(3) Bone marrow transplantation
(4) Enzyme replacement therapy

BBN238

23. Adenosine deaminase (ADA) deficiency can be treated by ___A___ and ___B___ but it is not fully curative. Here A and B can be -
(1) A-gene therapy, B-radiation therapy
(2) A-bone marrow transplantation, B-enzyme replacement therapy
(3) A-organ transplantation, B-hormone replacement
(4) A-radiation therapy, B-enzyme replacement therapy

BBN239

24. Why Bt toxin protein crystal does not kill the Bacillus?
(1) Bacteria encloses toxins in special sac
(2) Bacteria are resistant to toxin
(3) Toxin occurs as inactive protoxins in bacteria
(4) All of the above

BBN240

25. The main challenge for production of insulin using rDNA techniques was -
(1) Splitting A and B-peptide chains
(2) Addition of C-peptide to proinsulin
(3) Getting insulin assembled to mature form
(4) Removal of C-peptide from active insulin

BBN241

26. Exploitation of bioresources of a nation by multinational companies without authorisation from the concerned country is referred to -
(1) Bioethics (2) Bioweapon
(3) Biopiracy (4) Bioexploitation

BBN242

27. Which committee takes decision regarding the validity of GM research and the safety of introducing GM-organisms for public services?
(1) Indian Council of Medical Research (ICMR)
(2) Genetic Engineering Approval committee (GEAC)
(3) Indian Institute of Science Education and Research (IISER)
(4) Genetic Engineering Award committee (GEAC)

BBN243

28. Healthy plants can be recovered from diseased plant by :-
(1) Micro-propagation
(2) Somatic hybridization
(3) Meristem culture
(4) Anther culture

BBN244

29. The critical research areas of biotechnology are-
(1) Providing the best catalyst in the form of improved organism usually a microbe or pare enzyme.
(2) Creating optimal conditions for a catalyst of act
(3) Downstream processing technology.
(4) All of these.

BBN245

30. Match column I with column II

	Column I		**Column II**
(a)	Basmati rice	(i)	Second amendment
(b)	GEAC	(ii)	Use of bioresources without proper authorization
(c)	Indian patents bills	(iii)	Validity of GM research
(d)	Biopiracy	(iv)	27 varieties

(1) a-(ii); b-(i); c-(iii); d-(iv)
(2) a-(iv); b-(i); c-(iii); d-(ii)
(3) a-(iv); b-(iii); c-(i); d-(ii)
(4) a-(i); b-(ii); c-(iii); d-(iv)

BBN246

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	3	1	1	3	1	4	2	4	4	3	4	4	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	1	3	2	3	1	2	3	3	3	2	3	4	3

Exercise - II
Biotechnology and Its Applications

1. The Bt toxin protein -
(1) Obstruct a biosynthetic pathway
(2) Causes death of the insect
(3) Stops egg laying of adult
(4) Generating excessive heat

BBN247

2. Cry II Ab and Cry I Ab produces toxins that control -
(1) Cotton bollworms and corn borer resp.
(2) Cotton bollworms and budworms resp.
(3) Corn borer and cotton bollworms resp.
(4) Nematodes and tobacco budworms resp.

BBN248

3. Transposons are also known as -
(1) Silenced gene
(2) Pleiotropic genes
(3) Mobile genetic elements
(4) Both (1) and (3)

BBN249

4. The demerits of using Bovine insulin (from cow) and porcine insulin (from pig) in diabetic patients is -
(1) It leads to hypercalcaemia
(2) It may cause allergic reaction
(3) It is expensive
(4) All of the above

BBN250

5. In which of the following method, a probe is allowed to hybridise to its complementary DNA in the clone of cells ?
(1) Enzyme linked Immunosorbent Assay (ELISA)
(2) PCR
(3) Autoradiography
(4) Gene therapy

BBN251

6. Technique used to detect mutation in genes is known as-
(1) Gel electrophoresis
(2) PCR
(3) Gene therapy
(4) ELISA

BBN252

7. Which of the following technique is based on the principle of antigen-antibody interaction?
(1) PCR
(2) ELISA
(3) Recombinant DNA technology
(4) Gene therapy

BBN253

8. Animals whose DNA is manipulated to possess and express an extra (foreign) gene are known as
(1) Transgenic animals
(2) Hybrid animals
(3) Transferred animals
(4) All of the above

BBN254

9. 95% of all the existing transgenic animals are-
(1) Pigs (2) Cows
(3) Mice (4) All of these

BBN255

10. Which of the following transgenic human protein products development are used to treat emphysema?
(1) α-1 antitrypsin (2) α-1 trypsin
(3) α-1 albumin (4) α-1 globulin

BBN256

11. _____ are used in testing safety of polio vaccine before they are used on human.
(1) Transgenic pig
(2) Transgenic monkey
(3) Transgenic rabbits
(4) Transgenic mice

BBN257

12. A _____ is granted to a person who has either invented a new and useful product, made improvement existing product or invented a new process of making a product -
(1) bioethics
(2) patent
(3) bio piracy
(4) genetic recombination

BBN258

13. Basmati is unique for its aroma and flavour, whose __A__ documented varieties are cultivated in __B__.
(1) A-37, B-India (2) A-27, B-India
(3) A-27, B-USA (4) A-30, B-USA

BBN259

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14. Match the columns and select correct option.

Column-I		Column-II	
(A)	Golden rice	(i)	Increased shelf life
(B)	Flavr savr tomato	(ii)	HGH
(C)	Mouse	(iii)	Vitamin-A
(D)	Transgenic pig	(iv)	Organ transplantation

(1) A - iii, B - ii, C - i, D - iv
(2) A - iii, B - i, C - iv, D - ii
(3) A - iii, B - i, C - ii, D - iv
(4) A - i, B - iii, C - ii, D - iv

BBN260

15. Find out the correct match from the following table (regarding biotechnological applications):

Column-I		Column-II	Column-III
(i)	Humulin	Eli-lily firm	Treatment of insulin dependent diabetes
(ii)	*Bacillus thuringiensis*	Production of endotoxin	Biodegradable insecticide
(iii)	ADA deficiency	Maturation of T-lymphocytes affected	Duplication of ADA gene

(1) (i) only (2) (i) and (ii)
(3) (iii) only (4) (ii) and (iii)

BBN261

16. Match the following :-

Column-I		Column-II
(A)	Rosie	1. Human α-lactalbumin 2. Insulin like growth factors
(B)	Transgenic mice	3. α-1-antitrypsin 4. Polio vaccine
(C)	*Cry* endotoxin	5. Activates in alkaline medium 6. Eukaryotic cellular defence
(D)	RNA-i	7. AIDS vaccine 8. Activates in acidic medium

(1) A - 1, B - 7, C - 5, D - 6
(2) A - 1, B - 4, C - 5, D - 6
(3) A - 3, B - 4, C - 5, D - 6
(4) A - 3, B - 4, C - 8, D - 6

BBN262

17. Which of the following is correct match :-

Column-I		Column-II	
(A)	Nematode resistant tobacco	(i)	Antisense technology
(B)	Replacement of non-functional gene	(ii)	Gene silencing
(C)	Flavr-savr tomato	(iii)	Gene therapy

(1) A - i, B - ii, C - iii (2) A - iii, B - ii, C - i
(3) A - ii, B - iii, C - i (4) A - ii, B - i, C - iii

BBN263

18. Find incorrect statement with respect to genetically engineered insulin :-

(1) 'C' peptide is not present in mature insulin
(2) 'A' and 'B' peptide chain produced separately by Eli-Lilly
(3) Human insulin is produced by plasmid of transgenic *E. coli*
(4) Chain A and B, that are linked by H-bond (hydrogen bond)

BBN264

19. Match the following columns :-

Column-I		Column-II	
(A)	GMO	(i)	Prevents blood clotting
(B)	Hirudin	(ii)	Enhanced nutritional value of food e.g. Vitamin-A enriched rice
(C)	PCR	(iii)	Pest resistance and high yield
(D)	Bt cotton	(iv)	Amplifying a specific DNA molecule

(1) A - ii, B - i, C - iv, D - iii
(2) A - iv, B - iii, C - ii, D - i
(3) A - iii, B - i, C - iv, D - ii
(4) A - iv, B - i, C - ii, D – iii

BBN265

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20. Given below are two statements : One is labelled as assertion (A) and the other is labelled as reason (R).

Assertion (A) : Bt-toxin is produced by a bacterium called *Bacillus thuringiensis*.

Reason (R) : The Bt-toxin protein exist as inactive protoxins.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(4) (A) is not correct but (R) is correct.

BBN266

21. Given below are two statement

Statement-1 : PCR used to detect mutations in genes in suspected cancer patients.

Statement-2 : Very low concentration of bacteria or virus can be detected by amplification of their protein by PCR.

In light of above statement, choose the correct answer from the option given below.

(1) Both statement I and statement II are correct.
(2) Statement I is correct but statement II is incorrect.
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and statement II are incorrect

BBN267

22. Given below are two statement

Statement-1 : RNAi takes place in all eukaryotic organisms as a method of cellular defence.

Statement-2 : This method involves silencing of a specific m-RNA due to complementary dsRNA molecule that binds to and prevents transcription of the m-RNA.

In light of above statement, choose the correct answer from the option given below.

(1) Both statement I and statement II are correct.
(2) Statement I is correct but statement II is incorrect.
(3) Both statement I and statement II are incorrect
(4) Statement I is incorrect but statement II is correct.

BBN268

23. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : PCR is now routinely used to detect HIV in AIDS patients.

Reason (R) : It is being used to detect mutations in genes in suspected cancer patients too.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BBN269

24. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : Transgenic mice are being developed for use in testing the safety of vaccines before they are used on human.

Reason (R) : Transgenic mice are being used to test the safety of polio vaccine.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BBN270

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25. In reference to genetically modified plants, identify the correct uses of GM plant
(a) Crops less tolerant to abiotic stresses.
(b) Reduce reliance on chemical pesticides.
(c) Helps to increase post harvest losses.
(d) Increased efficiency of mineral usage by plants.
(e) Reduce nutritional value of food.
Choose the correct answer from options given below :
(1) a, b and c only (2) a, c and e only
(3) b and d only (4) a, d and e only

BBN271

26. Identify the correct set of statements :
(a) In 1997, the first transgenic cow Rosie was developed.
(b) Rosie produced human protein enriched milk.
(c) There are an estimated 20,000 varieties of rice in India alone.
(d) 27 Documented varieties of Basmati are grown in India.
Choose the correct answer from options given below :
(1) a, b and d only (2) a, c and d only
(3) b, c and d only (4) a, b, c and d all

BBN272

27. Transgenic models exist for human diseases except.
(a) Cancer
(b) Emphysema
(c) Cystic fibrosis
(d) Rheumatoid arthritis
(e) Alzheimer's
(f) SCID
(1) b, c and d (2) b and f
(3) c and d (4) d, e and f

BBN273

28. The first transgenic cow, Rosie, produced human protein enriched milk which contained _____, by weight its is _____.
(1) α1- antitrypsin, 24.4 gm/l
(2) α1- antitrypsin, 2.4 gm/l
(3) alpha lactalbumin, 24.4 gm/l
(4) alpha lactalbumin, 2.4 gm/l

BBN274

29. Match column-I with column-II.

Column-I		Column-II	
(a)	Basmati Rice in India	(i)	Second amendment
(b)	GEAC	(ii)	Use of bioresources without proper authorisation
(c)	Indian patents Bills	(iii)	Validity of GM Research
(d)	Biopiracy	(iv)	27 varieties

(1) a-(ii), b-(i), c-(iii), d-(iv)
(2) a-(iv), b-(i), c-(iii), d-(ii)
(3) a-(iv), b-(iii), c-(i), d-(ii)
(4) a-(i), b-(ii), c-(iii), d-(iv)

BBN275

30. **Assertion :** PCR serves the purpose of early diagnosis.
Reason : Very low concentration of a bacteria or virus can be detected by amplification of their nucleic acids by PCR.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true and Reason is false
(4) Both assertion and Reason are false.

BBN276

31. **Statement I :** RNAi mechanism takes place in all prokaryotic organism as method of cellular defence.
Statement II : *Meloidogyne incognita* infects the shoot of tobacco plants.
(1) Both statement I and II are correct
(2) Both statement I and II are incorrect
(3) Statement I is correct and statement II is correct
(4) Statement II is correct and statement I is correct

BBN277

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32. **Assertion :** In ELISA, infection by pathogen can be detected by the presence of antigens or by detecting the antibodies synthesized against the pathogen.

Reason : ELISA is based on the principle of antigen- antibody interaction.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.

(3) Assertion is true and Reason is false

(4) Both assertion and Reason are false.

BBN278

33. Which of the following statement is incorrect w.r.t. to recombinant insulin?

(1) In 1983, Eli lilly an American company produced human insulin.

(2) DNA sequence corresponding to A, B and C, chains of human insulin are introduced in plasmid of *E.coli*

(3) A and B chains were produced separately, extracted and combined by creating disulphide bond.

(4) Insulin used to cure diabetes was earlier extracted from pancreas of slaughtered cattle and pigs.

BBN279

34. The proteins by the genes __a__ and __b__ control the cotton bollworm, that of ___c__ controls corn borer.

	a	b	c
(1)	Cry II Ab	Cry II Ac	Cry I Ab
(2)	Cry I Ac	Cry I Ab	Cry II Ab
(3)	Cry I Ac	Cry II Ab	Cry I Ab
(4)	Cry I Ab	Cry II Ab	Cry I Ac

BBN280

35. **Statement I :** RNAi takes place in all eukaryotic organism as a method of cellular defense.

Statement II : This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

BBN281

36. **Statement I :** Bt toxin is produced by a bacterium *Bacillus thuringiensis.*

Statement II : A biopesticide has been created by expressing Bt toxin gene in plants.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

BBN282

37. **Assertion :** Toxicity testing is carried out on transgenic animals.

Reason : Toxicity testing results are obtained in less time using transgenic animals.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.

(3) Assertion is true and Reason is false

(4) Both assertion and Reason are false.

BBN283

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38. **Statement I :** At present, about 30 recombinant therapeutics have been approved world over for human use, out of which 12 are presently being marketed in India.

Statement II : Using retrovirus vectors, nematode specific gene were introduced into the host plants.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

BBN284

39. Which of the following is the incorrect match:

(1) Transgenic model — Rheumatoid arthritis

(2) Biological product as therapeutic — Alzheimer's disease

(3) Transgenic mice — Polio vaccine

(4) Emphysema — α – 1– antitrypsin.

BBN285

40. **Assertion :** The clone having the mutated gene do not appear on the photographic film.

Reason : The probe has the complementary with the mutated gene.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

BBN286

Exercise - II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	3	2	3	2	2	1	3	1	4	2	2	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	4	1	1	2	2	1	1	3	1	2	4	3	1
Question	31	32	33	34	35	36	37	38	39	40					
Answer	2	1	2	3	1	1	1	3	2	3					

Exercise - I — Human Health & Disease

1. Choose the correct pair :-
(1) Physical barrier – HCl in stomach, saliva in mouth
(2) Cellular barrier – PMNL, NK cells
(3) Physiological barrier – Mucosa of gut, and urinogenital tract
(4) Cytokine barrier – Exotoxins

BHD378

2. Thalassemic child needed repeated blood transfusions, got infected by HIV and it deteriorated child's immunity.
Read the following questions :-
(A) HIV virus attacks which types of cells ?
(B) Why did the increased number of the HIV virus deteriorate child's immunity ?
(C) Which cells is/are called as HIV factory ?
(D) Which test showed that the infective virus is HIV ?
Choose the option with all correct answers for A, B, C, D :-
(1) A - T_4 cells, B - HIV destroys T_4 cells and macrophages, C - macrophages, D - ELISA
(2) A - T_4 and macrophages, B - HIV depletes macrophages, C - Helper T-cells, D - ELISA
(3) A - T_4 cells with CD_4 receptors, B - HIV depletes T_4 cells count, C - Macrophage, D - ELISA
(4) A - T_4 cells, B - HIV depletes all T_4 cells, C - Helper T cells, D - Western blot

BHD379

3. The AIDS virus has :-
(1) ds DNA (2) ds RNA
(3) ss RNA (4) ss DNA

BHD380

4. Neoplastic transformation by DNA damage can be brought about by several factors. Choose the factors which lead to oncogenic transformation within exposed cells :-
(a) U.V. rays (b) X-rays
(c) Gamma-rays (d) Tobacco smoke
(e) Retrovirus
(1) a, b, c (2) a, b, c, d
(3) a, b, d (4) b, d

BHD381

5. Sexually transmitted disease is :-
(1) Hepatitis A (2) Hepatitis B
(3) Hepatitis E (4) Filariasis

BHD382

6. Match these columns with regard to innate immunity:

	EXAMPLE		BARRIER
(A)	Gastric juice, saliva tears & sweat	(i)	Cytokine
(B)	Interferons	(ii)	Physical
(C)	Mucosa & skin	(iii)	Cellular
(D)	PMNL-Neutrophils monocyte, NK-cell	(iv)	Physiological

(1) A-ii, B-i, C-iv, D-iii
(2) A-iii, B-i, C-ii, D-iv
(3) A-iv, B-i, C-ii, D-iii
(4) A-iv, B-i, C-iii, D-ii

BHD383

7. Find out the **incorrect** matching of disease, pathogen and mode of transmission:

	Disease	Pathogen	Transmission
(1)	Common cold	***Rhino virus***	Droplet infection
(2)	Malaria	***Plasmodium***	Female Aedes mosquito
(3)	Pneumonia	***Streptococcus pneumoniae***	Droplet infection
(4)	Typhoid	***Salmonella typhi***	Contaminated food & water

BHD384

8. Read the following statements (A-D)
(A) ***Haemophilus influenzae*** infects the alveoli of lungs.
(B) Intestinal perforation and death may occur in severe cases of typhoid
(C) Rhino virus infect the nose and respiratory passage but not the lungs
(D) Symptoms of amoebiasis are abdominal pain and cramps, stool with excess mucous and blood clots.

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How many of the above statements are correct ?
(1) One (2) Two
(3) Three (4) Four

BHD385

9. Withdrawal symptoms are characterized by :-
(a) Reckless behaviour (b) Anxiety
(c) Nausea (d) Vandalism
(e) Sweating (f) Violence
(g) Shakiness
(1) a, d, f (2) a, d, e, g
(3) b, c, e, g (4) d, e, f

BHD386

10. Choose the incorrect statement from the following :
(1) The cell mediated immune response is responsible for graft rejection
(2) Metastasis is the most feared property of malignant tumour
(3) Cocaine and L.S.D. are sedatives
(4) T-cells do not form antibody but help B-cells produce them

BHD387

11. Which one of the following statements is incorrect?
(1) The injection containing preformed antibodies provide passive immunity to the host.
(2) Hepatitis-B vaccine is produced from yeast by recombination DNA technology
(3) The causative agent of AIDS, HIV retrovirus enters helper T-lymphocytes thus reducing their number
(4) Cannabinoids are always taken by injection.

BHD388

12. Consider the given statements and select the true ones :-
(A) Health could be defined as a state of complete physical, mental and social well being.
(B) Pathogens can enter our body by various means, multiply and not interfere with normal vital activities.
(C) Salmonella typhi is a pathogenic virus which causes typhoid fever.
(D) Plague and diphtheria are bacterial diseases in man.
(1) A and B (2) C and D
(3) A and D (4) A, C and D

BHD389

13. How many diseases are caused by bacteria?
Common cold, Pneumonia, Malaria, Dengue, Leshmaniasis, Typhoid, Chikungunya, Tetanus
(1) 4 (2) 3
(3) 6 (4) 2

BHD390

14. Select the symptoms of the disease caused by round worm :-
(1) Internal bleeding and scaly lesions on various parts of body
(2) Muscular pain and sustained high fever
(3) Fever, anemia and internal bleeding
(4) Blockage of the intestinal passage and sustained high fever

BHD391

15. Which of the following diseases are caused by protozoans?
(1) Ringworm and amoebic dysentery
(2) Malaria and Amoebiasis
(3) Genital warts and Chlamydiasis
(4) Amoebic dysentry and genital herpes

BHD392

16. Identify the correct option :-

	Disease	Symptoms	Caused by
(1)	Typhoid	- increase in appetite - Chill and high fever	Salmonella typhi
(2)	Pneumonia and common cold	- cough and headache - in severe cases the lips may turn gray to bluish in color	Streptococcus
(3)	Filariasis	- slowly developing chronic inflammation of the organs	Wuchereria malayi
(4)	AIDS	Progressive decrease in the number of helper T-lymphocytes	Bacterium

BHD393

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17. Which of the following option is correct for cancer?

(1) Cells of malignant tumor grow very rapidly, invading and damaging the surrounding normal tissue

(2) Benign tumor does not exhibit the locality character

(3) Cancer cells show metagenesis

(4) All of the above

BHD394

18. Which of the following option is true for opioids ?

(1) They bind to specific opioid receptors present in nervous system

(2) Obtained from *Papaver somniferum*

(3) Obtained from coca plant

(4) Both (1) and (2)

BHD395

19. Choose the incorrect option :-

(1) Response of antibodies is also called humoral immune response

(2) Colostrum provide passive immunity

(3) B-Lymphocytes produce an army of protein in response to pathogens in our blood to fight against them.

(4) Spleen, tonsils and Peyer's patches of small intestine and appendix are primary lymphoid organs

BHD396

20. Match the following columns and choose the correct answer from the options given below:

	Column-I		Column-II
(A)	Physical barrier	(i)	PMNL
(B)	Physiological barrier	(ii)	Interferons
(C)	Cellular barrier	(iii)	Acid in stomach
(D)	Cytokine barrier	(iv)	Mucus coating of epithelial lining

	A	B	C	D
(1)	iv	iii	i	ii
(2)	iv	i	ii	iii
(3)	iii	ii	iv	i
(4)	ii	iv	iii	i

BHD397

21. Which of the following option properly define innate immunity?

(1) Immunity that is pathogen specific

(2) Immunity that is produced by artificial method

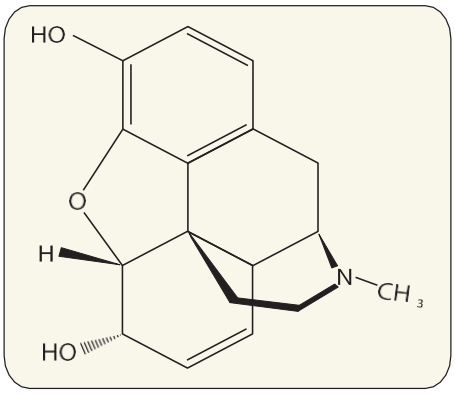
(3) Immunity that is nonspecific for pathogen

(4) Immunity that is specific but present at the time of birth

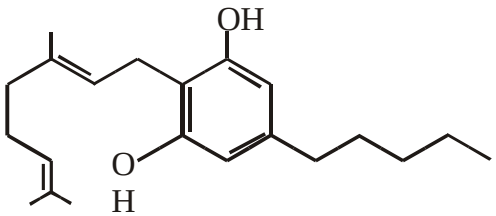
BHD398

22. Identify the molecules (a) and (b) given below and select the right option giving their source and use.

(a)



(b)



	Molecule	Source	Use
(1)	(a) Cocaine	*Erythroxylum coca*	Accelerates the transport of dopamine
(2)	(b) Heroin	*Cannabis sativa*	Depressant and slows down body functions
(3)	(b) Cannabinoid	*Atropa belladonna*	Produces hallucinations
(4)	(a) Morphine	*Papaver somniferum*	Sedative and pain killer

BHD399

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23. **Assertion:** Health means absence of disease.
Reason: Health can be defined as a state of complete physical well being only.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BHD400

24. **Assertion:** An Antibody is represented by H_2L_2.
Reason: Each antibody molecule has four peptide chains two light and two heavy chains.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BHD401

25. **Assertion:** Bone marrow and thymus are primary lymphoid organs.
Reason: Immature lymphocytes differentiate and proliferate in these organs.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BHD402

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	3	2	2	3	2	4	3	3	4	3	2	3	2
Question	16	17	18	19	20	21	22	23	24	25					
Answer	3	1	4	4	1	3	4	4	1	1					

Exercise - II
Human Health & Disease

1. Choose the correct statement about diseases.
(I) Disease adversely affects the functioning of one or more organs.
(II) A disease is characterized by the appearance of various signs and symptons.
(III) AIDS is a fatal non-infectious disease.
(IV) Cancer is an infectious disease.
(1) (I) and (II)
(2) (II) and (III)
(3) (III) and (IV)
(4) (I) and (IV)

BHD403

2. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Health	(1)	AIDS
(B)	Genetic disorders	(2)	Physical, mental and social well-being
(C)	Infectious disease	(3)	Cancer
(D)	Non-infectious disease	(4)	Inherited from parents from birth

	A	B	C	D
(1)	2	4	3	1
(2)	2	4	1	3
(3)	3	2	4	1
(4)	1	3	2	4

BHD404

3. Among the following, identify the infectious diseases.
(I) Cancer (II) Influenza
(III) Allergy (IV) Smallpox
(1) (I) and (II)
(2) (II) and (III)
(3) (III) and (IV)
(4) (II) and (IV)

BHD405

4. **Assertion :** Diseases are characterized by the appearance of various signs and symptoms.
Reason : Disease always adversely affects only one organ or system.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Both Assertion and Reason are false.

BHD406

5. **Assertion :** AIDS is an infectious disease.
Reason : AIDS can be easily transmitted from one person to another.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Both Assertion and Reason are false.

BHD407

6. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Typhoid	(1)	*Flavi arbo virus*
(B)	Pneumonia	(2)	*Rhino viruses*
(C)	Common cold	(3)	*Salmonella typhi*
(D)	Break bone fever	(4)	*Haemophilus influenzae*

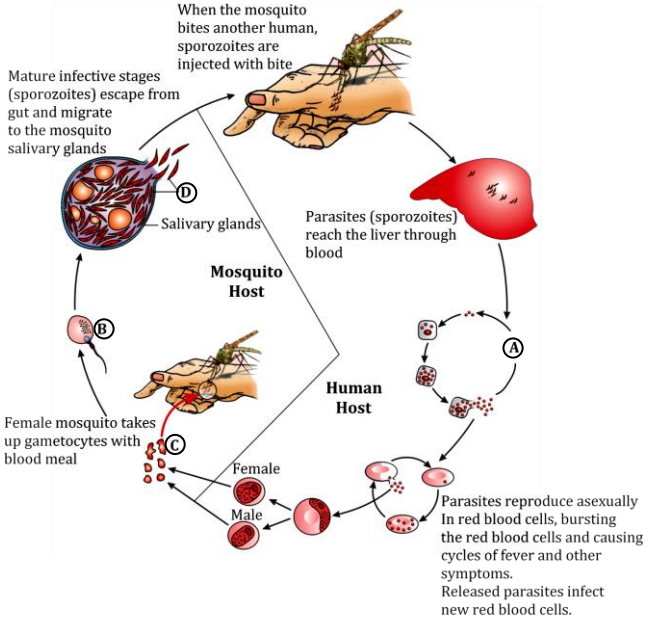
	A	B	C	D
(1)	3	1	4	2
(2)	3	4	2	1
(3)	2	3	1	4
(4)	1	4	2	3

BHD408

7. Refer to the given figure showing stages in the life cycle of use plasmodium. In the figure, which type of reproduction is occurring at stages A and B respectively. Also, identify C and D in the figure.

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	A	B	C	D
(A)	Asexual	Sexual	Gametocytes	Sporozoites
(B)	Sexual	Asexual	Gametocytes	Sporozoites
(C)	Asexual	Sexual	Sporozoites	Gametocytes
(D)	Asexual	Sexual	Gametocytes	Sporozoites

BHD409

8. Match Column-I (Diseases) with Column-II (causative agents) and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Typhoid	(1)	*Entamoeba histolytica*
(B)	Malaria	(2)	*Salmonella typhi*
(C)	Amoebiasis	(3)	*Rhino viruses*
(D)	Common cold	(4)	*Plasmodium vivax*

	A	B	C	D
(1)	2	4	1	3
(2)	4	3	2	1
(3)	1	2	4	3
(4)	3	1	2	4

BHD410

9. Choose the correct statements about amoebiasis.

(I) It is caused by the infection of *Entamoeba histolytica.*

(II) Its symptoms include loose motion, abdominal pain and cramps, stools with excess mucous and blood clots.

(III) Houseflies act as mechanical carriers for the parasite.

(IV) The main sources of its infection are drinking water and food contaminated by the fecal matter.

(1) (I) and (II) (2) (II) and (III)

(3) (I), (III) and (IV) (4) All of these

BHD411

10. Refer to the given figure.



Which disease is shown in the figure?

(1) Amoebiasis

(2) Filariasis

(3) Elephantiasis

(4) Both (2) and (3)

BHD412

11. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Ascaris	(1)	Lymphatic vessels of lower limbs
(B)	Wuchereria	(2)	Intestine
(C)	Trichopyton	(3)	Skin, scalp and nails
(D)	*Streptococcus pneumoniae*	(4)	Lungs

	A	B	C	D
(1)	2	1	3	4
(2)	1	2	4	3
(3)	3	2	1	4
(4)	4	3	2	1

BHD413

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12. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	*Entamoeba histolytica*	(1)	Sporozoa
(B)	*Plasmodium vivax*	(2)	Rhizopoda
(C)	*Ascaris lumbricoides*	(3)	Deuteromycetes
(D)	*Trichophyton rubrum*	(4)	Nematoda

	A	B	C	D
(1)	2	1	4	3
(2)	1	2	3	4
(3)	4	3	2	1
(4)	3	4	1	2

BHD414

13. Read the following statements carefully and choose the option that correctly identifies the true statements.

(I) Many infectious diseases can be prevented and controlled by maintaining personal and public hygiene.

(II) Proper disposal of waste and excreta is particularly essential for the airborne diseases.

(III) Malaria can be prevented by eliminating its vector and their breeding places.

(IV) Chikungunya is a vector-borne disease.

(1) (I), (II) and (III)

(2) (II), (III) and (IV)

(3) (IV), (II) and (I)

(4) (I), (III) and (IV)

BHD415

14. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Food-borne disease	(1)	Pneumonia
(B)	Air-borne disease	(2)	Amoebic dysentery
(C)	Vector-borne disease	(3)	Malaria
		(4)	Typhoid

	A	B	C
(1)	2,4	1	3
(2)	1,4	2	3
(3)	3	2,4	1
(4)	2	3	4,1

BHD416

15. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Amoebic dysentery	(1)	Female Anopheles mosquito
(B)	Dengue	(2)	Housefly
(C)	Malaria	(3)	Aedes mosquito

	A	B	C
(1)	3	2	1
(2)	2	3	1
(3)	1	2	3
(4)	2	1	3

BHD417

16. **Assertion :** Most of the parasites are pathogens.

Reason : Disease causing organisms are called pathogens hosts as parasite.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BHD418

17. **Assertion :** Pneumonia is caused by the infection of *Streptococcus pneumoniae.*

Reason : *Streptococcus pneumoniae* bacteria infect respiratory passage.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BHD419

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18. **Assertion :** The malarial parasite requires two hosts to complete its lifecycle.
Reason : These two hosts are human and mosquito.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Both Assertion and Reason are false.

BHD420

19. **Assertion :** *Wuchereria bancrofti* affects the blood vessels of the lower limbs.
Reason : This pathogen is transmitted to a healthy person through the bite of male mosquito vectors.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Both Assertion and Reason are false.

BHD421

20. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Physical barrier	(1)	Acid in the stomach
(B)	Physiological barrier	(2)	Monocytes
(C)	Cellular barrier	(3)	Interferon
(D)	Cytokine barrier	(4)	Mucus coating of the epithelium lining of urogenital tract

	A	B	C	D
(1)	4	1	2	3
(2)	1	3	4	2
(3)	2	4	3	1
(4)	3	2	1	4

BHD422

21. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Innate immunity	(1)	Antibodies
(B)	Acquired immunity	(2)	Non-specific immune response
(C)	Humoral immune response	(3)	T-lymphocytes
(D)	Cell-mediated immunity	(4)	Pathogen specific immune response

	A	B	C	D
(1)	4	2	3	1
(2)	2	4	1	3
(3)	2	4	3	1
(4)	1	3	4	2

BHD423

22. Consider the following statements and choose the correct statements.
(I) Active immunity is slow and takes time to give its full effective response.
(II) In passive immunity, ready-made antibodies are directly given.
(III) Colostrum contains IgE antibodies.
(IV) The foetus also receives some antibodies from its mother.
(1) (I), (II) and (III)
(2) (I), (II) and (IV)
(3) (I), (III) and (IV)
(4) All of these

BHD424

23. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Tuberculosis	(1)	Harmless virus
(B)	Whooping cough	(2)	Inactivated toxin
(C)	Diphtheria	(3)	Killed bacteria
(D)	Polio	(4)	Harmless bacteria

	A	B	C	D
(1)	3	2	4	1
(2)	4	3	2	1
(3)	1	2	4	3
(4)	2	1	3	4

BHD425

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24. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Colostrum	(1)	IgE
(B)	Allergy	(2)	IgA
(C)	Graft rejection	(3)	Passive immunisation
(D)	Preformed antibodies	(4)	Cell-mediated immunity

	A	B	C	D
(1)	3	2	4	1
(2)	4	3	2	1
(3)	1	2	4	3
(4)	2	1	4	3

BHD426

25. Choose the correct reason(s) for rheumatoid arthritis.

(I) Body attacks self-cells

(II) The ability of immune system to differentiate between self and non-self increases.

(III) The production of antibodies increases.

(IV) Immune system fails to discriminate between self and non-self cells.

(1) (I) and (IV) (2) (II) and (III)

(3) (II) and (IV) (4) (I) and (III)

BHD427

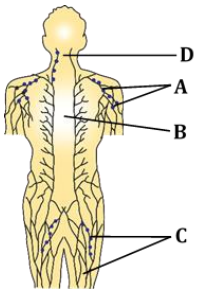
26. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Allergy	(1)	Inability to discriminate self non-self-cells
(B)	Autoimmunity	(2)	Introduction of killed/weakened pathogen
(C)	Active immunisation	(3)	Immune response against allergens
(D)	Passive immunisation	(4)	Introduction of antibodies

	A	B	C	D
(1)	3	1	2	4
(2)	4	3	2	1
(3)	3	1	4	2
(4)	2	4	3	1

BHD428

27. The figure given below is a diagrammatic representation of lymph nodes. In the figure some parts are labelled as A, B, C and D. Identify the part that serves to trap the antigens.



(1) B (2) A (3) C (4) D

BHD429

28. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Bone marrow	(1)	Maturation of T-Lymphocytes
(B)	Thymus	(2)	Production of blood cells
(C)	Spleen	(3)	Serve to trap antigens
(D)	Lymph Nodes	(4)	Reservoir of erythrocytes

	A	B	C	D
(1)	1	2	3	4
(2)	2	1	4	3
(3)	4	3	2	1
(4)	3	4	1	2

BHD430

29. **Assertion :** The immune response in which antibodies are formed is called humoral immune response.

Reason : Antibodies are found in blood.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BHD431

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30. **Assertion:** In passive immunization, preformed antibodies are given to the patients.

Reason : Polio vaccine also contains preformed antibodies.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BHD432

31. **Assertion** : Bone marrow is a secondary lymphoid organ.

Reason : In secondary lymphoid organs maturation of lymphocytes occur.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

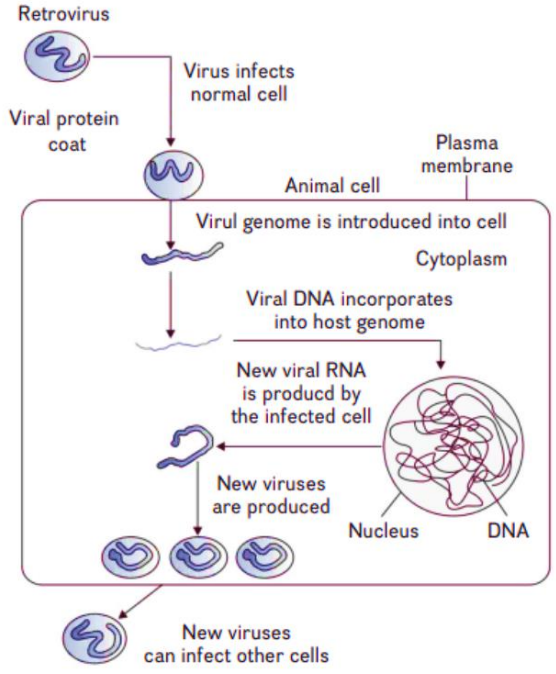
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BHD433

32. Refer to the given figure showing replication of retrovirus. How is it different from the replication of other viruses?



(1) Viral DNA is produced from viral RNA by reverse transcriptase.

(2) Viral RNA is produced dsRNA by RNA polymerase.

(3) Viral DNA is produced from viral RNA by DNA polymerase.

(4) Viral DNA is produced from host DNA by DNA polymerase.

BHD434

33. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	AIDS	(1)	Retrovirus
(B)	HIV	(2)	Enzyme
(C)	Reverse transcriptase	(3)	Diagnostic technique
(D)	ELISA	(4)	Immunodeficiency disorder

	A	B	C	D
(1)	4	1	2	3
(2)	1	2	4	3
(3)	3	4	1	2
(4)	2	3	4	1

BHD435

34. **Assertion** : AIDS is caused by Human Immunodeficiency Virus (HIV).

Reason : It is a member of group 'retroviruses'.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BHD436

35. **Assertion** : Transmission of HIV infection generally occurs by sexual contact with infected person.

Reason : HIV is not transmitted by the transfusion of contaminated blood and blood products.

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(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Both Assertion and Reason are false.

BHD437

36. **Assertion :** AIDS leads to a progressive decrease in number of helper T-lymphocytes in the infected person.
Reason : HIV virus replicated and produces progeny virus in helper T-lymphocytes which are released in blood.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Both Assertion and Reason are false.

BHD438

37. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Contact inhibition	(1)	Consists of neoplastic cells
(B)	Benign tumour	(2)	Property of normal cells to inhibit uncontrolled growth of other cells
(C)	Malignant tumour	(3)	Property of cancerous cells to form new tumour at distant sites
(D)	Metastasis	(4)	Remains confined to original location

	A	B	C	D
(1)	1	3	4	2
(2)	4	2	3	1
(3)	2	4	1	3
(4)	3	1	2	4

BHD439

38. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I (Type of Carcinogen)		Column-II (Example)
(A)	Physical agent	(1)	Oncogenic virus
(B)	Chemical agent	(2)	UV rays
(C)	Biological agent	(3)	Tobacco
		(4)	Gamma rays

	A	B	C
(1)	4,2	3	1
(2)	4,1	2	3
(3)	2	4,3	1
(4)	3	2	4,1

BHD440

39. Choose the correct statements about carcinogens.
(I) Carcinogens transform normal cells into cancerous cells.
(II) These carcinogens could be physical, chemical or biological.
(III) Inoizing radiations like UV-rays damage DNA leading to neoplastic transformations.
(1) (I) and (III) (2) (I) and (II)
(3) (III) and (IV) (4) All of these

BHD441

40. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		Column-II
(A)	Biopsy	(1)	Three dimensional image using X-rays
(B)	Radiography	(2)	Histopathological study
(C)	Computed tomography	(3)	Use of strong magnetic fields and non-ionising radiations
(D)	MRI	(4)	Use of X-rays

	A	B	C	D
(1)	4	2	3	1
(2)	2	4	1	3
(3)	3	1	4	2
(4)	1	3	2	4

BHD442

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41. **Assertion :** X-rays and γ-rays are called carcinogens.

Reason : Carcinogens transform normal cells into cancerous neoplastic cells.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BHD443

42. **Assertion :** Computed tomography can be used for the early detection of cancer of internal organs.

Reason : Computed tomography uses UV-rays to generate a three-dimensional image of the internals of an object.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BHD444

43. **Assertion :** The patients of cancer are given α-interferon.

Reason : α-interferon is a biological response modifier.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BHD445

44. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		**Column-II**
(A)	Smack	(1)	Hallucination
(B)	Cocaine	(2)	Depressant
(C)	Datura	(3)	Pain killer
(D)	Morphine	(4)	Stimulant

	A	**B**	**C**	**D**
(1)	2	4	1	3
(2)	3	2	4	1
(3)	1	3	2	4
(4)	4	1	3	2

BHD446

45. Choose the correct statements.

(I) Tobacco is smoked, chewed or used as a snuff.

(II) Tobacco contains nicotine, an alkaloid.

(III) Smoking decreases heart rate.

(IV) Tobacco chewing is associated with increased risk of cancer of the oral cavity.

(1) (I) and (II)

(2) (III) and (IV)

(3) (I), (II) and (IV)

(4) All of these

BHD447

46. Match Column-I with Column-II and choose the correct option from the codes given below.

	Column-I		**Column-II**
(A)	Adolescence	(1)	Lung cancer
(B)	Addiction	(2)	Abrupt discontinuation of regular close of drug
(C)	Smoking	(3)	Bridge linking childhood and adulthood
(D)	Withdrawal syndrome	(4)	Psychological euphoria associated with drugs

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	A	B	C	D
(1)	4	2	3	1
(2)	3	4	1	2
(3)	1	3	4	2
(4)	2	1	3	2

BHD448

47. **Assertion :** Cocaine is obtained from coca plant.

Reason : It has a potent stimulating action on central nervous system.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BHD449

48. **Assertion :** Withdrawal syndrome is characterised by anxiety, shakiness, nausea and sweating.

Reason : Withdrawal syndrome is not relieved even when use of drugs is resumed again.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BHD450

49. **Assertion :** Those who take drugs intravenously are much more likely to acquire AIDS and Hepatitis B.

Reason : The viruses of AIDS and Hepatitis B are transferred from one person to another by sharing of infected needles and syringes.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

BHD451

Exercise - II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	4	3	1	2	1	1	3	4	1	1	4	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	3	2	4	1	2	2	2	4	1	1	2	2	1	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	1	1	2	3	1	3	1	2	2	1	3	1	1	3
Question	46	47	48	49											
Answer	2	2	3	1											

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Exercise – I

Evolution

1. Theory of **natural selection** given by Darwin is based on :-
(1) Inheritance of acquired characters
(2) Random mutation
(3) Use and disuse of organs
(4) Enormous power of fertility in organisms, struggle for existence and survival of fittest

BEV510

2. Which among the following has largest brain capacity?
(1) *Australopithecus*
(2) *Homo habilis*
(3) Neanderthal man
(4) *Homo erectus*

BEV511

3. Select the correct option among the following statements :-
(a) The fitness, according to Darwin, refers ultimately and only to reproductive fitness
(b)Those who are better fit in an environment, leave more progeny than others.
(c) Alfred Wallace, a biochemist worked in Malay Archipelago.
(d) The geological history of earth closely correlates with the biological history of earth.
(1) Only a and c (2) Only a, b and c
(3) Only a, b and d (4) Only b, c and d

BEV512

4. **Assertion:** The Big Bang theory attempts to explain to us the origin of universe.
Reason: It talks of a singular huge explosion unimaginable in physical terms.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BEV513

5. All the following are examples of homologous organs except :-
(1) Arm of man and flipper of whale
(2) Brain of vertebrates
(3) Eye of an octopus and eye of a mammal
(4) Thorns of *Bougainvillea* and tendrils of *cucurbita*

BEV514

6. **Assertion:** Analogous organs are result of convergent evolution.
Reason: Different structures evolving for the same function and hence having similarity.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BEV515

7. **Assertion:** Tasmanian wolf and placental wolf are the good example of convergent evolution.
Reason: More than one adaptive radiation appeared to have occurred in an isolated geographical area (representing different habitats) one call this convergent evolution.
(1) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(2) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BEV516

8. **Assertion:** Excess use of herbicide and pesticide resulted in selection of resistant varieties.

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Reason: Resistant appeared due to anthropogenic action.

(1) If both the assertion and reason are false.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

BEV517

9. "Darwin's finches" are an excellent example of :-
(1) Connecting links
(2) Adaptive radiation
(3) Seasonal migration
(4) Brood parasitism

BEV518

10. Similarities in organism with different genotype indicates :-
(1) Micro evolution
(2) Macro evolution
(3) Divergent evolution
(4) Convergent evolution

BEV519

11. Select the correct matching pair from column-I and column-II :-

	Column-I	Column-II
(1)	Darwin's finches	Galapagos island of South America
(2)	Bandicoot	Placental mammal
(3)	Analogous	Similar in origin
(4)	Bobcat	Marsupial mammal

BEV520

12. Read the names of animals given below and select the correct option with respect to their category and the type of evolution they represent :-
Sugar glider, Koala, Wombat, Flying squirrel, Tiger cat and Bobcat :-
(1) 4 are marsupial - convergent evolution
(2) 5 are marsupial - parallel evolution
(3) 4 are marsupial - geographical isolation
(4) 4 are marsupial - adaptive radiation

BEV521

13. Read the following statements and answer as per instructions :
(a) Mutations are random and directionless
(b) According to Darwin, evolution is a single step process.
(c) Stabilising natural selection divides the population into two populations.
(d) Seed ferns are the ancestors of monocots and dicots.
How many of above statements are correct ?
(1) One (2) Two
(3) Four (4) Three

BEV522

14. Select the wrong one from the given statements.
(a) Some land reptiles went back into water to evolve into fish like reptiles probably 200 mya.
(b) Sauropsids rise to modern days reptiles and mammals.
(c) About 65mya the dinosaurs suddenly disappeared from the earth.
(d) The first mammals were like whales.
(1) a and b (2) c and d
(3) b and d (4) a and c

BEV523

15. Select the correct option for the given statements :-
(a) Natural selection is based on certain observations which are factual.
(b) The essence of Darwinian theory about evolution is natural selection
(c) Acquired characters do not inherit, according to Lamarck
(d) Hugo de-vries believed in single step large mutation
(1) Only a and b (2) Only c and d
(3) Only b and c (4) Only a, b and d

BEV524

16. Which of the following statement is correct ?
(1) *Homo erectus* lived in east and central Asia and used hides to protect their bodies
(2) Agriculture came around 18000 yrs back
(3) The skull of modern human resembles more closely to baby chimpanzee than adult chimpanzee
(4) *Australopithecus* probably lived in North America.

BEV525

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17. Find out correct matching pair between column-I and II :-

Column-I (Human ancestor)		Column-II (Cranial capcity)	
(a)	*Homo habilis*	(i)	900cc
(b)	Neanderthal man	(ii)	1600cc
(c)	*Homo erectus*	(iii)	650-800cc
(d)	Cromagnon man	(iv)	1400cc

(1) a-iv, b-ii, c-iii, d-i
(2) a-iii, b-iv, c-i, d-ii
(3) a-ii, b-iii, c-iv, d-i
(4) a-iii, b-iv, c-ii, d-i

BEV526

18. The story of evolution of modern man in respect to brain and language appears as :-
(1) Convergent evolution
(2) Adaptive radiation
(3) Divergent evolution
(4) Parallel evolution

BEV527

19. Read the following statements and select the correct one :-
(1) Evolution is not a stochastic process.
(2) Tyrannosaurus rex was about 10 feet in height.
(3) *Dryopithecus* was more ape-like while *Ramapithecus* was more man-like
(4) The evolution of fish like reptiles *Ichthyosaurs* took 300 mya

BEV528

20. Select the correct match between column I and II :-

	Column-I	Column-II
(1)	The first cellular form of life appeared	3000 million years ago
(2)	Neanderthal Man	Brain size-1400cc
(3)	Dinosaurs extinction	About 75 mya
(4)	Louis Pasteur	Theory of special creation

BEV529

21. Select the correct option which is right match between column-I and column-II :-

	Column-I		Column-II
(1)	Marsupials	(i)	Ancestor of mammals
(2)	Analogous organs	(ii)	Tool maker man
(3)	*Homo habilis*	(iii)	Similar in function
(4)	Synapsids	(iv)	Adaptive radiation

(1) A-iv, B-iii, C-ii, D-i
(2) A-iv, B-iii, C-i, D-ii
(3) A-iii, B-iv, C-ii, D-i
(4) A-i, B-ii, C-iv, D-iii

BEV530

22. Which one of the following factors cannot affect Hardy Weinberg's equilibrium?
(1) Natural selection (2) Mutation
(3) Genetic drift (4) Random mating

BEV531

23. **Assertion:** Australian marsupials can be taken as example of adaptive radiation.
Reason: A number of marsupials, each different from other evolved from an ancestral stock, but all within the Australian island continent.
(1) If the assertion is true but the reason is false.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(4) If both the assertion and reason are false.

BEV532

24. A population may get deviation from Hardy-Weinberg equilibrium, if it:
(1) Shows random mating
(2) Doesn't show gene migration
(3) Shows natural selection
(4) Doesn't show gene recombination

BEV533

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25. A process in which organisms of different origin, develop similar adaptations towards a common habitat, is called as:

(1) Divergent evolution

(2) Convergent evolution

(3) Adaptive radiation

(4) Natural selection

BEV534

26. If change in gene frequencies in a population occurs by chance, than it is called as:

(1) Natural selection

(2) Gene flow

(3) Gene migration

(4) Genetic drift

BEV535

27. Which of the following are analogous structures?

(1) Forelimbs of mammals

(2) Thorns of *Bougainvillea* and tendrils of *cucurbita*

(3) Eye of octopus and Eye of mammals.

(4) All of the above

BEV536

28. The Oparin-Haldane theory was experimentally proved by:

(1) Darwin (2) Miller

(3) Pasteur (4) deVries

BEV537

29. Which is a character of Neanderthal man?

(1) First human-like being

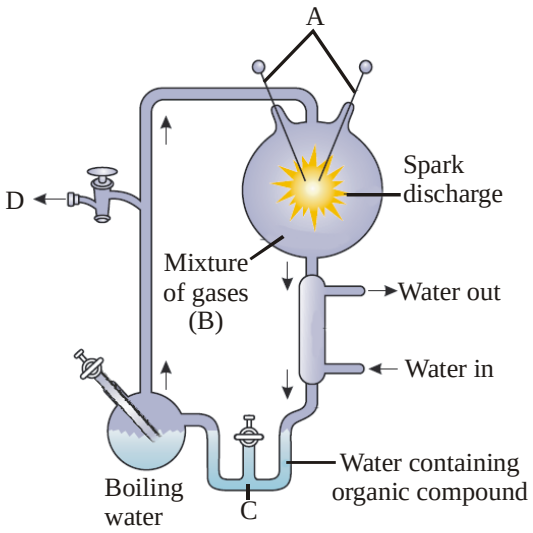
(2) Brain capacity 900cc

(3) They buried their dead

(4) They started agriculture

BEV538

30. The given diagram represents Miller's experiment. Choose the correct combination for labelling's A, B, C and D :-



(1) A-Electrodes, B-NH_3+H_2+H_2O+CH_4,
C-liquid water in trap, D-Vacuum pump

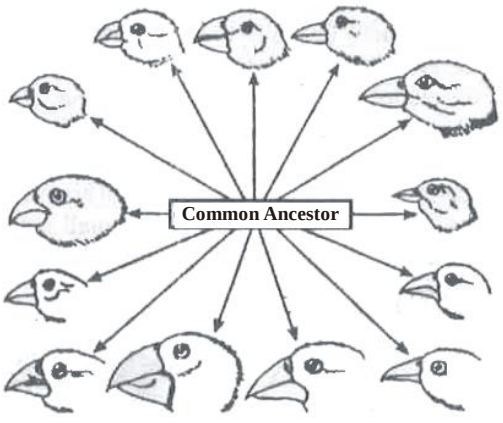
(2) A-Electrodes, B-NH_3+H_2+CO_2+CH_4,
C-liquid water in trap, D-Vacuum pump

(3) A-Electrodes, B-NH_3+H_2O,
C-liquid water in trap, D-Tap

(4) A-Electrodes, B-NH_3+O_2+H_2O + CH_4,
C-liquid water in trap, D-Vacuum pump

BEV539

31. The diversity within the wild bird species in the diagram below can be best explained by which process?



(1) Natural selection

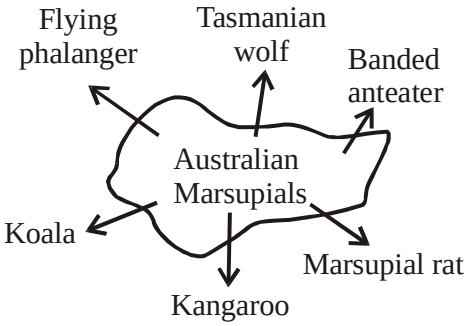
(2) Ecological succession

(3) Adaptive radiation

(4) Both (1) and (3)

BEV540

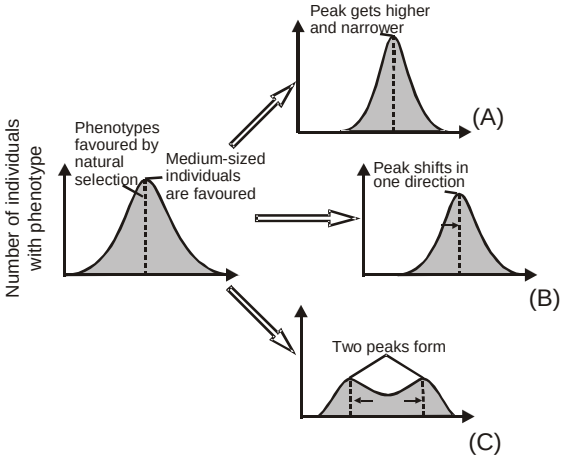
32. The given diagram of marsupials of Australia provides an example of :-



(1) Convergent evolution
(2) Parallel evolution
(3) Recapitulation
(4) Divergent evolution

BEV541

33. Following is the diagrammatic representation of the operation of natural selection of different traits. Which of the following option correctly identifies all the three graphs A, B and C.



	A	B	C
(1)	Directional	Stabilising	Disruptive
(2)	Stabilising	Directional	Disruptive
(3)	Disruptive	Stabilising	Directional
(4)	Directional	Disruptive	Stabilising

BEV542

Exercise - I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	3	2	3	1	2	4	2	4	1	4	2	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	2	4	3	2	1	4	3	3	2	4	3	2	3	1
Question	31	32	33												
Answer	4	4	2												

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Exercise – II
Evolution

1. Match the following columns.

Column-I		Column-II	
(a)	Origin of the universe	(1)	4.5 billion years ago
(b)	Origin of earth	(2)	4 billion years ago
(c)	Origin of life	(3)	1.5 billion years ago
(d)	Origin of first eukaryotes	(4)	20 billion years ago

Select the correct option.

	a	b	c	d
(1)	1	4	2	3
(2)	4	1	2	3
(3)	4	2	1	3
(4)	4	1	3	2

BEV543

2. Select the correct statement among the following.
(1) The universe is very old-almost 20,000 million years old.
(2) Huge clusters of galaxies comprise the universe. Galaxies contain stars and clouds of gas and dust.
(3) The Big Bang theory attempts to explain to us the origin of universe.
(4) All of these.

BEV544

3. How many statements are incorrect among the following?
(i) Oparin of England and Haldane of Russia proposed that the first form of life could have come from pre-existing non-living organic molecules (e.g. RNA, protein, etc.) and that formation of life was preceded by chemical evolution.
(ii) The conditions on earth were-high temperature, volcanic storms, reducing atmosphere containing CH_4, O_2, etc.
(iii) In 1952, S.L. Miller, an American scientist created similar conditions in a laboratory scale.
(iv) S.L. Miller created electric discharge in a closed flask containing CH_4, H_2, NH_3 and water vapour at 8000C.
(V) With limited evidence, the first part of the conjectured story, i.e., chemical evolution was more or less accepted.
(1) 2 (2) 3 (3) 4 (4) 5

BEV545

4. **Assertion:** New forms of life have arisen at different times in the history of earth.
Reason: life forms varied over time and certain life forms are restricted to certain geological time spans.
(1) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(2) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BEV546

5. The chronological order of human evolution from early to the recent is :
(1) Australopithecus > Ramapithecus > Homo habilis > Homo erectus
(2) Ramapithecus > Australopithecus > Homo habilis > Homo erectus
(3) Ramapithecus > Homo habilis > Australopithecus > Homo erectus
(4) Australopithecus > Homo habilis > Ramapithecus > Homo erectus

BEV547

6. Match the hominids with their correct brain size.

Column-I		Column-II	
(a)	Homo habilis	(1)	900 cc
(b)	Homo neanderthalensis	(2)	1450 cc
(c)	Homo erectus	(3)	650-800 cc
(d)	Homo sapiens	(4)	1400 cc

Select the correct option.

	a	b	c	d
(1)	4	1	3	2
(2)	3	4	1	2
(3)	4	1	2	3
(4)	2	4	1	3

BEV548

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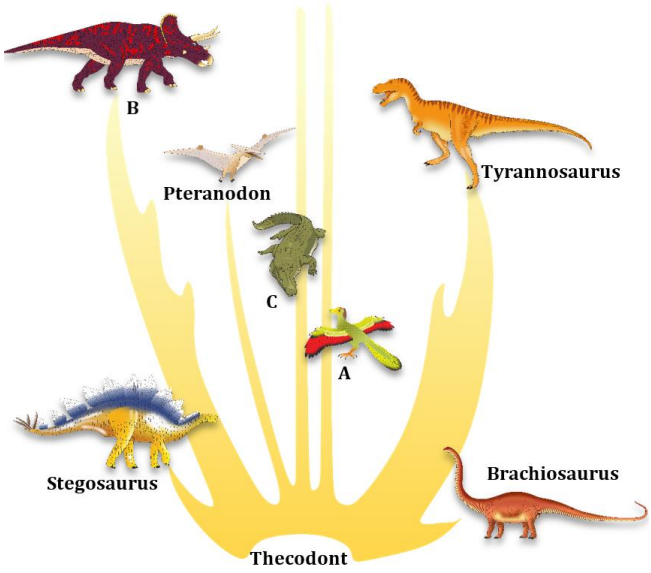
7. Consider the following statements about fossils :
(a) Rock sediments of different ages contain fossils of life forms that died during the formation of that particular layer of sediment.
(b) Fossil records are based on the sequence of occurrence of fossils on various strata of sedimentary rocks.
Select the correct option.
(1) Both a and b are true
(2) a is true, b is false
(3) Both a and b are false
(4) a is false, b is true

BEV549

8. Fossils serve as one of the evidence of evolution because :
(a) Fossil records show that various groups of organisms dominated earth during the different course of evolution.
(b) Many organisms are extinct today.
(c) Certain groups of organisms are restricted to a certain geological time period.
Select the option with all correct statements.
(1) Only a (2) Only b
(3) Both b and c (4) a, b and c

BEV550

9. Given below is the family tree of dinosaurs and their living present day counterpart organisms. Which of the given options correctly name A, B and C?



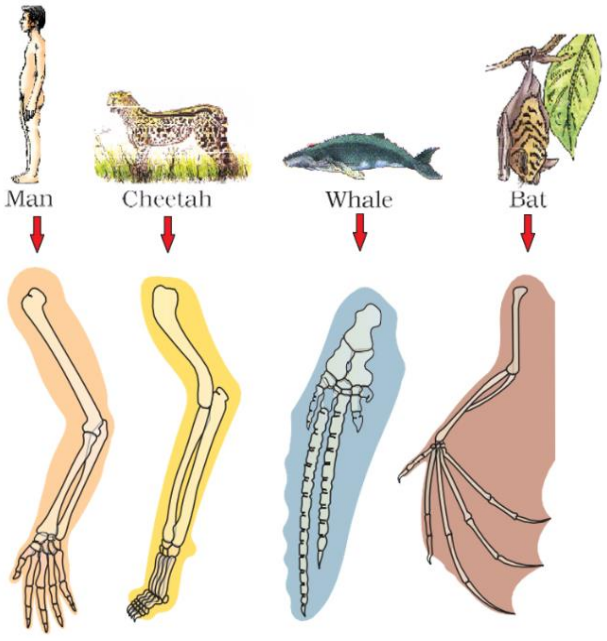
(1) A : Triceratops; B : Archaeopteryx; C : Crocodilian.
(2) A : Triceratops; B : Crocodilian; C : Archaeopteryx.
(3) A : Archaeopteryx; B : Triceratops; C : Crocodilian.
(4) A : Archaeopteryx; B : Crocodilian; C : Triceratops

BEV551

10. **Assertion:** Fitness is the end result of the ability to adapt and get selected by nature.
Reason: Some organisms are better adapted to survive in an otherwise hostile environment.
(1) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(2) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BEV552

11. The image shows the forelimbs of a whale, bat, cheeta and human (all mammals). Choose the correct option regarding the same.



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(1) The basic similarity in the arrangement of bones in forelimbs of whales, bats, cheetahs and humans reflect their common ancestry.
(2) The forelimbs of whales, bats, cheetahs and humans are analogous structures.
(3) The image represents structures that evolved from convergent evolution.
(4) The image represents vestigial structures.

BEV553

12. Match the Column-I with Column-II.

Column-I		Column-II	
(a)	Thorn of bougainvillea and pea tendril	(1)	Analogous structures
(b)	Wings of insects and birds	(2)	Vestigial structures
(c)	Hind limb bones of whales	(3)	Natural selection
(d)	Darwin's finches	(4)	Homologous structures

Select the correct option.

	a	b	c	d
(1)	1	4	2	3
(2)	4	1	2	3
(3)	4	2	1	3
(4)	4	1	3	2

BEV554

13. Match the Column-I with Column-II.

Column-I		Column-II	
(a)	Common proteins and genes present in diverse organisms	(1)	Artificial selection
(b)	Vertebrates heart or brain	(2)	Anatomical evidence of evolution
(c)	Animal husbandry and plant breeding	(3)	Natural selection
(d)	Industrial melanism	(4)	Biochemical evidence of evolution

Select the correct option.

	a	b	c	d
(1)	1	4	2	3
(2)	4	1	2	3
(3)	4	2	1	3
(4)	4	1	3	2

BEV555

14. The given image shows white-winged and dark-winged moths.



I II

(a) Image I represents the survival advantage of white winged moths in unpolluted areas.
(b) Image II represents the survival advantage of dark winged moths in polluted areas.

Select the correct option.
(1) Both a and b are true
(2) a is true and b is false
(3) Both a and b are false
(4) a is false but b is true

BEV556

15. Match the Column-I with Column-II.

Column-I		Column-II	
(a)	Herbicide resistant varieties	(1)	20 feet
(b)	Tyrannosaurus	(2)	Connecting link between reptile and birds
(c)	Archaeopteryx	(3)	A flying reptile
(d)	Pteranodon	(4)	Evolution by anthropogenic action

Select the correct option.

	a	b	c	d
(1)	1	4	2	3
(2)	4	1	2	3
(3)	4	2	1	3
(4)	4	1	3	2

BEV557

16. **Assertion :** Archaeopteryx is a connecting link between reptiles and birds.
Reason : Archaeopteryx is the earliest known bird.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.

BEV558

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17. **Assertion :** Evolution is not a directed process in the sense of determinism.
Reason : Evolution is based on chance events that occur in nature.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.

BEV559

18. Consider the following statements :
(a) Darwin compared the animals and plant species of Galapagos Islands with those of mainland of South America.
(b) He found observable and distinct similarities and differences among them.
(c) Flora and fauna of the Galapagos Islands do not exhibit resemblances to that of the South American mainland.
Select the correct option.
(1) Both a and b are true
(2) a is true but b, c are false
(3) Both a and b are false
(4) All are false

BEV560

19. Consider the following statements :
(a) Adaptive radiation occurs when numerous unexploited ecological opportunities are available.
(b) Islands have a large number of empty ecological niche as compared to the mainland present in the same geographical location.
Select the correct option.
(1) Both a and b are true
(2) a is true but b, c are true
(3) Both a and b are false
(4) All are false

BEV561

20. **Assertion:** Natural selection is the mechanism of evolution.
Reason: Those who better fit in an environment, leave more progeny than other, therefore will survive more and hence selected by nature.
(1) If the assertion is true but the reason is false.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(4) If both the assertion and reason are false.

BEV562

21. When more than one (i) appeared to have occurred in an isolated geographical area (representing different habitats), one can call this (ii).
(1) (i) Adaptive radiation;
(ii) Divergent evolution
(2) (i) Adaptive radiation;
(ii) Convergent evolution
(3) (i) Convergent evolution;
(ii) Adaptive radiation
(4) (i) Divergent evolution;
(ii) Adaptive radiation

BEV563

22. Match the placental mammals given in Column-I to their corresponding Australian marsupials given in Column-II.

Column-I		Column-II	
(a)	Mole	(1)	Marsupial mole
(b)	Anteater	(2)	Marsupial mouse
(c)	Mouse	(3)	Spotted cuscus
(d)	Lemur	(4)	Numbat

Select the correct option.

	a	b	c	d
(1)	1	4	2	3
(2)	4	1	2	3
(3)	4	2	1	3
(4)	4	1	3	2

BEV564

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23. Match the placental mammals given in Column-I to their corresponding Australian marsupials given in Column-II.

Column-I		Column-II	
(a)	Flying squirrel	(1)	Tasmanian tiger cat
(b)	Bobcat	(2)	Tasmanian wolf
(c)	Wolf	(3)	Spotted cuscus
(d)	Lemur	(4)	Flying phalanger

Select the correct option.

	a	b	c	d
(1)	1	4	2	3
(2)	4	1	2	3
(3)	4	2	1	3
(4)	4	1	3	2

BEV565

24. Consider the following statements :

(a) Evolution by natural selection started with the origin of cellular life forms with differences in metabolic capability.

(b) The theory of natural selection was given by Lamarck.

Select the correct option.

(1) Both a and b are true

(2) a is true but b us false

(3) Both a and b is false

(4) a is false but b is true

BEV566

25. Match the Column-I with Column-II.

Column-I		Column-II	
(a)	Branching descent	(1)	New species from a common descendant
(b)	Natural selection	(2)	Source of new genes
(c)	Mutation	(3)	Lamarck
(d)	Inheritance of acquired features	(4)	Tendency of organisms to become adapted to the environment

Select the correct option.

	a	b	c	d
(1)	4	1	3	2
(2)	1	4	2	3
(3)	4	1	2	3
(4)	4	2	1	3

BEV567

26. Match the Column-I with Column-II.

Column-I		Column-II	
(a)	Inheritance of acquired characters	(1)	Use and disuse
(b)	An Essay on the Principle of Population	(2)	Thomas Malthus
(c)	Evolution of long necked giraffe population	(3)	Natural selection
(d)	Industrial melanism	(4)	Lamarck

Select the correct option.

	a	b	c	d
(1)	4	1	3	2
(2)	4	1	2	3
(3)	1	4	2	3
(4)	4	2	1	3

BEV568

27. Match the aspects of Darwin's theory of natural selection in Column-I with the correct description in Column-II.

Column-I		Column-II	
(a)	Variation	(1)	Competition for limited available resources
(b)	Overproduction	(2)	Make resource utilization better
(c)	Struggle for existence	(3)	More offspring
(d)	Differential reproductive success	(4)	Geometric increase in population size

Select the correct option.

	a	b	c	d
(1)	2	4	1	3
(2)	4	1	3	2
(3)	4	1	2	3
(4)	4	2	1	3

BEV569

28. Consider the following statements :

(a) For Darwin, evolution was gradual and driven by small genetic variations.

(b) For de Vries, evolution was a single step process driven by large mutation.

Select the correct option.

(1) Both a and b are true

(2) a is true but b is false

(3) Both a and b are false

(4) a is false but b is true

BEV570

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29. **Assertion :** Evolution refers to modification in the lines of descent.
Reason : Natural selection is the equal survival success of individuals of a population.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.
BEV571

30. **Assertion :** Darwin's theory of evolution was based on the inheritance of adaptive genetic variations.
Reason : He could not recognize the mechanism of inheritance.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.
BEV572

31. Select the incorrect statement among the following :
(1) $p^2+2pq+q^2 = 1$. This is binomial expansion of $(p + q)^2$.
(2) When frequency measured, differs from expected values, the difference (direction) indicates the extent of evolutionary change.
(3) Hardy-Weinberg principle says that phenotype frequencies in a population are stable and is constant from generation to generation.
(4) The gene pool (total genes and their alleles in a population) remains a constant. This is called genetic equilibrium. Sum total of all the allelic frequencies is 1.
BEV573

32. **Assertion:** The rate of appearance of new forms is linked to the life cycle or the life span of organisms.
Reason: The organism with short life span have the ability to produce more progenies in shorter time and that have chance to more exposure with nature.
(1) If the assertion is true but the reason is false.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(4) If both the assertion and reason are false.
BEV574

33. Consider the following statements :
(a) Allele frequencies are supposed to remain constant over generations.
(b) Hardy Weinberg equation is the algebraic representation of the allele frequencies of a population.
Select the correct option.
(1) Both a and b are true
(2) a is true but b is false
(3) Both a and b are false
(4) a is false but b is true
BEV575

34. Select the incorrect match:
(1) Frequency of a dominant allele in the population-p.
(2) Frequency of a recessive allele in the population-q
(3) Frequency of a heterozygous dominant genotype-2pq.
(4) Frequency of homozygous dominant genotype-q2.
BEV576

35. Consider the following statements :
(a) Gene flow, genetic drift, mutation, natural selection and genetic recombination deviate the population from the Hardy-Weinberg equilibrium.
(b) Small population size is required to maintain constant allele frequencies over generations.

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Select the correct option.
(1) Both a and b are true
(2) a is true but b is false
(3) Both a and B are false
(4) a is false but b is true

BEV577

36. Select the correctly matched option.
(1) Gene flow : Changes in allele frequencies by a chance event.
(2) Mutation : Migration of individuals of the population.
(3) Random mating : Equal chances of each individual in a population to mate with any other individual.
(4) Natural selection : Selection of mate on the basis of morphological features.

BEV578

37. Match the terms in Column-I with the correct description in Column-II.

Column-I		Column-II	
(a)	Genetic drift	(1)	Genetic drift due to small colonizing population
(b)	Founder effect	(2)	Change in allele frequency by a chance event
(c)	Bottleneck effect	(3)	A major factor to introduce variations in sexually reproducing populations
(d)	Genetic combination	(4)	Adverse environmental factors

Select the correct option.

	a	b	c	d
(1)	1	4	2	3
(2)	2	1	4	3
(3)	3	2	4	1
(4)	4	1	3	2

BEV579

38. Natural selection in which more individuals acquire mean character value is known as ___ while the one wherein more individuals acquire peripheral character value at both ends of the distribution curve is _____.
(1) stabilizing selection; disruptive selection
(2) disruptive selection; stabilizing selection
(3) disruptive selection; directional selection
(4) stabilizing selection; directional selection

BEV580

39. **Assertion:** Homo sapiens arose in Africa and moved across continents and developed into distinct races.
Reason: After development of agriculture human settlement started.
(1) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(2) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BEV581

40. Match the Column-I with Column-II.

Column-I		Column-II	
(a)	Australopithecus	(1)	First human like being
(b)	Homo habilis	(2)	Java man
(c)	Homo erectus	(3)	Speech centre develop
(d)	Neanderthal man	(4)	Connecting link between apes and man

Select the correct option.

	a	b	c	d
(1)	1	4	2	3
(2)	4	1	2	3
(3)	2	4	1	3
(4)	4	1	3	2

BEV582

41. Consider the following statements :
(a) Homo habilis is also known as 'handy man' and could cook food using fire.
(b) Homo erectus was upright hominid and exhibited improved hunting skills.
Select the correct option.
(1) Both a and b are true
(2) a is true but b is false
(3) Both a and b are false
(4) a is false but b is true

BEV583

42. **Assertion :** Ramapithecus was more man-like while Dryopithecus was more ape-like.
Reason : Ramapithecus has a brain size of 900 cc.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.

BEV584

43. **Assertion :** Evolution of modern humans exhibited a gradual increase in brain size.
Reason : Homo habilis had the smallest brain capacity among prehistoric man.
(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.

BEV585

Exercise - II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	2	2	2	2	1	4	3	1	1	2	3	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	1	1	3	2	1	2	2	2	4	1	1	3	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43		
Answer	3	3	1	4	2	3	2	1	1	2	1	3	2		

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